CALCULUS - I (MMT226J)

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Chapter 1

LIMITS, CONTINUITY, AND DIFFERENTIABILITY

Unit I establishes the foundational concepts of differential calculus for single-variable and multivariable real functions. It begins with the rigorous analysis of limits, continuity, classifications of discontinuities, and the fundamental properties of differentiability. The unit further explores higher-order successive differentiation governed by Leibnitz's Theorem for product functions. Finally, it extends differentiation into multivariable calculus through partial differentiation, total differentials, and Euler's Theorem on homogeneous functions.

1.1 Limits of Functions and Infinitesimals

The concept of a limit is fundamental to mathematical analysis and calculus. It describes the behavior of a function near a point rather than at that point itself.

Definition 1.1.1 (ϵ–δ Definition of a Limit)**.** Let $f : A \to \mathbb{R}$ be a function defined on a subset $A \subseteq \mathbb{R}$, and let a be a limit point of A. We say that the limit of $f(x)$ as x approaches a is L, written as

$$\lim_{x \to a} f(x) = L,$$

if for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$

Definition 1.1.2 (One-Sided Limits)**.** (i) **Left-Hand Limit (LHL):** Written as $\lim_{x \to a^-} f(x) = L_1$, defined by:

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } a - \delta < x < a \implies |f(x) - L_1| < \epsilon.$$

(ii) **Right-Hand Limit (RHL):** Written as $\lim_{x \to a^+} f(x) = L_2$, defined by:

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } a < x < a + \delta \implies |f(x) - L_2| < \epsilon.$$

A two-sided limit $\lim_{x \to a} f(x)$ exists if and only if both L_1 and L_2 exist and $L_1 = L_2 = L$.

Definition 1.1.3 (Infinitesimals)**.** A function $\alpha(x)$ is called an **infinitesimal** as $x \to a$ if

$$\lim_{x \to a} \alpha(x) = 0.$$

If $\alpha(x)$ and $\beta(x)$ are two infinitesimals as $x \to a$, we say that:

- α is an infinitesimal of **higher order** than β if $\lim_{x\to a} \dfrac{\alpha(x)}{\beta(x)} = 0$.
- α and β are of the **same order** if $\lim_{x\to a} \dfrac{\alpha(x)}{\beta(x)} = c \neq 0$.

Example 1.1.4. Prove using the ϵ–δ definition that $\lim_{x\to 5}(3x-4) = 11$.

Solution. Let $\epsilon > 0$. We seek $\delta > 0$ such that, whenever $0 < |x-5| < \delta$, then $|(3x-4)-11| < \epsilon$.

We first estimate the error:

$$|(3x-4)-11| = |3x-15| = 3|x-5|.$$

It is therefore enough to require

$$3|x-5| < \epsilon \iff |x-5| < \frac{\epsilon}{3}.$$

Choose $\delta = \dfrac{\epsilon}{3}$.

Verification: If $0 < |x-5| < \delta$, then

$$|(3x-4)-11| = 3|x-5| < 3\cdot\frac{\epsilon}{3} = \epsilon.$$

Hence, $\lim_{x\to 5}(3x-4) = 11$. ■

Example 1.1.5. Prove using the ϵ–δ definition that $\lim_{x\to 3}(x^2-2x) = 3$.

Solution. Let $\epsilon > 0$. Since

$$|(x^2-2x)-3| = |(x-3)(x+1)| = |x-3|\,|x+1|,$$

we first control the factor $|x+1|$. If $|x-3| < 1$, then $2 < x < 4$, and hence $|x+1| < 5$. Therefore, whenever $|x-3| < 1$,

$$|(x^2-2x)-3| < 5|x-3|.$$

Choose $\delta = \min\{1, \epsilon/5\}$. Then $0 < |x-3| < \delta$ gives

$$|(x^2-2x)-3| < 5|x-3| < 5\frac{\epsilon}{5} = \epsilon.$$

Thus $\lim_{x\to 3}(x^2-2x) = 3$. ■

Example 1.1.6. Using the ϵ–δ definition, prove that $\lim_{x\to 2} \dfrac{1}{x} = \dfrac{1}{2}$.

Solution. Let $\epsilon > 0$. We have

$$\left|\frac{1}{x} - \frac{1}{2}\right| = \frac{|x-2|}{2|x|}.$$

If $|x-2| < 1$, then $1 < x < 3$, so $|x| > 1$. Hence

$$\left|\frac{1}{x} - \frac{1}{2}\right| < \frac{|x-2|}{2}.$$

Choose $\delta = \min\{1, 2\epsilon\}$. Then $0 < |x-2| < \delta$ implies

$$\left|\frac{1}{x} - \frac{1}{2}\right| < \frac{|x-2|}{2} < \epsilon.$$

Therefore $\lim_{x\to 2} \dfrac{1}{x} = \dfrac{1}{2}$. ■

Proving Non-Existence of Limits via ϵ–δ Definition

To prove that $\lim_{x\to a} f(x)$ does **not** exist from the ϵ–δ definition, it is often convenient to argue by contradiction.

Suppose there exists a real number L such that $\lim_{x\to a} f(x) = L$. By definition, for every $\epsilon > 0$, there exists a $\delta > 0$ such that $0 < |x-a| < \delta \implies |f(x) - L| < \epsilon$. If we can choose a specific $\epsilon > 0$ and find two points x_1, x_2 within $0 < |x-a| < \delta$ such that the triangle inequality

$$|f(x_1) - f(x_2)| \le |f(x_1) - L| + |f(x_2) - L| < 2\epsilon$$

is violated, we obtain a contradiction, proving that no such limit L exists.

Example 1.1.7. Using the ϵ–δ definition, prove that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist.

Solution. Assume by contradiction that $\lim_{x\to 0} \frac{|x|}{x} = L$ for some real number L.

Choose $\epsilon = 1$. By the ϵ–δ definition of a limit, there exists some $\delta > 0$ such that for all x satisfying $0 < |x| < \delta$, we have:

$$\left| \frac{|x|}{x} - L \right| < 1.$$

Choose the two points in this punctured neighborhood $0 < |x| < \delta$:

$$x_1 = \frac{\delta}{2} \quad \text{and} \quad x_2 = -\frac{\delta}{2}.$$

Both points satisfy $0 < |x_1| < \delta$ and $0 < |x_2| < \delta$. At these points,

$$f(x_1) = \frac{|\delta/2|}{\delta/2} = 1 \quad \text{and} \quad f(x_2) = \frac{|-\delta/2|}{-\delta/2} = -1.$$

Applying the triangle inequality:

$$2 = |1 - (-1)| = |f(x_1) - f(x_2)| = |(f(x_1) - L) + (L - f(x_2))| \le |f(x_1) - L| + |f(x_2) - L|.$$

Since both $|f(x_1) - L| < 1$ and $|f(x_2) - L| < 1$, we get:

$$2 < 1 + 1 = 2,$$

a contradiction. Hence, no real number L exists, and $\lim_{x\to 0} \frac{|x|}{x}$ does not exist. ■

Example 1.1.8 (Infinitely Oscillating Function)**.** Using the ϵ–δ definition, prove that $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist.

Solution. Assume by contradiction that $\lim_{x\to 0} \sin\left(\frac{1}{x}\right) = L$ for some $L \in \mathbb{R}$.

Choose $\epsilon = \frac{1}{2}$. By the limit definition, there must exist a $\delta > 0$ such that:

$$0 < |x| < \delta \implies \left| \sin\left(\frac{1}{x}\right) - L \right| < \frac{1}{2}.$$

By the Archimedean property of real numbers, we can choose a sufficiently large integer $n \in \mathbb{N}$ such that both points:

$$x_1 = \frac{1}{\frac{\pi}{2} + 2n\pi} \quad \text{and} \quad x_2 = \frac{1}{\frac{3\pi}{2} + 2n\pi}$$

satisfy $0 < |x_1| < \delta$ and $0 < |x_2| < \delta$.

At x_1 and x_2,

$$f(x_1) = \sin\left(\frac{\pi}{2} + 2n\pi\right) = 1, \quad \text{and} \quad f(x_2) = \sin\left(\frac{3\pi}{2} + 2n\pi\right) = -1.$$

Using the triangle inequality:

$$2 = |1 - (-1)| = |f(x_1) - f(x_2)| \leq |f(x_1) - L| + |f(x_2) - L| < \frac{1}{2} + \frac{1}{2} = 1,$$

a contradiction. Hence, no such limit L exists, and $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist. ■

Example 1.1.9. Let $f : \mathbb{R} \to \mathbb{R}$ be the Dirichlet function defined by:

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Prove using the ϵ–δ definition that $\lim_{x\to 0} f(x)$ does not exist.

Solution. Assume by contradiction that $\lim_{x\to 0} f(x) = L$ for some real number L.

Choose $\epsilon = \frac{1}{2}$. By definition of a limit, there exists some $\delta > 0$ such that:

$$0 < |x| < \delta \implies |f(x) - L| < \frac{1}{2}.$$

By the density of rational and irrational numbers in $\mathbb{R}$, the punctured neighborhood $(-\delta, \delta) \setminus \{0\}$ contains at least one rational number $q \in \mathbb{Q}$ and at least one irrational number $r \in \mathbb{R} \setminus \mathbb{Q}$.

Evaluating f at these points yields $f(q) = 1$ and $f(r) = 0$. Applying the triangle inequality:

$$1 = |1 - 0| = |f(q) - f(r)| \leq |f(q) - L| + |f(r) - L| < \frac{1}{2} + \frac{1}{2} = 1,$$

a contradiction. Therefore, no real number L can satisfy the definition, and $\lim_{x\to 0} f(x)$ does not exist. ■

Theorem 1.1.10. *If the limit of a function $f(x)$ as x approaches a exists, then this limit is unique.*

Proof. Assume, for the sake of contradiction, that $\lim_{x\to a} f(x) = L_1$ and $\lim_{x\to a} f(x) = L_2$ where $L_1 \neq L_2$.

By the definition of a limit, there exists a $\delta > 0$ such that for any x satisfying $0 < |x - a| < \delta$, we must have both:

$$|f(x) - L_1| < \epsilon \quad \text{and} \quad |f(x) - L_2| < \epsilon$$

Using the triangle inequality, we can write:

$$\begin{aligned} |L_1 - L_2| &= |(L_1 - f(x)) + (f(x) - L_2)| \\ &\leq |f(x) - L_1| + |f(x) - L_2| \\ &< \epsilon + \epsilon = 2\epsilon \end{aligned}$$

Let $\epsilon = \frac{|L_1 - L_2|}{2}$. Since $L_1 \neq L_2$, we know that $\epsilon > 0$.

Substituting our choice of ϵ:

$$|L_1 - L_2| < 2\left(\frac{|L_1 - L_2|}{2}\right) = |L_1 - L_2|$$

This results in the contradiction $|L_1 - L_2| < |L_1 - L_2|$. This contradiction proves that the limit is unique. ■

Theorem 1.1.11 (The Algebra of Limits). *Let c be a real number, and let $f(x)$ and $g(x)$ be two functions such that*

$$\lim_{x\to c} f(x) = L \quad \textit{and} \quad \lim_{x\to c} g(x) = M$$

where L and M are finite real numbers. Then the following properties hold:

1. ***Sum Rule:*** *The limit of the sum is the sum of the limits.*
$$\lim_{x\to c}[f(x) + g(x)] = L + M$$
2. ***Difference Rule:*** *The limit of the difference is the difference of the limits.*
$$\lim_{x\to c}[f(x) - g(x)] = L - M$$
3. ***Product Rule:*** *The limit of the product is the product of the limits.*
$$\lim_{x\to c}[f(x)g(x)] = LM$$
4. ***Quotient Rule:*** *The limit of the quotient is the quotient of the limits, provided the limit of the denominator is not zero.*
$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad \textit{provided } M \neq 0$$

Proof of the Sum Rule. Let $\epsilon > 0$ be given. We want to find a $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|(f(x) + g(x)) - (L + M)| < \epsilon$.

By the definition of the limits for f and g, we know that for $\epsilon/2 > 0$, there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that:

- if $0 < |x - c| < \delta_1$, then $|f(x) - L| < \epsilon/2$.
- if $0 < |x - c| < \delta_2$, then $|g(x) - M| < \epsilon/2$.

Let us choose $\delta = \min(\delta_1, \delta_2)$. Then for any x satisfying $0 < |x - c| < \delta$, both of the above inequalities hold. Using the triangle inequality, we have:

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

The difference rule follows by the same argument, replacing g by $-g$.

■

Proof of the Product Rule. Let $\epsilon > 0$ be given. We want to show $|f(x)g(x) - LM| < \epsilon$.

$$\begin{aligned} |f(x)g(x) - LM| &= |f(x)g(x) - Lg(x) + Lg(x) - LM| \\ &= |g(x)(f(x) - L) + L(g(x) - M)| \\ &\leq |g(x)||f(x) - L| + |L||g(x) - M| \end{aligned}$$

Since $\lim_{x\to c} g(x) = M$, there exists a $\delta_1 > 0$ such that if $0 < |x - c| < \delta_1$, then $|g(x) - M| < 1$, which implies $|g(x)| < |M| + 1$. Let $K = |M| + 1$. Now, for a given $\epsilon > 0$, we can find δ_2 and δ_3 such that:

- if $0 < |x - c| < \delta_2$, then $|f(x) - L| < \dfrac{\epsilon}{2K}$.

- if $0 < |x-c| < \delta_3$, then $|g(x) - M| < \dfrac{\epsilon}{2(|L|+1)}$. (We use $|L|+1$ to avoid division by zero if $L = 0$).

Choose $\delta = \min(\delta_1, \delta_2, \delta_3)$. If $0 < |x-c| < \delta$, then:

$$\begin{aligned} |f(x)g(x) - LM| &\leq |g(x)||f(x) - L| + |L||g(x) - M| \\ &< K \cdot \frac{\epsilon}{2K} + |L| \cdot \frac{\epsilon}{2(|L|+1)} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

Therefore, $\lim_{x \to c}[f(x)g(x)] = LM$. ■

Proof of the Quotient Rule. It suffices to first prove that $\lim_{x\to c} \dfrac{1}{g(x)} = \dfrac{1}{M}$ for $M \neq 0$, and then apply the Product Rule to $f(x) \cdot \dfrac{1}{g(x)}$.

Let $\epsilon > 0$ be given. We want to show $\left| \dfrac{1}{g(x)} - \dfrac{1}{M} \right| < \epsilon$.

$$\left| \frac{M - g(x)}{Mg(x)} \right| = \frac{|g(x) - M|}{|M||g(x)|}$$

Since $M \neq 0$, $|M|/2 > 0$. There exists a $\delta_1 > 0$ such that if $0 < |x-c| < \delta_1$, then $|g(x) - M| < |M|/2$. By the reverse triangle inequality, this implies $|g(x)| > |M| - |g(x) - M| > |M| - |M|/2 = |M|/2$. Therefore, $\dfrac{1}{|g(x)|} < \dfrac{2}{|M|}$. Now, there exists a $\delta_2 > 0$ such that if $0 < |x-c| < \delta_2$, then $|g(x) - M| < \dfrac{\epsilon|M|^2}{2}$. Choose $\delta = \min(\delta_1, \delta_2)$. If $0 < |x-c| < \delta$, then:

$$\frac{|g(x) - M|}{|M||g(x)|} < \frac{|g(x) - M|}{|M|(|M|/2)} = \frac{2|g(x) - M|}{|M|^2} < \frac{2}{|M|^2} \cdot \frac{\epsilon|M|^2}{2} = \epsilon$$

So, $\lim_{x\to c} \dfrac{1}{g(x)} = \dfrac{1}{M}$. By the Product Rule, $\lim_{x\to c} \dfrac{f(x)}{g(x)} = \left(\lim_{x\to c} f(x)\right)\left(\lim_{x\to c} \dfrac{1}{g(x)}\right) = L \cdot \dfrac{1}{M} = \dfrac{L}{M}$. ■

1.2 Continuity and Types of Discontinuities

Definition 1.2.1 (Continuity at a Point). A function $f : A \to \mathbb{R}$ is said to be **continuous at a point** $a \in A$ if:

(i) $f(a)$ is defined.

(ii) $\lim_{x\to a} f(x)$ exists finitely.

(iii) $\lim_{x\to a} f(x) = f(a)$.

In ϵ–δ terms: f is continuous at a if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$,

$$|x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

Definition 1.2.2 (Continuity on an Interval).

- **Open Interval** (a, b)**:** f is continuous on (a, b) if it is continuous at every point $c \in (a, b)$.
- **Closed Interval** $[a, b]$**:** f is continuous on $[a, b]$ if it is continuous on (a, b), right-continuous at a ($\lim_{x\to a^+} f(x) = f(a)$), and left-continuous at b ($\lim_{x\to b^-} f(x) = f(b)$).

1.2.1 Classification of Discontinuities

If $f(x)$ fails to be continuous at $x = a$, then a is called a **point of discontinuity**. Discontinuities are classified into the following types:

1. **Removable Discontinuity:** The limit $\lim_{x\to a} f(x) = L$ exists as a finite real number, but either $f(a)$ is undefined or $L \neq f(a)$. The discontinuity can be "removed" by defining/redefining $f(a) = L$.

2. **Discontinuity of the First Kind (Jump Discontinuity):** Both one-sided limits $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ exist finitely, but are unequal:
$$\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x).$$
The non-zero quantity $J = |\lim_{x\to a^+} f(x) - \lim_{x\to a^-} f(x)|$ is called the **jump** of f at a.

3. **Discontinuity of the Second Kind (Essential / Infinite Discontinuity):** At least one of the one-sided limits $\lim_{x\to a^-} f(x)$ or $\lim_{x\to a^+} f(x)$ does not exist or becomes infinite ($\pm\infty$).

4. **Oscillatory Discontinuity:** The function values oscillate endlessly between two bounds as $x \to a$, preventing a unique limit from being reached.

Example 1.2.3 (See Figure 1 in the Atlas)**.** Classify the discontinuity of $f(x) = \dfrac{x^2-4}{x-2}$ at $x = 2$.

Solution. The function is undefined at $x = 2$. Simplify for $x \neq 2$:
$$f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2.$$
The limit exists:
$$\lim_{x\to 2} f(x) = \lim_{x\to 2}(x+2) = 4.$$
Since the limit exists but $f(2)$ is undefined, the discontinuity is **removable**. ∎

Example 1.2.4 (See Figure 2 in the Atlas)**.** Classify the discontinuity of the function
$$f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ x+3 & \text{if } x \geq 2 \end{cases}$$
at $x = 2$.

Solution. The one-sided limits are
$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} x^2 = 2^2 = 4$$
$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} (x+3) = 2 + 3 = 5$$
Since $4 \neq 5$, the left-hand and right-hand limits exist but are not equal.
Function value: $f(2) = 2 + 3 = 5$.
The discontinuity at $x = 2$ is a **jump discontinuity**. ∎

Example 1.2.5 (See Figure 3 in the Atlas)**.** Classify the discontinuity of $g(x) = \dfrac{1}{(x-3)^2}$ at $x = 3$.

Solution. Examine the limit as x approaches 3:

$$\lim_{x\to 3} \frac{1}{(x-3)^2} = +\infty$$

The function is undefined at $x = 3$. As $x \to 3$, the function values increase without bound. This is an **infinite discontinuity** (vertical asymptote at $x = 3$). ■

Example 1.2.6 (See Figure 4 in the Atlas)**.** Classify the discontinuity of $h(x) = \sin\left(\frac{1}{x}\right)$ at $x = 0$.

Solution. Consider the behavior as $x \to 0$:

- As $x \to 0^+$, $\frac{1}{x} \to +\infty$ and $\sin(1/x)$ oscillates between -1 and 1
- As $x \to 0^-$, $\frac{1}{x} \to -\infty$ and $\sin(1/x)$ oscillates between -1 and 1

The limit $\lim_{x\to 0} \sin(1/x)$ does not exist because oscillations become increasingly rapid.
This is an **oscillating discontinuity** at $x = 0$. ■

Example 1.2.7. Examine the following function for continuity at the origin.

$$f(x) = \begin{cases} \dfrac{xe^{1/x}}{1+e^{1/x}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Solution. $\lim_{x\to 0-} f(x) = \lim_{x\to 0-} \frac{xe^{1/x}}{1+e^{1/x}} = 0$ and $\lim_{x\to 0+} f(x) = \lim_{x\to 0+} \frac{x}{e^{-1/x}+1} = 0$ Also, $f(0) = 0$. Thus, the function is continuous at the origin. ■

Example 1.2.8 (See Figures 1–4 in the Atlas)**.** Classify the discontinuities of the following functions at $x = 0$:

(i) $f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

(ii) $g(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}$

(iii) $h(x) = \dfrac{1}{x^2}, \quad x \neq 0$

(iv) $k(x) = \begin{cases} \sin\left(\dfrac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$

Solution. (i) Since $\lim_{x\to 0} \frac{\sin x}{x} = 1$, but $f(0) = 0 \neq 1$, the limit exists finitely but differs from $f(0)$. Thus, $x = 0$ is a **removable discontinuity**.

(ii) Evaluating the one-sided limits:

$$\lim_{x\to 0^+} g(x) = 1 \quad \text{and} \quad \lim_{x\to 0^-} g(x) = -1.$$

Both one-sided limits exist finitely, but $\lim_{x\to 0^+} g(x) \neq \lim_{x\to 0^-} g(x)$. Hence, $x = 0$ is a **discontinuity of the first kind (jump discontinuity)** with jump $J = |1 - (-1)| = 2$.

(iii) As $x \to 0$, $\lim_{x\to 0} h(x) = \lim_{x\to 0} \frac{1}{x^2} = +\infty$. Since the limit is infinite, $x = 0$ is a **discontinuity of the second kind (infinite discontinuity)**.

(iv) As $x \to 0$, $1/x \to \pm\infty$, causing $\sin(1/x)$ to oscillate infinitely between -1 and 1. Neither one-sided limit exists. Thus, $x = 0$ is an **oscillatory discontinuity**.

■

Example 1.2.9. Find the value of k so that the function $f(x)$ is continuous at $x = 3$:

$$f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

Solution. For $f(x)$ to be continuous at $x = 3$, we require:

$$\lim_{x\to 3} f(x) = f(3).$$

Evaluate the limit as $x \to 3$:

$$\lim_{x\to 3} \frac{x^2 - 9}{x - 3} = \lim_{x\to 3} \frac{(x-3)(x+3)}{x - 3} = \lim_{x\to 3}(x + 3) = 6.$$

Since $f(3) = k$, setting $\lim_{x\to 3} f(x) = f(3)$ yields:

$$k = 6.$$

■

Example 1.2.10 (Dirichlet Function – Everywhere Discontinuous)**.** Consider Dirichlet's function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Show that $f(x)$ is discontinuous at every real point $x \in \mathbb{R}$.

Solution. Let $a \in \mathbb{R}$ be any real number. Choose $\epsilon = \frac{1}{2}$.

By the density of rational and irrational numbers in $\mathbb{R}$, for any $\delta > 0$:

- The neighborhood $(a - \delta, a + \delta)$ contains at least one rational $q \in \mathbb{Q}$, where $f(q) = 1$.
- The neighborhood $(a - \delta, a + \delta)$ contains at least one irrational $r \in \mathbb{R} \setminus \mathbb{Q}$, where $f(r) = 0$.

Case 1 (a is rational): Here $f(a) = 1$. Choose $r \in \mathbb{R} \setminus \mathbb{Q}$ within $(a - \delta, a + \delta)$. Then:

$$|f(r) - f(a)| = |0 - 1| = 1 > \frac{1}{2} = \epsilon.$$

Case 2 (a is irrational): Here $f(a) = 0$. Choose $q \in \mathbb{Q}$ within $(a - \delta, a + \delta)$. Then:

$$|f(q) - f(a)| = |1 - 0| = 1 > \frac{1}{2} = \epsilon.$$

In both cases, no $\delta > 0$ can satisfy the ϵ–δ definition of continuity. Thus, $\lim_{x\to a} f(x)$ does not exist for any $a \in \mathbb{R}$, making f discontinuous everywhere. ■

Example 1.2.11 (Thomae's Function / Popcorn Function)**.** Consider Thomae's function $f : (0,1) \to \mathbb{R}$ defined by:

$$f(x) = \begin{cases} \frac{1}{q}, & \text{if } x = \frac{p}{q} \in \mathbb{Q} \text{ in lowest terms } (p, q \in \mathbb{N},\ \gcd(p,q) = 1) \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Show that f is continuous at every irrational point and discontinuous at every rational point in $(0,1)$.

Solution. **Discontinuity at Rational Points:** Let $a = \frac{p}{q} \in \mathbb{Q} \cap (0,1)$. Then $f(a) = \frac{1}{q} > 0$. Every neighborhood of a contains irrational numbers r where $f(r) = 0$. Choosing $\epsilon = \frac{1}{2q}$, for any $\delta > 0$ there exists an irrational $r \in (a - \delta, a + \delta)$ such that:

$$|f(r) - f(a)| = \left|0 - \frac{1}{q}\right| = \frac{1}{q} > \frac{1}{2q} = \epsilon.$$

Hence, f is discontinuous at every rational point in $(0,1)$.

Continuity at Irrational Points: Let $a \in (0,1) \setminus \mathbb{Q}$, so $f(a) = 0$. Given $\epsilon > 0$, choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$. In the interval $(0,1)$, there are only finitely many rational numbers with denominators less than N.

Let $\delta > 0$ be the distance from a to the nearest rational number with denominator less than N. If $0 < |x - a| < \delta$:

- If x is irrational, $|f(x) - f(a)| = |0 - 0| = 0 < \epsilon$.
- If $x = p'/q'$ is rational, its denominator q' must be $\geq N$ (by our choice of δ). Thus:

$$|f(x) - f(a)| = \left|\frac{1}{q'} - 0\right| = \frac{1}{q'} \leq \frac{1}{N} < \epsilon.$$

Therefore, $\lim_{x \to a} f(x) = 0 = f(a)$, proving that f is continuous at every irrational point. ■

1.2.2 Piecewise Continuity

A function $f(x)$ is said to be piecewise continuous in an interval I, if the interval can be subdivided into a finite number of subintervals such that $f(x)$ is continuous in each of the subintervals and the limits of $f(x)$ as x approaches the end points of each subinterval are finite. For example, the greatest integer function $f(x) = [x]$ defined on $[-1,3]$ is piecewise continuous on $[-1,3]$.

1.3 Differentiability of Functions

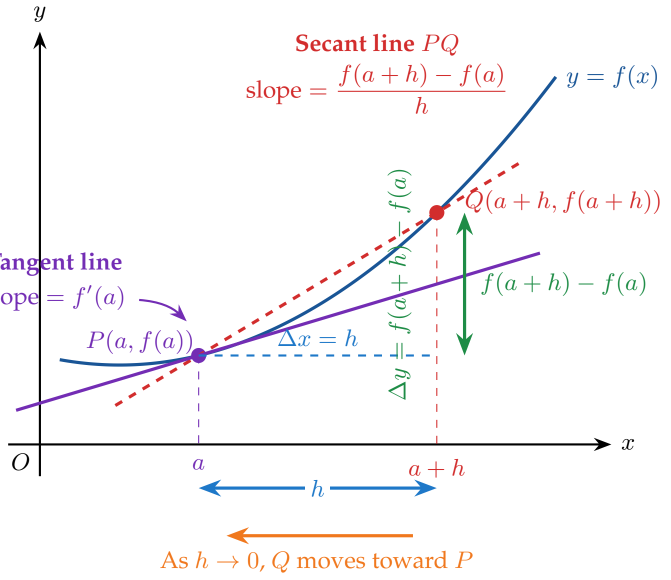
Definition 1.3.1 (Derivative at a Point)**.** Let $f : I \to \mathbb{R}$ be defined on an open interval I. The function f is said to be **differentiable** at $a \in I$ if the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists as a finite real number. The value $f'(a)$ is called the **derivative** of f at a.

See Figure 5 in the Atlas for the geometric interpretation of the derivative.

Definition 1.3.2 (Left-Hand and Right-Hand Derivatives)**.** Let $h > 0$. The one-sided derivatives at $x = a$ are defined as:



Difference Quotient

$$\frac{f(a+h)-f(a)}{h}$$

gives the slope of the secant line PQ.

Derivative at $x = a$

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

As $h \to 0$, the secant line approaches the tangent line. Thus $f'(a)$ is the **slope of the tangent line** at P.

Figure 1.1: Geometric interpretation of the derivative of a single-variable function.

- **Left-Hand Derivative (LHD):**

$$Lf'(a) = f'(a^-) = \lim_{h\to 0^+} \frac{f(a-h)-f(a)}{-h} = \lim_{x\to a^-} \frac{f(x)-f(a)}{x-a}.$$

- **Right-Hand Derivative (RHD):**

$$Rf'(a) = f'(a^+) = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h} = \lim_{x\to a^+} \frac{f(x)-f(a)}{x-a}.$$

A function f is differentiable at $x = a$ if and only if both $Lf'(a)$ and $Rf'(a)$ exist, are finite, and are equal:

$$Lf'(a) = Rf'(a) = f'(a).$$

Interpretation 1.3.3 (Geometric Significance of Differentiability)**.** Geometrically, $f'(a)$ represents the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$. See Figure 5 in the Atlas.

- $Lf'(a)$ and $Rf'(a)$ represent the slopes of the secant lines approaching $(a, f(a))$ from the left and right, respectively.
- If $Lf'(a) \neq Rf'(a)$, the graph exhibits a sharp turn or "corner point" at $x = a$ (e.g., $y = |x|$ at $x = 0$).
- If $f'(a) = \pm\infty$, the graph possesses a vertical tangent at $x = a$ (e.g., $y = x^{1/3}$ at $x = 0$).

Theorem 1.3.4 (Differentiability Implies Continuity)**.** *If a function f is differentiable at $x = a$, then f is continuous at $x = a$.*

Proof. Assume f is differentiable at $x = a$. For any $x \neq a$, we can express the difference $f(x) - f(a)$ as:

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

Taking the limit as $x \to a$ on both sides:

$$\lim_{x \to a}[f(x) - f(a)] = \left(\lim_{x \to a} \frac{f(x) - f(a)}{x - a}\right) \cdot \left(\lim_{x \to a}(x - a)\right).$$

Since f is differentiable at a, the first limit is the finite number $f'(a)$, and the second limit is 0:

$$\lim_{x \to a}[f(x) - f(a)] = f'(a) \cdot 0 = 0 \implies \lim_{x \to a} f(x) = f(a).$$

Hence $\lim_{x \to a} f(x) = f(a)$, so f is continuous at a. ■

Remark 1.3.5. The converse is **false**. A function may be continuous at a point without being differentiable there.

Example 1.3.6 (Corner Point – Continuous but Not Differentiable; See Figure 6 in the Atlas)**.** Show that the function $f(x) = |x|$ is continuous at $x = 0$, but fails to be differentiable at $x = 0$.

Solution. **Continuity at $x = 0$:**

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} |x| = 0 = f(0).$$

Hence, $f(x)$ is continuous at $x = 0$.

Differentiability at $x = 0$: Evaluate the left-hand and right-hand derivatives using $h > 0$:

$$Rf'(0) = \lim_{h \to 0^+} \frac{|0 + h| - |0|}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1,$$

$$Lf'(0) = \lim_{h \to 0^+} \frac{|0 - h| - |0|}{-h} = \lim_{h \to 0^+} \frac{h}{-h} = -1.$$

Since $Lf'(0) = -1 \neq 1 = Rf'(0)$, the derivative $f'(0)$ does not exist. ■

Example 1.3.7. Find the values of constants a and b such that the function

$$f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$$

is differentiable at $x = 1$.

Solution. For $f(x)$ to be differentiable at $x = 1$, it must first be continuous at $x = 1$.

Continuity at $x = 1$.

$$\lim_{x \to 1^-} f(x) = 1^2 + 3(1) + a = 4 + a,$$
$$\lim_{x \to 1^+} f(x) = b(1) + 2 = b + 2.$$

Equating the two one-sided limits and $f(1)$:

$$4 + a = b + 2 \implies b - a = 2. \tag{1.1}$$

Differentiability at $x = 1$. The one-sided derivatives are

$$Lf'(1) = \lim_{h\to 0^+} \frac{f(1-h) - f(1)}{-h} = \frac{d}{dx}(x^2 + 3x + a)\bigg|_{x=1} = 2(1) + 3 = 5,$$

$$Rf'(1) = \lim_{h\to 0^+} \frac{f(1+h) - f(1)}{h} = \frac{d}{dx}(bx + 2)\bigg|_{x=1} = b.$$

Equating $Lf'(1) = Rf'(1)$ gives:

$$b = 5.$$

Substituting $b = 5$ into equation (1.1):

$$5 - a = 2 \implies a = 3.$$

Hence, $a = 3$ and $b = 5$. ■

Example 1.3.8 (Differentiable Function with Discontinuous Derivative)**.** Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by:

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

(i) Show that $f(x)$ is differentiable at $x = 0$ and find $f'(0)$.

(ii) Show that $f'(x)$ is discontinuous at $x = 0$.

Solution. **(i) Differentiability at $x = 0$:** By definition of the derivative at $x = 0$:

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h} = \lim_{h\to 0} h \sin\left(\frac{1}{h}\right).$$

Since $-1 \leq \sin(1/h) \leq 1$, we have:

$$-|h| \leq h \sin\left(\frac{1}{h}\right) \leq |h|.$$

Since $\lim_{h\to 0}(-|h|) = \lim_{h\to 0} |h| = 0$, by the Squeeze Theorem:

$$f'(0) = 0.$$

Thus, f is differentiable everywhere on $\mathbb{R}$.

(ii) Discontinuity of $f'(x)$ at $x = 0$: For $x \neq 0$, apply the product and chain rules to differentiate $f(x)$:

$$f'(x) = \frac{d}{dx}\left[x^2 \sin\left(\frac{1}{x}\right)\right] = 2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right).$$

Now, evaluate $\lim_{x\to 0} f'(x)$:

$$\lim_{x\to 0} f'(x) = \lim_{x\to 0} \left[2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)\right].$$

While $\lim_{x\to 0} 2x \sin(1/x) = 0$, the limit $\lim_{x\to 0} \cos(1/x)$ does not exist due to infinite oscillation between -1 and 1. Consequently, $\lim_{x\to 0} f'(x)$ does not exist, so $f'(x)$ is discontinuous at $x = 0$. ■

Example 1.3.9 (Vertical Tangent Line; See Figure 8 in the Atlas)**.** Show that $f(x) = x^{1/3}$ is continuous at $x = 0$, but not differentiable at $x = 0$.

Solution. **Continuity:** $\lim_{x\to 0} x^{1/3} = 0 = f(0)$. Thus, f is continuous at $x = 0$.

Differentiability: Using the definition of derivative,

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{h^{1/3} - 0}{h} = \lim_{h\to 0} \frac{1}{h^{2/3}} = +\infty.$$

Since the limit is infinite (indicating a vertical tangent line at $x = 0$), $f'(0)$ does not exist as a finite real number. Hence, $f(x)$ is not differentiable at $x = 0$. ■

1.3.1 Geometric Concepts of Differentiability

Geometrically, a function f is differentiable at $x = c$ if the graph $y = f(x)$ is **locally smooth** at $(c, f(c))$ and possesses a unique, non-vertical tangent line.

Theorem 1.3.10 (Geometric Results for Differentiability). *Let $f : I \to \mathbb{R}$ be a function defined on an open interval I.*

Result 1. Existence of Unique Non-Vertical Tangent Line: *f is differentiable at $x = c$ if and only if the secant line passing through $(c, f(c))$ and $(c+h, f(c+h))$ converges to a single, unique line with finite slope $m = f'(c)$ as $h \to 0$.*

Result 2. Local Linearity (Best Linear Approximation): *f is differentiable at $x = c$ if and only if the graph $y = f(x)$ is **locally linear** at $(c, f(c))$. That is, zooming in indefinitely around $(c, f(c))$ makes the graph indistinguishable from the tangent line:*

$$y = f(c) + f'(c)(x - c).$$

Result 3. Smooth Joining Condition: *Two differentiable curves $y = f_1(x)$ $(x \le c)$ and $y = f_2(x)$ $(x > c)$ meet smoothly at $x = c$ if and only if they share both the same point and the same tangent slope:*

$$f_1(c) = f_2(c) \quad \textit{and} \quad f_1'(c) = f_2'(c).$$

Interpretation 1.3.11 (Geometric Obstructions to Differentiability)**.** Even if a graph is continuous (unbroken), it can fail to be differentiable at $x = c$ due to three primary geometric features:

1. **Corner Point (Kink):** The left-hand tangent slope $Lf'(c)$ and right-hand tangent slope $Rf'(c)$ exist and are finite, but $Lf'(c) \neq Rf'(c)$. The curve makes an abrupt change in direction.
2. **Cusp:** The secant slopes from the left and right approach opposite infinities ($+\infty$ and $-\infty$). The curve comes to a sharp point where left and right tangents become vertical in opposite directions.
3. **Vertical Tangent Line:** The secant slopes from both sides approach the same infinity ($+\infty$ or $-\infty$). The tangent line exists but is completely vertical, meaning its slope is undefined.

Example 1.3.12 (Geometric Analysis of a Corner Point; See Figure 6 in the Atlas)**.** Analyze the function $f(x) = |x - 2| + 1$ geometrically at $x = 2$ for continuity and differentiability.

Solution. Rewrite $f(x)$ as a piecewise function:

$$f(x) = \begin{cases} -(x-2) + 1 = 3 - x, & x < 2 \\ (x-2) + 1 = x - 1, & x \ge 2 \end{cases}$$

1. Geometric Analysis of Continuity:

- As $x \to 2^-$, the graph follows the line $y = 3 - x$, approaching $(2, 1)$.
- As $x \to 2^+$, the graph follows the line $y = x - 1$, approaching $(2, 1)$.
- Since $f(2) = 1$, both branches meet at $(2, 1)$ with no gap or hole. Hence, the graph is continuous at $x = 2$.

2. Geometric Analysis of Differentiability:

- To the left of $x = 2$, the graph is a straight line with constant slope $Lf'(2) = -1$.
- To the right of $x = 2$, the graph is a straight line with constant slope $Rf'(2) = +1$.

Since $Lf'(2) = -1 \neq 1 = Rf'(2)$, the graph possesses a sharp **corner point** at $(2, 1)$. Consequently, no unique tangent line exists at $x = 2$, making f non-differentiable at $x = 2$. ■

Example 1.3.13 (Geometric Analysis of a Cusp; See Figure 7 in the Atlas)**.** Examine the continuity and differentiability of $f(x) = x^{2/3}$ at $x = 0$ using geometric concepts.

Solution. **1. Geometric Continuity:** $\lim_{x\to 0} x^{2/3} = 0 = f(0)$. The curve is continuous and unbroken across $x = 0$, forming a symmetric V-like shape with curved sides touching the origin $(0, 0)$.

2. Geometric Differentiability (Secant Slopes): Compute the one-sided derivatives at $x = 0$:

$$Rf'(0) = \lim_{h\to 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0^+} \frac{h^{2/3}}{h} = \lim_{h\to 0^+} \frac{1}{h^{1/3}} = +\infty,$$
$$Lf'(0) = \lim_{h\to 0^+} \frac{f(0-h) - f(0)}{-h} = \lim_{h\to 0^+} \frac{(-h)^{2/3}}{-h} = \lim_{h\to 0^+} \frac{h^{2/3}}{-h} = \lim_{h\to 0^+} \left(-\frac{1}{h^{1/3}}\right) = -\infty.$$

Geometric Conclusion: As $x \to 0$ from the right, the tangent line becomes infinitely steep with a positive slope ($+\infty$). As $x \to 0$ from the left, the tangent line becomes infinitely steep with a negative slope ($-\infty$). The two sides meet at a sharp point directed downwards at the origin $(0, 0)$. This geometric feature is a **cusp**, so $f(x)$ is not differentiable at $x = 0$. ■

Example 1.3.14 (Smooth Curve Fitting / Geometric Tangency)**.** Find the values of a and b such that a quadratic parabola $y = ax^2 + b$ joins smoothly with the line $y = 4x - 3$ at $x = 1$.

Solution. For a smooth join at $x = 1$, the curves must agree both in position and in tangent slope:

1. **Position Match (Continuity):** The two curves must intersect at $x = 1$ (no gap).
2. **Slope Match (Differentiability):** The two curves must share a common tangent line at $x = 1$ (no corner).

Let $f_1(x) = ax^2 + b$ and $f_2(x) = 4x - 3$.
Condition 1: Position Match at $x = 1$

$$f_1(1) = f_2(1) \implies a(1)^2 + b = 4(1) - 3 \implies a + b = 1.$$

Condition 2: Slope Match at $x = 1$ Compute derivatives:

$$f_1'(x) = 2ax, \quad f_2'(x) = 4.$$

Equating slopes at $x = 1$:

$$f_1'(1) = f_2'(1) \implies 2a(1) = 4 \implies a = 2.$$

Substituting $a = 2$ into $a + b = 1$:

$$2 + b = 1 \implies b = -1.$$

Geometric Conclusion: When $a = 2$ and $b = -1$, the parabola $y = 2x^2 - 1$ meets the line $y = 4x - 3$ at the point $(1, 1)$, and both share the exact same tangent slope $m = 4$ at that point, ensuring a completely smooth geometric transition. ∎

1.4 Successive Differentiation and Leibnitz's Theorem

Successive differentiation is the process of differentiating a function repeatedly to obtain higher-order derivatives: $y_1 = y'$, $y_2 = y''$, $\ldots$, $y_n = y^{(n)}$.

1.4.1 Standard Formulas for n-th Derivatives

1. $\dfrac{d^n}{dx^n}(ax+b)^m = \dfrac{m!}{(m-n)!}a^n(ax+b)^{m-n} \quad (m \geq n)$.

2. $\dfrac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$.

3. $\dfrac{d^n}{dx^n}\sin(ax+b) = a^n \sin\left(ax + b + \frac{n\pi}{2}\right)$.

4. $\dfrac{d^n}{dx^n}\cos(ax+b) = a^n \cos\left(ax + b + \frac{n\pi}{2}\right)$.

Theorem 1.4.1 (Leibnitz's Theorem). *If $u(x)$ and $v(x)$ are two functions possessing derivatives up to n-th order, then the n-th derivative of their product $y = u \cdot v$ is given by:*

$$y_n = (uv)_n = \binom{n}{0}u_n v + \binom{n}{1}u_{n-1}v_1 + \binom{n}{2}u_{n-2}v_2 + \cdots + \binom{n}{r}u_{n-r}v_r + \cdots + \binom{n}{n}uv_n,$$

where $u_r = \dfrac{d^r u}{dx^r}$ and $v_r = \dfrac{d^r v}{dx^r}$.

Proof by Mathematical Induction. **Base Case ($n = 1$):** By the standard product rule:

$$(uv)_1 = u_1 v + uv_1 = \binom{1}{0}u_1 v + \binom{1}{1}uv_1.$$

The result holds for $n = 1$.

Inductive Hypothesis: Assume the theorem holds for $n = k$:

$$(uv)_k = u_k v + \binom{k}{1}u_{k-1}v_1 + \binom{k}{2}u_{k-2}v_2 + \cdots + \binom{k}{r}u_{k-r}v_r + \cdots + uv_k.$$

Inductive Step: Differentiating both sides with respect to x:

$$\begin{aligned}(uv)_{k+1} &= (u_{k+1}v + u_k v_1) + \binom{k}{1}(u_k v_1 + u_{k-1}v_2) + \cdots + \binom{k}{r}(u_{k-r+1}v_r + u_{k-r}v_{r+1}) + \cdots + (u_1 v_k + uv_{k+1}) \\ &= u_{k+1}v + \left[1 + \binom{k}{1}\right]u_k v_1 + \left[\binom{k}{1} + \binom{k}{2}\right]u_{k-1}v_2 + \cdots + \left[\binom{k}{r-1} + \binom{k}{r}\right]u_{k-r+1}v_r + \cdots + uv_{k+1}.\end{aligned}$$

Using Pascal's identity $\binom{k}{r-1} + \binom{k}{r} = \binom{k+1}{r}$:

$$(uv)_{k+1} = \binom{k+1}{0} u_{k+1}v + \binom{k+1}{1} u_k v_1 + \cdots + \binom{k+1}{r} u_{k-r+1} v_r + \cdots + \binom{k+1}{k+1} uv_{k+1}.$$

Thus, the result holds for $n = k+1$. By induction, Leibnitz's Theorem is valid for all $n \in \mathbb{N}$. ■

Example 1.4.2. If $y = x^3 e^{2x}$, find y_n using Leibnitz's Theorem.

Solution. Let $u = e^{2x}$ and $v = x^3$. Then:

$$u_r = 2^r e^{2x}, \quad v_1 = 3x^2, \quad v_2 = 6x, \quad v_3 = 6, \quad v_r = 0 \text{ for } r \geq 4.$$

By Leibnitz's Theorem:

$$y_n = \binom{n}{0} u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \binom{n}{3} u_{n-3} v_3.$$

Substituting the derivatives:

$$\begin{aligned} y_n &= (2^n e^{2x})(x^3) + n(2^{n-1}e^{2x})(3x^2) + \frac{n(n-1)}{2}(2^{n-2}e^{2x})(6x) + \frac{n(n-1)(n-2)}{6}(2^{n-3}e^{2x})(6) \\ &= 2^{n-3}e^{2x}\left[8x^3 + 12nx^2 + 12n(n-1)x + 8n(n-1)(n-2)\right]. \end{aligned}$$

■

Example 1.4.3. Find the n-th derivative of $y = x^3 \sin x$.

Solution. Let $u = \sin x$ and $v = x^3$.

The k-th derivative of $u = \sin x$ is given by the standard formula:

$$u_k = \sin\left(x + \frac{k\pi}{2}\right).$$

The derivatives of $v = x^3$ are:

$$v_0 = x^3, \quad v_1 = 3x^2, \quad v_2 = 6x, \quad v_3 = 6, \quad v_k = 0 \text{ for } k \geq 4.$$

Applying Leibnitz's Rule:

$$y_n = \binom{n}{0} u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \binom{n}{3} u_{n-3} v_3.$$

Substituting the derivatives and binomial coefficients:

$$\begin{aligned} y_n = x^3 \sin\left(x + \frac{n\pi}{2}\right) &+ 3nx^2 \sin\left(x + \frac{(n-1)\pi}{2}\right) \\ &+ 3n(n-1)x \sin\left(x + \frac{(n-2)\pi}{2}\right) \\ &+ n(n-1)(n-2) \sin\left(x + \frac{(n-3)\pi}{2}\right). \end{aligned}$$

■

Example 1.4.4. Find the n-th derivative of $y = x^2 \ln x$ for $n \geq 3$.

Solution. Let $u = \ln x$ and $v = x^2$.
The k-th derivative of $u = \ln x$ is:

$$u_k = \frac{(-1)^{k-1}(k-1)!}{x^k}.$$

The derivatives of $v = x^2$ are:

$$v_0 = x^2, \quad v_1 = 2x, \quad v_2 = 2, \quad v_k = 0 \text{ for } k \geq 3.$$

Applying Leibnitz's Rule for $n \geq 3$:

$$y_n = \binom{n}{0} u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2.$$

Substituting the terms:

$$y_n = (1)\left[\frac{(-1)^{n-1}(n-1)!}{x^n}\right] x^2 + n\left[\frac{(-1)^{n-2}(n-2)!}{x^{n-1}}\right](2x) + \frac{n(n-1)}{2}\left[\frac{(-1)^{n-3}(n-3)!}{x^{n-2}}\right](2).$$

Simplifying powers of x and factoring out $\frac{(-1)^{n-3}(n-3)!}{x^{n-2}}$:

$$\begin{aligned} y_n &= \frac{(-1)^{n-3}(n-3)!}{x^{n-2}}\Big[(-1)^2(n-1)(n-2) + 2n(-1)^1(n-2) + n(n-1)\Big] \\ &= \frac{(-1)^{n-3}(n-3)!}{x^{n-2}}\Big[(n^2-3n+2) - (2n^2-4n) + (n^2-n)\Big] \\ &= \frac{(-1)^{n-3}(n-3)!}{x^{n-2}}[2]. \end{aligned}$$

Thus:

$$y_n = \frac{2(-1)^{n-3}(n-3)!}{x^{n-2}} \quad (n \geq 3).$$

■

Example 1.4.5 (Legendre Polynomial Differential Relation)**.** If $y = (x^2-1)^n$, prove that $(x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$.

Solution. Given $y = (x^2-1)^n$. Differentiating once with respect to x:

$$y_1 = n(x^2-1)^{n-1}(2x).$$

Multiplying both sides by (x^2-1):

$$(x^2-1)y_1 = 2nx(x^2-1)^n \implies (x^2-1)y_1 = 2nxy.$$

Now differentiate this identity $(n+1)$ times with respect to x using Leibnitz's Rule.
Differentiating the left-hand side $D^{n+1}[(x^2-1)y_1]$ with $u = y_1$ and $v = x^2-1$:

$$D^{n+1}\Big[(x^2-1)y_1\Big] = \binom{n+1}{0} y_{n+2}(x^2-1) + \binom{n+1}{1} y_{n+1}(2x) + \binom{n+1}{2} y_n(2).$$

Simplifying the LHS:

$$(x^2-1)y_{n+2} + 2(n+1)xy_{n+1} + (n+1)ny_n.$$

Differentiating the right-hand side $D^{n+1}[2nxy]$ with $u = y$ and $v = 2nx$:

$$D^{n+1}[2nxy] = 2n\left[\binom{n+1}{0}y_{n+1}x + \binom{n+1}{1}y_n(1)\right] = 2nxy_{n+1} + 2n(n+1)y_n.$$

Equating both sides:

$$(x^2-1)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 2nxy_{n+1} + 2n(n+1)y_n.$$

Subtracting $2nxy_{n+1}$ and $2n(n+1)y_n$ from both sides:

$$(x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0.$$

This completes the proof. ∎

Example 1.4.6. If $y = e^{a\sin^{-1}(x)}$, prove the following:

(a) $(1-x^2)y_2 - xy_1 - a^2y = 0$, where $y_1 = \dfrac{dy}{dx}$ and $y_2 = \dfrac{d^2y}{dx^2}$.

(b) Hence, using Leibniz's Theorem, show that the following recurrence relation holds:

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

Solution. We are given the function $y = e^{a\sin^{-1}(x)}$. Using the chain rule, we differentiate y with respect to x:

$$y_1 = \frac{dy}{dx} = e^{a\sin^{-1}(x)} \cdot \frac{d}{dx}(a\sin^{-1}(x))$$
$$y_1 = e^{a\sin^{-1}(x)} \cdot \frac{a}{\sqrt{1-x^2}}$$

that is,

$$y_1 = \frac{ay}{\sqrt{1-x^2}}$$

because $y = e^{a\sin^{-1}(x)}$.

Squaring both sides of the equation:

$$y_1^2 = \frac{a^2y^2}{1-x^2}$$

Now, multiply both sides by $(1-x^2)$ to clear the fraction:

$$(1-x^2)y_1^2 = a^2y^2 \quad (*).$$

We differentiate the entire equation $(*)$ with respect to x.

$$\frac{d}{dx}\left[(1-x^2)y_1^2\right] = \frac{d}{dx}\left[a^2y^2\right]$$
$$\left(\frac{d}{dx}(1-x^2)\right)\cdot y_1^2 + (1-x^2)\cdot\left(\frac{d}{dx}(y_1^2)\right) = a^2\cdot\left(\frac{d}{dx}(y^2)\right)$$
$$(-2x)y_1^2 + (1-x^2)(2y_1y_2) = a^2(2yy_1)$$

Divide the entire equation by $2y_1$:

$$-xy_1 + (1-x^2)y_2 = a^2y$$

Rearranging the terms gives the desired differential equation:

$$(1-x^2)y_2 - xy_1 - a^2y = 0$$

Applying Leibniz's Theorem to find the Recurrence Relation

We now differentiate the equation $(1-x^2)y_2 - xy_1 - a^2y = 0$ successively n times with respect to x.

$$\frac{d^n}{dx^n}\left[(1-x^2)y_2 - xy_1 - a^2y\right] = 0$$

By linearity of the derivative, we can differentiate each term separately:

$$\underbrace{\frac{d^n}{dx^n}\left[(1-x^2)y_2\right]}_{\text{Term A}} - \underbrace{\frac{d^n}{dx^n}\left[xy_1\right]}_{\text{Term B}} - \underbrace{\frac{d^n}{dx^n}\left[a^2y\right]}_{\text{Term C}} = 0$$

For Term: $D^n[(1-x^2)y_2]$ Let $u = y_2$ and $v = 1-x^2$. We apply Leibniz's Theorem. The derivatives of v terminate quickly:

- $v = 1-x^2$
- $v' = -2x$
- $v'' = -2$
- $v''' = 0$

The derivatives of u are $u^{(k)} = (y_2)^{(k)} = y_{k+2}$. The Leibniz expansion is:

$$\binom{n}{0}vu^{(n)} + \binom{n}{1}v'u^{(n-1)} + \binom{n}{2}v''u^{(n-2)} + \dots$$

Substituting our functions (only the first three terms are non-zero):

$$(1)(1-x^2)y_{n+2} + (n)(-2x)y_{n+1} + \frac{n(n-1)}{2}(-2)y_n$$

Simplifying gives: $(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n$.
For Term: $D^n[xy_1]$ Let $u = y_1$ and $v = x$. The derivatives of v are $v' = 1$ and $v'' = 0$. The Leibniz expansion has only two non-zero terms:

$$\binom{n}{0}vu^{(n)} + \binom{n}{1}v'u^{(n-1)}$$

Substituting our functions:

$$(1)(x)y_{n+1} + (n)(1)y_n = xy_{n+1} + ny_n$$

For Term: $D^n[a^2y]$ Since a^2 is a constant, this is straightforward: a^2y_n.
Combining the results: Now we substitute the expanded terms back into our main equation:

$$[(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n] - [xy_{n+1} + ny_n] - [a^2y_n] = 0$$

Finally, we group the terms by the order of the derivative (y_{n+2}, y_{n+1}, y_n):

$$(1-x^2)y_{n+2} + (-2nx - x)y_{n+1} + (-n(n-1) - n - a^2)y_n = 0$$

Simplifying gives:

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

■

1.5 Partial Differentiation and Euler's Theorem

1.5.1 Partial and Total Differentiation

Definition 1.5.1 (Partial Derivatives)**.** Let $z = f(x, y)$ be a real-valued function of two variables defined on an open region $D \subseteq \mathbb{R}^2$.

(i) The **partial derivative of** f **with respect to** x at (x, y) is obtained by treating y as a constant and differentiating f with respect to x:

$$\frac{\partial z}{\partial x} = f_x(x, y) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}.$$

(ii) The **partial derivative of** f **with respect to** y at (x, y) is obtained by treating x as a constant and differentiating f with respect to y:

$$\frac{\partial z}{\partial y} = f_y(x, y) = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

Definition 1.5.2 (Total Differential)**.** Let $z = f(x, y)$ be a differentiable function at (x, y). The **total differential** dz (also written as df) is the linear function of the independent differentials dx and dy defined by:

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = f_x(x, y) dx + f_y(x, y) dy.$$

For a function of three variables $w = f(x, y, z)$, the total differential generalizes to:

$$dw = \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz.$$

Chain Rule for Total Derivatives

If $z = f(x, y)$ with $x = g(t)$, $y = h(t)$, then:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

Remark 1.5.3 (Geometric and Analytical Interpretation).

- **Geometric Slope:** Geometrically, $f_x(x_0, y_0)$ represents the slope of the tangent line to the curve formed by the intersection of the surface $z = f(x, y)$ with the vertical plane $y = y_0$.
- **Linear Approximation:** The total differential dz measures the principal linear change in z. For small increments $\Delta x = dx$ and $\Delta y = dy$, dz provides the **best linear approximation** to the actual change Δz:

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y) \approx dz.$$

Example 1.5.4 (See Figure 9 in the Atlas)**.** Find all first-order and second-order partial derivatives of the function

$$f(x, y) = x^3 y^2 + \sin(xy).$$

Solution. To compute the first-order partial derivative with respect to x, treat y as a constant:

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(x^3 y^2 + \sin(xy)\right) = 3x^2 y^2 + y\cos(xy).$$

To compute the first-order partial derivative with respect to y, treat x as a constant:

$$f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(x^3y^2 + \sin(xy)\right) = 2x^3y + x\cos(xy).$$

Now, differentiate f_x and f_y to obtain the second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\Big(3x^2y^2 + y\cos(xy)\Big) = 6xy^2 - y^2\sin(xy),$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\Big(2x^3y + x\cos(xy)\Big) = 2x^3 - x^2\sin(xy).$$

Finally, differentiate f_x with respect to y to get the mixed partial derivative:

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\Big(3x^2y^2 + y\cos(xy)\Big) = 6x^2y + \cos(xy) - xy\sin(xy).$$

■

Example 1.5.5. Verify that $f_{xy} = f_{yx}$ for the function $f(x, y) = x^3e^{2y} + y\ln x$.

Solution. Differentiating $f(x, y)$ partially with respect to x:

$$f_x = \frac{\partial f}{\partial x} = 3x^2e^{2y} + \frac{y}{x}.$$

Differentiating f_x partially with respect to y gives:

$$f_{xy} = \frac{\partial}{\partial y}\left(3x^2e^{2y} + \frac{y}{x}\right) = 6x^2e^{2y} + \frac{1}{x}.$$

Next, differentiating $f(x, y)$ partially with respect to y:

$$f_y = \frac{\partial f}{\partial y} = 2x^3e^{2y} + \ln x.$$

Differentiating f_y partially with respect to x gives:

$$f_{yx} = \frac{\partial}{\partial x}\left(2x^3e^{2y} + \ln x\right) = 6x^2e^{2y} + \frac{1}{x}.$$

Comparing the two expressions, $f_{xy} = f_{yx} = 6x^2e^{2y} + \frac{1}{x}$, confirming Clairaut's theorem for this function. ■

Example 1.5.6 (See Figure 10 in the Atlas)**.** Find the total differential dz for the function

$$z = x^2y^3 - 3xy + \ln(x^2 + y^2).$$

Solution. The total differential formula for a two-variable function $z = f(x, y)$ is:

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy.$$

Compute the partial derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$:

$$\frac{\partial z}{\partial x} = 2xy^3 - 3y + \frac{2x}{x^2 + y^2},$$

$$\frac{\partial z}{\partial y} = 3x^2y^2 - 3x + \frac{2y}{x^2 + y^2}.$$

Substituting these partial derivatives into the total differential formula yields:

$$dz = \left(2xy^3 - 3y + \frac{2x}{x^2+y^2}\right) dx + \left(3x^2y^2 - 3x + \frac{2y}{x^2+y^2}\right) dy.$$

■

Example 1.5.7. If $z = x^2y + y^3$, where $x = e^{2t}$ and $y = \sin(3t)$, find the total derivative $\frac{dz}{dt}$ and evaluate it at $t = 0$.

Solution. By the multivariable chain rule, the total derivative of z with respect to parameter t is:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Calculate the required partial derivatives and ordinary derivatives:

$$\frac{\partial z}{\partial x} = 2xy, \qquad \frac{\partial z}{\partial y} = x^2 + 3y^2,$$

$$\frac{dx}{dt} = 2e^{2t}, \qquad \frac{dy}{dt} = 3\cos(3t).$$

Substitute these into the chain rule formula:

$$\frac{dz}{dt} = (2xy)(2e^{2t}) + (x^2+3y^2)(3\cos(3t)) = 4xye^{2t} + 3(x^2+3y^2)\cos(3t).$$

To evaluate at $t = 0$, compute the corresponding values of x and y:

$$x(0) = e^0 = 1 \quad \text{and} \quad y(0) = \sin 0 = 0.$$

Substituting $t = 0, x = 1, y = 0$:

$$\left.\frac{dz}{dt}\right|_{t=0} = 4(1)(0)e^0 + 3\left(1^2 + 3(0)^2\right)\cos(0) = 0 + 3(1)(1) = 9.$$

Thus, the total rate of change at $t = 0$ is 9. ■

Example 1.5.8. Consider the function $f(x, y)$ defined as:

$$f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Show that the mixed partial derivatives are not equal.

Solution. To compute the mixed partials at the origin, we first need the first-order partial derivatives evaluated along the axes. We use the limit definition of the derivative.

The partial derivative with respect to x at $(0, y)$ is:

$$f_x(0,y) = \lim_{h\to 0} \frac{f(h,y) - f(0,y)}{h} = \lim_{h\to 0} \frac{1}{h}\left(\frac{hy(h^2-y^2)}{h^2+y^2} - 0\right) = \lim_{h\to 0} \frac{y(h^2-y^2)}{h^2+y^2} = \frac{-y^3}{y^2} = -y$$

The partial derivative with respect to y at $(x, 0)$ is:

$$f_y(x,0) = \lim_{k\to 0} \frac{f(x,k) - f(x,0)}{k} = \lim_{k\to 0} \frac{1}{k}\left(\frac{xk(x^2-k^2)}{x^2+k^2} - 0\right) = \lim_{k\to 0} \frac{x(x^2-k^2)}{x^2+k^2} = \frac{x^3}{x^2} = x$$

At the origin itself, both first partials are zero:

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = 0 \quad \text{and} \quad f_y(0,0) = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = 0$$

Mixed Partial Derivatives at the Origin. We now compute the mixed partials at $(0,0)$ using the limit definition and the results above.

The derivative of f_y with respect to x at $(0,0)$ is:

$$f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(h,0) - f_y(0,0)}{h} = \lim_{h\to 0} \frac{h-0}{h} = \lim_{h\to 0} 1 = \mathbf{1}$$

The derivative of f_x with respect to y at $(0,0)$ is:

$$f_{xy}(0,0) = \lim_{k\to 0} \frac{f_x(0,k) - f_x(0,0)}{k} = \lim_{k\to 0} \frac{-k-0}{k} = \lim_{k\to 0} -1 = -\mathbf{1}$$

Conclusion. Since $f_{yx}(0,0) = 1$ and $f_{xy}(0,0) = -1$, we have shown that for this function, the mixed partial derivatives are not equal at the origin:

$$f_{yx}(0,0) \neq f_{xy}(0,0)$$

■

A theorem that guarantees when mixed partial derivatives are equal.

Theorem 1.5.9 (Clairaut's Theorem)**.** *If $f(x,y)$ and its partial derivatives f_x, f_y, f_{xy}, and f_{yx} are defined on an open set containing (a,b) and are continuous at (a,b), then:*

$$\frac{\partial^2 f}{\partial x \partial y}(a,b) = \frac{\partial^2 f}{\partial y \partial x}(a,b)$$

Proof. For sufficiently small $h, k \neq 0$, define the auxiliary function:

$$\Delta(h,k) = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a,b).$$

Analyze via f_{yx}
Define $g(y) = f(a+h, y) - f(a,y)$. Then:

$$\Delta(h,k) = g(b+k) - g(b).$$

Since f_y exists, thus, $g'(y)$ exists, so by the Mean Value Theorem (MVT), there exists d between b and $b+k$ such that:

$$g(b+k) - g(b) = k \cdot g'(d) = k \left[\frac{\partial f}{\partial y}(a+h, d) - \frac{\partial f}{\partial y}(a,d) \right].$$

Apply MVT to $h(x) = \dfrac{\partial f}{\partial y}(x,d)$ on $[a, a+h]$. There exists c_1 between a and $a+h$ such that:

$$\frac{\partial f}{\partial y}(a+h,d) - \frac{\partial f}{\partial y}(a,d) = h \cdot \frac{\partial^2 f}{\partial x \partial y}(c_1, d).$$

Thus:

$$\Delta(h,k) = hk \cdot \frac{\partial^2 f}{\partial x \partial y}(c_1, d).$$

Analyze via f_{xy}
Define $r(x) = f(x, b+k) - f(x,b)$. Then:

$$\Delta(h,k) = r(a+h) - r(a).$$

By MVT, there exists e between a and $a+h$ such that:

$$r(a+h) - r(a) = h \cdot r'(e) = h \left[\frac{\partial f}{\partial x}(e, b+k) - \frac{\partial f}{\partial x}(e, b) \right].$$

Apply MVT to $s(y) = \frac{\partial f}{\partial x}(e, y)$ on $[b, b+k]$. There exists c_2 between b and $b+k$ such that:

$$\frac{\partial f}{\partial x}(e, b+k) - \frac{\partial f}{\partial x}(e, b) = k \cdot \frac{\partial^2 f}{\partial y \partial x}(e, c_2).$$

Thus:

$$\Delta(h, k) = hk \cdot \frac{\partial^2 f}{\partial y \partial x}(e, c_2).$$

Equate and take limits
From Steps 1 and 2:

$$hk \cdot \frac{\partial^2 f}{\partial x \partial y}(c_1, d) = hk \cdot \frac{\partial^2 f}{\partial y \partial x}(e, c_2).$$

For $hk \neq 0$, we have:

$$\frac{\partial^2 f}{\partial x \partial y}(c_1, d) = \frac{\partial^2 f}{\partial y \partial x}(e, c_2).$$

As $(h, k) \to (0, 0)$:

$$(c_1, d) \to (a, b) \quad \text{and} \quad (e, c_2) \to (a, b).$$

By continuity of f_{xy} and f_{yx} at (a, b):

$$\lim_{(h,k)\to(0,0)} \frac{\partial^2 f}{\partial x \partial y}(c_1, d) = \frac{\partial^2 f}{\partial x \partial y}(a, b),$$
$$\lim_{(h,k)\to(0,0)} \frac{\partial^2 f}{\partial y \partial x}(e, c_2) = \frac{\partial^2 f}{\partial y \partial x}(a, b).$$

Therefore:

$$\frac{\partial^2 f}{\partial x \partial y}(a, b) = \frac{\partial^2 f}{\partial y \partial x}(a, b)$$

■

1.5.2 Homogeneous Functions and Euler's Theorem

Definition 1.5.10 (Homogeneous Function)**.** A function $f(x, y)$ is called **homogeneous of degree n** if

$$f(tx, ty) = t^n f(x, y) \quad \forall t > 0.$$

Theorem 1.5.11 (Euler's Theorem on Homogeneous Functions)**.** *If $u = f(x, y)$ is a homogeneous function of degree n with continuous partial derivatives, then:*

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu.$$

Furthermore, for second-order partial derivatives:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u.$$

Proof. **(i).**
Since $u(x, y)$ is homogeneous of degree n, it can be expressed as:

$$u(x,y) = x^n F\left(\frac{y}{x}\right).$$

Differentiating u partially with respect to x:

$$\frac{\partial u}{\partial x} = nx^{n-1}F\left(\frac{y}{x}\right) + x^n F'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right) = nx^{n-1}F\left(\frac{y}{x}\right) - x^{n-2}yF'\left(\frac{y}{x}\right).$$

Multiply both sides by x:

$$x\frac{\partial u}{\partial x} = nx^n F\left(\frac{y}{x}\right) - x^{n-1}yF'\left(\frac{y}{x}\right). \tag{1.2}$$

Next, differentiating u partially with respect to y:

$$\frac{\partial u}{\partial y} = x^n F'\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) = x^{n-1}F'\left(\frac{y}{x}\right).$$

Multiply both sides by y:

$$y\frac{\partial u}{\partial y} = x^{n-1}yF'\left(\frac{y}{x}\right). \tag{1.3}$$

Adding equations (1.2) and (1.3):

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nx^n F\left(\frac{y}{x}\right) = nu.$$

(ii). Since $u = f(x, y)$ is a homogeneous function of degree n, by the first-order Euler's Theorem, we have:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu. \tag{1.4}$$

Differentiating equation (1.4) partially with respect to x using the product rule:

$$\frac{\partial}{\partial x}\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial x}(nu)$$

$$\left(1\cdot\frac{\partial u}{\partial x} + x\frac{\partial^2 u}{\partial x^2}\right) + y\frac{\partial^2 u}{\partial x \partial y} = n\frac{\partial u}{\partial x}.$$

Rearranging terms by subtracting $\frac{\partial u}{\partial x}$ from both sides:

$$x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = (n-1)\frac{\partial u}{\partial x}.$$

Multiplying the entire equation by x:

$$x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y} = (n-1)x\frac{\partial u}{\partial x}. \tag{1.5}$$

Next, differentiating equation (1.4) partially with respect to y using the product rule:

$$\frac{\partial}{\partial y}\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}(nu)$$

$$x\frac{\partial^2 u}{\partial y \partial x} + \left(1 \cdot \frac{\partial u}{\partial y} + y\frac{\partial^2 u}{\partial y^2}\right) = n\frac{\partial u}{\partial y}.$$

Since the second-order partial derivatives are continuous, by Clairaut's Theorem we have $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$. Rearranging terms gives:

$$x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} = (n-1)\frac{\partial u}{\partial y}.$$

Multiplying the entire equation by y:

$$xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (n-1)y\frac{\partial u}{\partial y}. \tag{1.6}$$

Adding equations (1.5) and (1.6):

$$\left(x^2\frac{\partial^2 u}{\partial x^2} + xy\frac{\partial^2 u}{\partial x \partial y}\right) + \left(xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}\right) = (n-1)x\frac{\partial u}{\partial x} + (n-1)y\frac{\partial u}{\partial y}.$$

Grouping terms on both sides:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (n-1)\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right).$$

Substituting equation (1.4) ($x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$) into the right-hand side yields:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (n-1)(nu) = n(n-1)u.$$

■

Example 1.5.12. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.

Solution. We can rewrite the expression as:

$$z = \tan u = \frac{x^3+y^3}{x-y}.$$

To find the degree of homogeneity of $z(x, y)$:

$$z(tx, ty) = \frac{(tx)^3 + (ty)^3}{(tx) - (ty)} = \frac{t^3(x^3+y^3)}{t(x-y)} = t^2 z(x, y).$$

Thus, $z = \tan u$ is a homogeneous function of degree $n = 2$.
Applying Euler's Theorem to z:

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z.$$

Since $z = \tan u$:

$$\frac{\partial z}{\partial x} = \sec^2 u\frac{\partial u}{\partial x}, \quad \text{and} \quad \frac{\partial z}{\partial y} = \sec^2 u\frac{\partial u}{\partial y}.$$

Substitute these back into Euler's formula:

$$x\left(\sec^2 u\frac{\partial u}{\partial x}\right)+y\left(\sec^2 u\frac{\partial u}{\partial y}\right)=2\tan u$$

$$\sec^2 u\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)=2\tan u$$

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{2\tan u}{\sec^2 u}=2\sin u\cos u=\sin 2u.$$

■

Example 1.5.13. Verify Euler's Theorem for the homogeneous function

$$u(x,y)=\frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}.$$

Solution. To determine the degree of homogeneity n, test $u(tx,ty)$ for $t>0$:

$$u(tx,ty)=\frac{(tx)^{1/4}+(ty)^{1/4}}{(tx)^{1/5}+(ty)^{1/5}}=\frac{t^{1/4}(x^{1/4}+y^{1/4})}{t^{1/5}(x^{1/5}+y^{1/5})}=t^{\frac{1}{4}-\frac{1}{5}}\frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}=t^{1/20}u(x,y).$$

Thus, $u(x,y)$ is a homogeneous function of degree $n=\dfrac{1}{20}$.

Applying Euler's Theorem directly gives:

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{20}u.$$

We can verify this result by direct partial differentiation. Differentiating u with respect to x:

$$\frac{\partial u}{\partial x}=\frac{\frac{1}{4}x^{-3/4}(x^{1/5}+y^{1/5})-\frac{1}{5}x^{-4/5}(x^{1/4}+y^{1/4})}{(x^{1/5}+y^{1/5})^2}.$$

Differentiating u with respect to y:

$$\frac{\partial u}{\partial y}=\frac{\frac{1}{4}y^{-3/4}(x^{1/5}+y^{1/5})-\frac{1}{5}y^{-4/5}(x^{1/4}+y^{1/4})}{(x^{1/5}+y^{1/5})^2}.$$

Multiplying $\dfrac{\partial u}{\partial x}$ by x and $\dfrac{\partial u}{\partial y}$ by y, then adding:

$$\begin{aligned}x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}&=\frac{\frac{1}{4}(x^{1/4}+y^{1/4})(x^{1/5}+y^{1/5})-\frac{1}{5}(x^{1/5}+y^{1/5})(x^{1/4}+y^{1/4})}{(x^{1/5}+y^{1/5})^2}\\&=\left(\frac{1}{4}-\frac{1}{5}\right)\frac{(x^{1/4}+y^{1/4})(x^{1/5}+y^{1/5})}{(x^{1/5}+y^{1/5})^2}=\frac{1}{20}\frac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}}=\frac{1}{20}u.\end{aligned}$$

This confirms Euler's Theorem. ■

Example 1.5.14. If $u=\sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that:

(i) $x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=\dfrac{1}{2}\tan u$

(ii) $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x \partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = -\dfrac{\sin u \cos 2u}{4\cos^3 u}$

Solution. Rewrite the function as $z = \sin u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$.

Testing z for homogeneity:

$$z(tx, ty) = \frac{t(x+y)}{\sqrt{t}(\sqrt{x}+\sqrt{y})} = t^{1/2}z(x,y).$$

Thus, $z = \sin u$ is homogeneous of degree $n = \dfrac{1}{2}$.

Part (i): Applying Euler's Theorem to z:

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{1}{2}z.$$

Since $\dfrac{\partial z}{\partial x} = \cos u\dfrac{\partial u}{\partial x}$ and $\dfrac{\partial z}{\partial y} = \cos u\dfrac{\partial u}{\partial y}$:

$$x\left(\cos u\frac{\partial u}{\partial x}\right) + y\left(\cos u\frac{\partial u}{\partial y}\right) = \frac{1}{2}\sin u \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\tan u.$$

Part (ii): Let $G(u) = \dfrac{1}{2}\tan u$. Differentiating $xu_x + yu_y = G(u)$ with respect to x and y leads to the standard second-order relation:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = G(u)\left(G'(u) - 1\right).$$

Compute $G'(u)$:

$$G'(u) = \frac{d}{du}\left(\frac{1}{2}\tan u\right) = \frac{1}{2}\sec^2 u.$$

Substituting $G(u)$ and $G'(u)$:

$$\begin{aligned} G(u)\left(G'(u) - 1\right) &= \frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u - 1\right) = \frac{\sin u}{2\cos u}\left(\frac{1 - 2\cos^2 u}{2\cos^2 u}\right) \\ &= \frac{\sin u(-\cos 2u)}{4\cos^3 u} = -\frac{\sin u \cos 2u}{4\cos^3 u}. \end{aligned}$$

■

Example 1.5.15. If $u = \ln\left(\dfrac{x^4+y^4+z^4}{x+y+z}\right)$, prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 3$.

Solution. Transform the function using exponentials: $f = e^u = \dfrac{x^4+y^4+z^4}{x+y+z}$.

Test $f(x,y,z)$ for homogeneity in three variables:

$$f(tx,ty,tz) = \frac{(tx)^4+(ty)^4+(tz)^4}{(tx)+(ty)+(tz)} = \frac{t^4(x^4+y^4+z^4)}{t(x+y+z)} = t^3 f(x,y,z).$$

Thus, $f = e^u$ is a homogeneous function of degree $n = 3$.

Applying Euler's Theorem for three variables to f:

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = 3f.$$

Compute partial derivatives using the chain rule $\frac{\partial f}{\partial x} = e^u \frac{\partial u}{\partial x}$, $\frac{\partial f}{\partial y} = e^u \frac{\partial u}{\partial y}$, and $\frac{\partial f}{\partial z} = e^u \frac{\partial u}{\partial z}$:

$$x\left(e^u \frac{\partial u}{\partial x}\right) + y\left(e^u \frac{\partial u}{\partial y}\right) + z\left(e^u \frac{\partial u}{\partial z}\right) = 3e^u.$$

Factoring out e^u (since $e^u \neq 0$):

$$e^u\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}\right) = 3e^u \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 3.$$

■

Example 1.5.16 (See Figure 8 in the Atlas)**.** If $u = \csc^{-1}\left(\frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)$, prove that:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{36}\left(\tan^2 u + 7\right).$$

Solution. Rewrite the given function as $v = \csc u = \frac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}$.

Test $v(x, y)$ for homogeneity:

$$v(tx, ty) = \frac{(tx)^{1/2}+(ty)^{1/2}}{(tx)^{1/3}+(ty)^{1/3}} = \frac{t^{1/2}(x^{1/2}+y^{1/2})}{t^{1/3}(x^{1/3}+y^{1/3})} = t^{\frac{1}{2}-\frac{1}{3}}v(x,y) = t^{1/6}v(x,y).$$

Thus, $v = \csc u$ is a homogeneous function of degree $n = \frac{1}{6}$.

Applying the first-order Euler's Theorem to v:

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = \frac{1}{6}v.$$

Using the chain rule, $\frac{\partial v}{\partial x} = -\csc u \cot u \frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial y} = -\csc u \cot u \frac{\partial u}{\partial y}$:

$$x\left(-\csc u \cot u \frac{\partial u}{\partial x}\right) + y\left(-\csc u \cot u \frac{\partial u}{\partial y}\right) = \frac{1}{6}\csc u.$$

Dividing both sides by $-\csc u \cot u$:

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{\frac{1}{6}\csc u}{-\csc u \cot u} = -\frac{1}{6\cot u} = -\frac{1}{6}\tan u.$$

Let $G(u) = -\frac{1}{6}\tan u$. For the second-order partial derivatives, apply the second-order Euler operator identity:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = G(u)\left(G'(u) - 1\right).$$

Calculate the derivative $G'(u)$:

$$G'(u) = \frac{d}{du}\left(-\frac{1}{6}\tan u\right) = -\frac{1}{6}\sec^2 u.$$

Substitute $G(u)$ and $G'(u)$ into the second-order relation:

$$\begin{aligned}
G(u)\left(G'(u)-1\right) &= \left(-\frac{1}{6}\tan u\right)\left(-\frac{1}{6}\sec^2 u - 1\right) \\
&= \frac{1}{6}\tan u\left(\frac{1}{6}\sec^2 u + 1\right) \\
&= \frac{\tan u}{36}\left(\sec^2 u + 6\right).
\end{aligned}$$

Using the Pythagorean identity $\sec^2 u = 1 + \tan^2 u$:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{36}\left(1 + \tan^2 u + 6\right) = \frac{\tan u}{36}\left(\tan^2 u + 7\right).$$

This completes the proof. ■

Example 1.5.17. If $u = \sin^{-1}\left(\dfrac{x^2+y^2+z^2}{x+y+z}\right)$, prove that:

$$x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + z^2\frac{\partial^2 u}{\partial z^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + 2yz\frac{\partial^2 u}{\partial y\partial z} + 2zx\frac{\partial^2 u}{\partial z\partial x} = \tan^3 u.$$

Solution. Transform the given equation as $v = \sin u = \dfrac{x^2+y^2+z^2}{x+y+z}$.

Testing $v(x,y,z)$ for homogeneity in three variables:

$$v(tx,ty,tz) = \frac{(tx)^2+(ty)^2+(tz)^2}{(tx)+(ty)+(tz)} = \frac{t^2(x^2+y^2+z^2)}{t(x+y+z)} = t^1 v(x,y,z).$$

Thus, $v = \sin u$ is a homogeneous function of degree $n = 1$ in x, y, and z.

Applying Euler's Theorem for three variables to v:

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} + z\frac{\partial v}{\partial z} = 1\cdot v = \sin u.$$

Substitute $\dfrac{\partial v}{\partial x} = \cos u\dfrac{\partial u}{\partial x}$, $\dfrac{\partial v}{\partial y} = \cos u\dfrac{\partial u}{\partial y}$, and $\dfrac{\partial v}{\partial z} = \cos u\dfrac{\partial u}{\partial z}$:

$$\cos u\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}\right) = \sin u \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = \frac{\sin u}{\cos u} = \tan u.$$

Let $G(u) = \tan u$. Define the linear differential operator $\mathbf{D} = x\dfrac{\partial}{\partial x} + y\dfrac{\partial}{\partial y} + z\dfrac{\partial}{\partial z}$. The first-order relation is $\mathbf{D}(u) = G(u)$.

Applying the operator $\mathbf{D}$ to both sides:

$$\mathbf{D}(\mathbf{D}(u)) = \mathbf{D}(G(u)).$$

Expanding the left-hand side yields the second-order partial derivative sum plus $\mathbf{D}(u)$:

$$\left(x^2u_{xx} + y^2u_{yy} + z^2u_{zz} + 2xyu_{xy} + 2yzu_{yz} + 2zxu_{zx}\right) + \mathbf{D}(u) = G'(u)\mathbf{D}(u).$$

Substitute $\mathbf{D}(u) = G(u)$:

$$x^2u_{xx} + y^2u_{yy} + z^2u_{zz} + 2xyu_{xy} + 2yzu_{yz} + 2zxu_{zx} = G(u)\left(G'(u)-1\right).$$

Since $G(u) = \tan u$, we have $G'(u) = \sec^2 u$. Substituting these into the RHS:

$$G(u)\left(G'(u) - 1\right) = \tan u \left(\sec^2 u - 1\right).$$

Using the identity $\sec^2 u - 1 = \tan^2 u$:

$$\tan u \left(\tan^2 u\right) = \tan^3 u.$$

Hence the entire second-order operator sum equals $\tan^3 u$. ■

Exercise 1.5.18. Solve the following exercises:

1. Prove using the formal ϵ–δ definition of a limit that

$$\lim_{x\to 2}(3x^2 - 4x + 1) = 5.$$

2. Prove using the formal ϵ–δ definition that

$$\lim_{x\to 4} \frac{1}{\sqrt{x}} = \frac{1}{2}.$$

3. Test the continuity and classify the type of discontinuity at $x = 0$ for the function:

$$f(x) = \begin{cases} \dfrac{e^{1/x} - 1}{e^{1/x} + 1}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

4. Determine the nature of discontinuity of $f(x) = \lfloor x \rfloor + \lfloor -x \rfloor$ at all integer points $x = n \in \mathbb{Z}$.

5. Find the values of constants a, b, and c such that $f(x)$ is continuous on $\mathbb{R}$:

$$f(x) = \begin{cases} \dfrac{\sin(a+1)x + \sin x}{x}, & x < 0 \\ c, & x = 0 \\ \dfrac{\sqrt{x + bx^2} - \sqrt{x}}{bx^{3/2}}, & x > 0 \end{cases}$$

6. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by: $f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q} \\ 1 - x, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$ Show that $f(x)$ is continuous only at $x = \frac{1}{2}$ and discontinuous at all other points in $\mathbb{R}$.

7. Test the differentiability of $f(x) = |x - 1| + |x - 2|$ at $x = 1$ and $x = 2$.

8. Show that the function defined below is differentiable at $x = 0$, but its derivative $f'(x)$ is discontinuous at $x = 0$:

$$f(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

9. Find the constants a and b so that $f(x)$ is differentiable at $x = 1$: $f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$

10. Find the n-th derivative y_n of the function: $y = \frac{x^2}{(x-1)^2(x+2)}$.

11. Find y_n if $y = e^{ax}\sin(bx+c)$.

12. If $y = \left(x+\sqrt{1+x^2}\right)^m$, prove using Leibnitz's theorem that:

$$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0.$$

13. If $y = e^{a\sin^{-1}x}$, show that: $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$.

14. If $y = \tan^{-1}x$, evaluate the n-th derivative at the origin $y_n(0)$ for both even and odd values of n.

15. If $u = \ln(x^3+y^3+z^3-3xyz)$, show that:

(i) $\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} + \dfrac{\partial u}{\partial z} = \dfrac{3}{x+y+z}$

(ii) $\left(\dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}\right)^2 u = -\dfrac{9}{(x+y+z)^2}$

16. If $z = f(x,y)$ where $x = e^u\cos v$ and $y = e^u\sin v$, prove that:

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right].$$

17. Compute the total differential df for the function: $f(x,y) = \tan^{-1}\left(\frac{y}{x}\right) + \ln\sqrt{x^2+y^2}$.

18. If $u = \tan^{-1}\left(\dfrac{x^3+y^3}{x-y}\right)$, prove that:

(i) $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \sin 2u$

(ii) $x^2\dfrac{\partial^2 u}{\partial x^2} + 2xy\dfrac{\partial^2 u}{\partial x\partial y} + y^2\dfrac{\partial^2 u}{\partial y^2} = 2\sin u\cos 3u$

19. If $u = \csc^{-1}\left(\dfrac{x^{1/2}+y^{1/2}}{x^{1/3}+y^{1/3}}\right)$, show that:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144}\left(13+\tan^2 u\right).$$

20. If $u = f(x,y)$ is a homogeneous function of degree n, show that: $x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x\partial y} = (n-1)\frac{\partial u}{\partial x}$.

Advanced Problems based on Unit I

Exercise 1.5.19. 1. Using the formal ϵ–δ definition of a limit, prove that:

$$\lim_{x\to 0}\left[\sin\left(\frac{1}{x}\right)\cos\left(\frac{1}{x}\right)\right] \quad \text{does not exist.}$$

2. Let $f:\mathbb{R}\to\mathbb{R}$ be defined by:

$$f(x) = \begin{cases} x^2, & \text{if } x\in\mathbb{Q} \\ -x^2, & \text{if } x\in\mathbb{R}\setminus\mathbb{Q} \end{cases}$$

Determine all points at which $f(x)$ is continuous, and prove your claim using the ϵ–δ definition.

3. Determine the values of constants a, b, and c such that the function

$$f(x) = \begin{cases} ax^2 + bx + c, & x \leq 0 \\ \dfrac{\sin(2x) - 2x}{x^3}, & x > 0 \end{cases}$$

is twice differentiable at $x = 0$.

4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by:

$$f(x) = \begin{cases} x^\alpha \sin\left(x^{-\beta}\right), & x > 0 \\ 0, & x \leq 0 \end{cases}$$

where $\alpha, \beta > 0$. Derive the exact conditions on α and β such that:

(i) $f'(0)$ exists.

(ii) $f'(x)$ is continuous at $x = 0$.

5. If $y = \left(x + \sqrt{x^2+1}\right)^m$, prove that:

$$(1 + x^2)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0.$$

Using this relation, determine the value of the n-th derivative at the origin, $y_n(0)$, for all $n \in \mathbb{N}$.

6. If $y = (\sin^{-1} x)^2$, show that:

$$(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0.$$

Hence, evaluate $y_n(0)$ for both even and odd values of n.

7. Consider the function defined by:

$$f(x, y) = \begin{cases} \dfrac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Prove using the limit definition of partial derivatives that $f_{xy}(0, 0) = -1$ and $f_{yx}(0, 0) = 1$. Explain why this result does not violate Clairaut's Theorem.

8. If $z = f(x, y)$ is defined implicitly by the surface equation $x^3 + y^3 + z^3 - 3xyz = a^3$, find the total differential dz, and show that:

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z - \frac{a^3}{z^2 - xy}.$$

9. If $u = \csc^{-1}\left(\dfrac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}\right)$, prove that:

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{36}\left(\tan^2 u + 7\right).$$

10. If $u = \ln\left(\dfrac{x^4 + y^4 + z^4}{x + y + z}\right)$, prove that the 3D second-order differential expression evaluates to:

$$x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + z^2\frac{\partial^2 u}{\partial z^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + 2yz\frac{\partial^2 u}{\partial y \partial z} + 2zx\frac{\partial^2 u}{\partial z \partial x} = -3.$$

Chapter 2

Indeterminate Forms, Tangents, and Normals

2.1 Indeterminate Forms and L'Hôpital's Rule

When evaluating limits of the form $\lim_{x \to a} \frac{f(x)}{g(x)}$, direct substitution may lead to expressions such as $\frac{0}{0}$ or $\frac{\infty}{\infty}$. These are called **indeterminate forms** because the limit cannot be determined solely from the limits of the individual functions.

Definition 2.1.1 (The Seven Indeterminate Forms)**.** The seven fundamental indeterminate forms in analysis are:

- **Quotient Forms:** $\frac{0}{0}, \quad \frac{\infty}{\infty}$
- **Product Form:** $0 \cdot \infty$
- **Difference Form:** $\infty - \infty$
- **Exponential / Power Forms:** $0^0, \quad 1^\infty, \quad \infty^0$

Theorem 2.1.2 (L'Hôpital's Rule)**.** *Let f and g be differentiable functions on an open interval containing a (except possibly at a itself). Suppose that*

$$\lim_{x \to a} f(x) = 0 \quad \textit{and} \quad \lim_{x \to a} g(x) = 0$$

or

$$\lim_{x \to a} f(x) = \pm\infty \quad \textit{and} \quad \lim_{x \to a} g(x) = \pm\infty.$$

If $g'(x) \neq 0$ near a (for $x \neq a$) and the limit $\lim_{x \to a} \frac{f'(x)}{g'(x)}$ exists (or is $\pm\infty$), then:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

Remark 2.1.3. L'Hôpital's rule applies directly **only** to the quotient forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$. All other indeterminate forms must first be algebraically transformed into one of these two forms.

2.1.1 Quotient Forms: $\frac{0}{0}$ and $\frac{\infty}{\infty}$

Example 2.1.4. Evaluate $\lim_{x\to 0} \frac{e^{2x}-1}{\sin x}$.

Solution. Direct substitution yields $\frac{e^0-1}{\sin 0} = \frac{0}{0}$ (Indeterminate form).

Applying L'Hôpital's Rule:

$$\lim_{x\to 0} \frac{e^{2x}-1}{\sin x} = \lim_{x\to 0} \frac{\frac{d}{dx}(e^{2x}-1)}{\frac{d}{dx}(\sin x)} = \lim_{x\to 0} \frac{2e^{2x}}{\cos x} = \frac{2e^0}{\cos 0} = \frac{2(1)}{1} = 2.$$

■

Example 2.1.5. Evaluate $\lim_{x\to\infty} \frac{\ln x}{x^2}$.

Solution. As $x \to \infty$, we get $\frac{\infty}{\infty}$ (Indeterminate form).

Applying L'Hôpital's Rule:

$$\lim_{x\to\infty} \frac{\ln x}{x^2} = \lim_{x\to\infty} \frac{1/x}{2x} = \lim_{x\to\infty} \frac{1}{2x^2} = 0.$$

■

Example 2.1.6. Evaluate $\lim_{x\to 0} \frac{x-\sin x}{x^3}$.

Solution. Direct substitution yields $\frac{0-0}{0} = \frac{0}{0}$ (Indeterminate form).

Applying L'Hôpital's Rule:

$$\lim_{x\to 0} \frac{x-\sin x}{x^3} = \lim_{x\to 0} \frac{1-\cos x}{3x^2} \quad \left(\text{Still } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule a second time:

$$\lim_{x\to 0} \frac{\sin x}{6x} \quad \left(\text{Still } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule a third time:

$$\lim_{x\to 0} \frac{\cos x}{6} = \frac{\cos 0}{6} = \frac{1}{6}.$$

■

Example 2.1.7. Evaluate $\lim_{x\to\infty} \frac{x^2}{e^x}$.

Solution. As $x \to \infty$, we get $\frac{\infty}{\infty}$ (Indeterminate form).

Applying L'Hôpital's Rule:

$$\lim_{x\to\infty} \frac{x^2}{e^x} = \lim_{x\to\infty} \frac{2x}{e^x} \quad \left(\text{Still } \frac{\infty}{\infty}\right).$$

Applying L'Hôpital's Rule a second time:

$$\lim_{x\to\infty} \frac{2}{e^x} = 0.$$

■

2.1.2 Product Form: $0 \cdot \infty$

To evaluate $f(x) \cdot g(x)$ where $f(x) \to 0$ and $g(x) \to \infty$, transform the product into a quotient:

$$f(x) \cdot g(x) = \frac{f(x)}{1/g(x)} \quad (\text{form } \tfrac{0}{0}) \quad \text{or} \quad f(x) \cdot g(x) = \frac{g(x)}{1/f(x)} \quad (\text{form } \tfrac{\infty}{\infty}).$$

Example 2.1.8. Evaluate $\lim\limits_{x \to 0^+} x \ln x$.

Solution. As $x \to 0^+$, $x \ln x \to 0 \cdot (-\infty)$. Rewrite as a quotient:

$$\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \frac{\ln x}{1/x} \quad \left(\text{Form } \frac{-\infty}{\infty}\right).$$

Apply L'Hôpital's Rule:

$$\lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0.$$

■

Example 2.1.9. Evaluate $\lim\limits_{x \to 0^+} \sin x \cdot \ln x$.

Solution. As $x \to 0^+$, we have $\sin 0 \cdot \ln(0^+) \to 0 \cdot (-\infty)$.
Rewrite the product as a quotient:

$$\lim_{x \to 0^+} \frac{\ln x}{\csc x} \quad \left(\text{Form } \frac{-\infty}{\infty}\right).$$

Applying L'Hôpital's Rule:

$$\lim_{x \to 0^+} \frac{1/x}{-\csc x \cot x} = \lim_{x \to 0^+} \frac{1/x}{-\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}} = \lim_{x \to 0^+} \frac{-\sin^2 x}{x \cos x}.$$

Separate into standard limits:

$$\lim_{x \to 0^+} \left(\frac{\sin x}{x}\right) \cdot \left(\frac{-\sin x}{\cos x}\right) = (1) \cdot (0) = 0.$$

■

2.1.3 Difference Form: $\infty - \infty$

This form is resolved by finding a common denominator, factoring, or rationalizing to convert the expression into a single quotient.

Example 2.1.10. Evaluate $\lim\limits_{x \to 1^+} \left(\frac{1}{x-1} - \frac{1}{\ln x}\right)$.

Solution. Form: $\infty - \infty$. Combine into a single fraction:

$$L = \lim_{x \to 1^+} \frac{\ln x - (x-1)}{(x-1)\ln x} \quad \left(\text{Form } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule:

$$L = \lim_{x \to 1^+} \frac{\frac{1}{x} - 1}{\ln x + 1 - \frac{1}{x}} \quad \left(\text{Still } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule a second time:

$$L = \lim_{x \to 1^+} \frac{-1/x^2}{\frac{1}{x} + \frac{1}{x^2}} = \frac{-1}{1+1} = -\frac{1}{2}.$$

■

Example 2.1.11. Evaluate $\lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{1}{x}\right)$.

Solution. As $x \to 0$, we have $\infty - \infty$. Combine into a single fraction:

$$\lim_{x\to 0}\frac{x-\sin x}{x\sin x} \quad \left(\text{Form } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule:

$$\lim_{x\to 0}\frac{1-\cos x}{\sin x + x\cos x} \quad \left(\text{Still } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule a second time:

$$\lim_{x\to 0}\frac{\sin x}{\cos x+\cos x - x\sin x} = \frac{0}{1+1-0} = \frac{0}{2} = 0.$$

■

2.1.4 Exponential Indeterminate Forms: $1^\infty, 0^0, \infty^0$

For limit expressions $L = \lim_{x\to a}[f(x)]^{g(x)}$, take the natural logarithm:

$$\ln L = \lim_{x\to a}\Big[g(x)\ln f(x)\Big].$$

Evaluate this product limit $K = \ln L$, then recover the original limit using $L = e^K$.

Example 2.1.12. Evaluate $\lim_{x\to\infty}\left(1+\frac{3}{x}\right)^{2x}$.

Solution. Form: 1^∞. Let $L = \lim_{x\to\infty}\left(1+\frac{3}{x}\right)^{2x}$. Taking the natural logarithm:

$$\ln L = \lim_{x\to\infty} 2x\ln\left(1+\frac{3}{x}\right) = \lim_{x\to\infty}\frac{2\ln(1+3/x)}{1/x} \quad \left(\text{Form } \frac{0}{0}\right).$$

Apply L'Hôpital's Rule:

$$\ln L = \lim_{x\to\infty}\frac{2\cdot\frac{1}{1+3/x}\cdot\left(-\frac{3}{x^2}\right)}{-\frac{1}{x^2}} = \lim_{x\to\infty}\frac{6}{1+3/x} = 6.$$

Since $\ln L = 6$, we obtain $L = e^6$. ■

Example 2.1.13. Evaluate $\lim_{x\to 0^+} x^x$.

Solution. Form: 0^0. Let $L = \lim_{x\to 0^+} x^x$. Taking natural logarithms:

$$\ln L = \lim_{x\to 0^+} x\ln x = 0 \quad (\text{using the result from } 0\cdot\infty).$$

Since $\ln L = 0$, we get $L = e^0 = 1$. ■

Example 2.1.14. Evaluate $\lim_{x\to 0}(\cos x)^{1/x^2}$.

Solution. As $x \to 0$, $\cos 0 \to 1$ and $1/x^2 \to \infty$. Form: 1^∞.

Let $L = \lim_{x\to 0}(\cos x)^{1/x^2}$. Taking natural logarithms:

$$\ln L = \lim_{x\to 0} \frac{\ln(\cos x)}{x^2} \quad \left(\text{Form } \frac{0}{0}\right).$$

Applying L'Hôpital's Rule:

$$\ln L = \lim_{x\to 0} \frac{\frac{-\sin x}{\cos x}}{2x} = \lim_{x\to 0} \left(-\frac{\tan x}{2x}\right) = \lim_{x\to 0} \left(-\frac{1}{2} \cdot \frac{\tan x}{x}\right) = -\frac{1}{2}(1) = -\frac{1}{2}.$$

Since $\ln L = -\frac{1}{2}$, we obtain:

$$L = e^{-1/2} = \frac{1}{\sqrt{e}}.$$

■

Example 2.1.15. Evaluate $\lim_{x\to 0^+} (\sin x)^x$.

Solution. As $x \to 0^+$, we have $(\sin 0)^0 \to 0^0$.

Let $L = \lim_{x\to 0^+} (\sin x)^x$. Taking natural logarithms:

$$\ln L = \lim_{x\to 0^+} x \ln(\sin x) \quad (\text{Form } 0 \cdot (-\infty)).$$

Rewrite as a quotient:

$$\ln L = \lim_{x\to 0^+} \frac{\ln(\sin x)}{1/x} \quad \left(\text{Form } \frac{-\infty}{\infty}\right).$$

Applying L'Hôpital's Rule:

$$\ln L = \lim_{x\to 0^+} \frac{\cot x}{-1/x^2} = \lim_{x\to 0^+} \frac{-x^2 \cos x}{\sin x} = \lim_{x\to 0^+} \left(\frac{x}{\sin x}\right) \cdot (-x\cos x) = (1) \cdot (0) = 0.$$

Since $\ln L = 0$, the final limit is:

$$L = e^0 = 1.$$

■

Example 2.1.16. Evaluate $\lim_{x\to (\pi/2)^-} (\tan x)^{\cos x}$.

Solution. As $x \to (\pi/2)^-$, $\tan x \to \infty$ and $\cos x \to 0$. Form: ∞^0.

Let $L = \lim_{x\to (\pi/2)^-} (\tan x)^{\cos x}$. Taking natural logarithms:

$$\ln L = \lim_{x\to (\pi/2)^-} \cos x \cdot \ln(\tan x) \quad (\text{Form } 0 \cdot \infty).$$

Rewrite as a quotient:

$$\ln L = \lim_{x\to (\pi/2)^-} \frac{\ln(\tan x)}{\sec x} \quad \left(\text{Form } \frac{\infty}{\infty}\right).$$

Applying L'Hôpital's Rule:

$$\ln L = \lim_{x\to (\pi/2)^-} \frac{\frac{\sec^2 x}{\tan x}}{\sec x \tan x} = \lim_{x\to (\pi/2)^-} \frac{\sec x}{\tan^2 x} = \lim_{x\to (\pi/2)^-} \frac{\frac{1}{\cos x}}{\frac{\sin^2 x}{\cos^2 x}} = \lim_{x\to (\pi/2)^-} \frac{\cos x}{\sin^2 x} = \frac{0}{1^2} = 0.$$

Since $\ln L = 0$, we get:

$$L = e^0 = 1.$$

■

2.2 Tangents and Normals in Polar Coordinates

For a polar curve $r = f(\theta)$, let $P(r, \theta)$ be any point on the curve and let O denote the pole. The line OP is called the **radius vector** of the point P, and it makes an angle θ with the initial line.

Let the tangent to the curve at P make an angle ψ with the initial line. If ϕ denotes the angle between the radius vector OP and the tangent at P, then $\psi = \theta + \phi$.

To obtain a relation for ϕ, write the Cartesian coordinates of P as $x = r\cos\theta$ and $y = r\sin\theta$. Differentiating with respect to θ, we get

$$\frac{dx}{d\theta} = \frac{dr}{d\theta}\cos\theta - r\sin\theta, \qquad \frac{dy}{d\theta} = \frac{dr}{d\theta}\sin\theta + r\cos\theta.$$

Hence, the slope of the tangent is

$$\tan\psi = \frac{dy}{dx} = \frac{\dfrac{dr}{d\theta}\sin\theta + r\cos\theta}{\dfrac{dr}{d\theta}\cos\theta - r\sin\theta}.$$

Since $\psi = \theta + \phi$, using the formula for $\tan(\theta + \phi)$ and simplifying, we obtain

$$\tan\phi = \frac{r}{dr/d\theta} \qquad \text{or equivalently} \qquad \cot\phi = \frac{1}{r}\frac{dr}{d\theta}.$$

Hence the angle between the radius vector and the tangent is determined directly by the polar equation.

The **normal** at P is the straight line through P perpendicular to the tangent. Therefore, if the tangent makes an angle ψ with the initial line, the normal makes an angle $\psi + \frac{\pi}{2}$ with the initial line.

Now let OM be drawn perpendicular from the pole O to the tangent at P, and let $OM = p$. Since OMP is a right-angled triangle and the angle between OP and the tangent is ϕ, we have

$$\sin\phi = \frac{OM}{OP} = \frac{p}{r}, \qquad \text{and hence} \qquad p = r\sin\phi.$$

Accordingly, the two fundamental relations represented in Figure 2.1 are

$$\tan\phi = \frac{r}{dr/d\theta}, \qquad p = r\sin\phi.$$

See Figure 12 in the Atlas for the geometry of the polar tangent, normal, radius vector, and perpendicular from the pole.

2.2.1 Perpendicular from Pole to the Tangent

Let $P(r, \theta)$ be a point on the polar curve $r = f(\theta)$, and let the tangent at P meet the perpendicular drawn from the pole O at M. If $OM = p$ and ϕ is the angle between the radius vector OP and the tangent at P, then p represents the perpendicular distance from the pole to the tangent.

Since triangle OMP is right-angled at M, we have $p = r\sin\phi$.

For a polar curve, $\cot\phi = \dfrac{1}{r}\dfrac{dr}{d\theta}$. Hence $\csc^2\phi = 1 + \cot^2\phi = 1 + \dfrac{1}{r^2}\left(\dfrac{dr}{d\theta}\right)^2$.

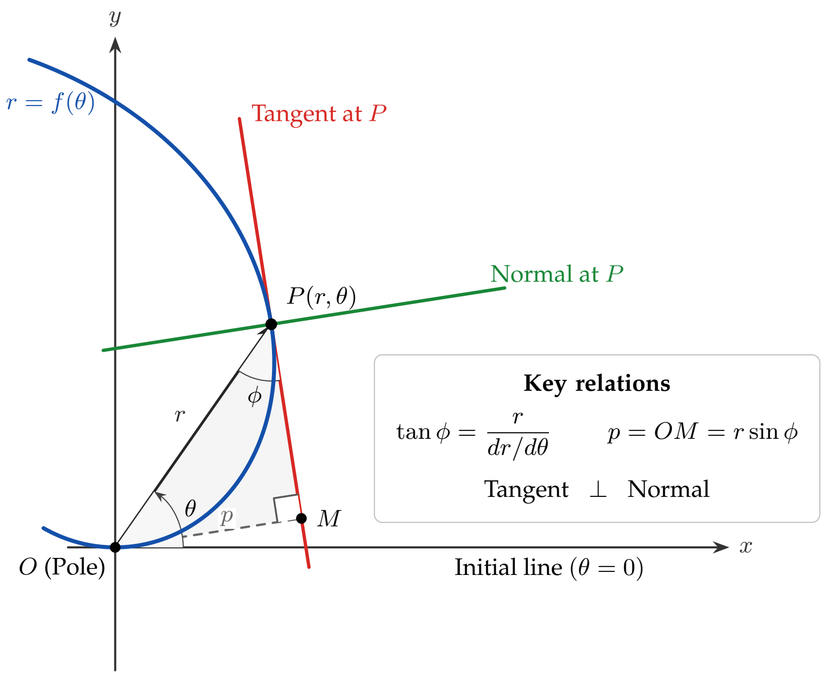


Figure 2.1: Geometry of the tangent and normal to the polar curve $r = f(\theta)$ at $P(r, \theta)$. Here ϕ is the angle between the radius vector and the tangent, and $p = OM$ is the perpendicular distance from the pole to the tangent.

Therefore,

$$p = \frac{r^2}{\sqrt{r^2 + \left(\dfrac{dr}{d\theta}\right)^2}}.$$

Thus, the length of the perpendicular from the pole to the tangent at any point of the polar curve can be found directly from r and $\dfrac{dr}{d\theta}$.

Example 2.2.1 (See Figure 17 in the Atlas)**.** Find the length of the perpendicular from the pole to the tangent to the cardioid $r = a(1 + \cos\theta)$.

Solution. For the given curve, $\dfrac{dr}{d\theta} = -a\sin\theta$. Also, $r^2 + \left(\dfrac{dr}{d\theta}\right)^2 = a^2(1+\cos\theta)^2 + a^2\sin^2\theta = 2a^2(1+\cos\theta) = 2ar$.

Using the formula for the perpendicular from the pole to the tangent,

$$p = \frac{r^2}{\sqrt{2ar}} = \sqrt{\frac{r^3}{2a}}.$$

Hence, the required perpendicular is $p = \sqrt{\dfrac{r^3}{2a}}$. ■

Example 2.2.2 (See Figure 15 in the Atlas)**.** Find the length of the perpendicular from the pole to the tangent to the circle $r = 2a\cos\theta$.

Solution. Differentiating, $\dfrac{dr}{d\theta} = -2a\sin\theta$. Therefore, $r^2 + \left(\dfrac{dr}{d\theta}\right)^2 = 4a^2\cos^2\theta + 4a^2\sin^2\theta = 4a^2$.

Hence,

$$p = \frac{r^2}{2a}.$$

Thus, the required perpendicular from the pole to the tangent is $p = \dfrac{r^2}{2a}$. ∎

Example 2.2.3 (See Figure 25 in the Atlas)**.** Find the length of the perpendicular from the pole to the tangent to the equiangular spiral $r = ae^{\theta \cot \alpha}$.

Solution. Differentiating, $\dfrac{dr}{d\theta} = r \cot \alpha$. Hence, $r^2 + \left(\dfrac{dr}{d\theta}\right)^2 = r^2(1 + \cot^2 \alpha) = r^2 \csc^2 \alpha$.

Therefore,

$$p = \frac{r^2}{r \csc \alpha} = r \sin \alpha.$$

Thus, the perpendicular from the pole to the tangent is $p = r \sin \alpha$. ∎

Example 2.2.4 (See Figure 21 in the Atlas)**.** Find the length of the perpendicular from the pole to the tangent to the lemniscate $r^2 = a^2 \cos 2\theta$.

Solution. Differentiating $r^2 = a^2 \cos 2\theta$, we get $2r\dfrac{dr}{d\theta} = -2a^2 \sin 2\theta$, so that $\dfrac{dr}{d\theta} = -\dfrac{a^2 \sin 2\theta}{r}$.

Since $\cos 2\theta = \dfrac{r^2}{a^2}$, we have $a^4 \sin^2 2\theta = a^4 - r^4$. Hence $\left(\dfrac{dr}{d\theta}\right)^2 = \dfrac{a^4 - r^4}{r^2}$.

It follows that $r^2 + \left(\dfrac{dr}{d\theta}\right)^2 = \dfrac{a^4}{r^2}$. Therefore,

$$p = \frac{r^2}{a^2/r} = \frac{r^3}{a^2}.$$

Hence, the required perpendicular is $p = \dfrac{r^3}{a^2}$. ∎

Example 2.2.5 (See Figure 27 in the Atlas)**.** Find the length of the perpendicular from the pole to the tangent to the parabola $\dfrac{2a}{r} = 1 - \cos \theta$.

Solution. Writing $r = \dfrac{2a}{1 - \cos \theta}$ and differentiating, we obtain $\dfrac{dr}{d\theta} = -r \cot \dfrac{\theta}{2}$.

Therefore, $r^2 + \left(\dfrac{dr}{d\theta}\right)^2 = r^2 \left(1 + \cot^2 \dfrac{\theta}{2}\right) = r^2 \csc^2 \dfrac{\theta}{2}$.

Hence,

$$p = r \sin \frac{\theta}{2}.$$

Now $1 - \cos \theta = 2 \sin^2 \dfrac{\theta}{2}$ and $\dfrac{2a}{r} = 1 - \cos \theta$. Therefore, $\sin^2 \dfrac{\theta}{2} = \dfrac{a}{r}$. Consequently, $p^2 = r^2 \dfrac{a}{r} = ar$.

Thus, the required perpendicular from the pole to the tangent is $p = \sqrt{ar}$. ∎

2.2.2 Angle of Intersection of Two Polar Curves

Let two polar curves $r = f_1(\theta)$ and $r = f_2(\theta)$ intersect at point P. Let ϕ_1 and ϕ_2 be the angles made by their respective tangents with the common radius vector OP.

Definition 2.2.6. The angle of intersection α between two polar curves is:

$$\alpha = |\phi_1 - \phi_2|.$$

The acute angle α is given by:

$$\tan\alpha = \left|\frac{\tan\phi_1 - \tan\phi_2}{1 + \tan\phi_1\tan\phi_2}\right|.$$

The curves intersect **orthogonally** ($\alpha = 90°$) if:

$$\tan\phi_1 \cdot \tan\phi_2 = -1 \quad \text{or} \quad \cot\phi_1 \cdot \cot\phi_2 = -1.$$

Example 2.2.7 (See Figures 13 and 17 in the Atlas)**.** Find the angle of intersection between the cardioid $r = a(1 + \cos\theta)$ and the circle $r = a$.

Solution. Equating the two curve equations to determine the points of intersection:

$$a(1 + \cos\theta) = a \implies 1 + \cos\theta = 1 \implies \cos\theta = 0.$$

In the interval $[0, 2\pi)$, this gives $\theta = \dfrac{\pi}{2}$ and $\theta = \dfrac{3\pi}{2}$. Evaluating at $\theta = \dfrac{\pi}{2}$:

For the cardioid $r_1 = a(1 + \cos\theta)$, differentiating with respect to θ yields:

$$\frac{dr_1}{d\theta} = -a\sin\theta.$$

The angle ϕ_1 between the radius vector and the tangent is given by:

$$\cot\phi_1 = \frac{1}{r_1}\frac{dr_1}{d\theta} = \frac{-a\sin\theta}{a(1+\cos\theta)} = -\tan\left(\frac{\theta}{2}\right).$$

At $\theta = \dfrac{\pi}{2}$, we obtain:

$$\cot\phi_1 = -\tan\left(\frac{\pi}{4}\right) = -1 \implies \phi_1 = \frac{3\pi}{4} = 135°.$$

For the circle $r_2 = a$, differentiating with respect to θ gives:

$$\frac{dr_2}{d\theta} = 0 \implies \cot\phi_2 = \frac{1}{r_2}\frac{dr_2}{d\theta} = 0 \implies \phi_2 = \frac{\pi}{2} = 90°.$$

Applying the angle of intersection formula $\alpha = |\phi_1 - \phi_2|$:

$$\alpha = \left|\frac{3\pi}{4} - \frac{\pi}{2}\right| = \frac{\pi}{4} = 45°.$$

Thus, the cardioid and the circle intersect at an acute angle of $45°$. ■

Example 2.2.8 (See Figures 13, 15, and 16 in the Atlas)**.** Show that the circles $r = 2a\cos\theta$ and $r = 2a\sin\theta$ intersect orthogonally.

Solution. Equating the two equations to find the intersection point:

$$2a\cos\theta = 2a\sin\theta \implies \tan\theta = 1 \implies \theta = \frac{\pi}{4}.$$

For the first circle $r_1 = 2a\cos\theta$:

$$\frac{dr_1}{d\theta} = -2a\sin\theta \implies \cot\phi_1 = \frac{1}{r_1}\frac{dr_1}{d\theta} = \frac{-2a\sin\theta}{2a\cos\theta} = -\tan\theta.$$

At $\theta = \frac{\pi}{4}$, we have $\cot\phi_1 = -1$, which implies $\tan\phi_1 = -1$.

For the second circle $r_2 = 2a\sin\theta$:

$$\frac{dr_2}{d\theta} = 2a\cos\theta \implies \cot\phi_2 = \frac{1}{r_2}\frac{dr_2}{d\theta} = \frac{2a\cos\theta}{2a\sin\theta} = \cot\theta.$$

At $\theta = \frac{\pi}{4}$, we have $\cot\phi_2 = 1$, which implies $\tan\phi_2 = 1$.

Their product is

$$\tan\phi_1 \cdot \tan\phi_2 = (-1)\cdot(1) = -1.$$

Since $\tan\phi_1 \cdot \tan\phi_2 = -1$, the two circles intersect **orthogonally** ($\alpha = 90°$). ■

Example 2.2.9 (See Figures 13, 17, and 18 in the Atlas)**.** Show that the cardioids $r = a(1+\cos\theta)$ and $r = b(1-\cos\theta)$ intersect orthogonally at all points of intersection.

Solution. For the first cardioid $r_1 = a(1+\cos\theta)$, differentiating with respect to θ gives:

$$\frac{dr_1}{d\theta} = -a\sin\theta \implies \cot\phi_1 = \frac{-a\sin\theta}{a(1+\cos\theta)} = \frac{-2\sin(\theta/2)\cos(\theta/2)}{2\cos^2(\theta/2)} = -\tan\left(\frac{\theta}{2}\right).$$

For the second cardioid $r_2 = b(1-\cos\theta)$, differentiating with respect to θ gives:

$$\frac{dr_2}{d\theta} = b\sin\theta \implies \cot\phi_2 = \frac{b\sin\theta}{b(1-\cos\theta)} = \frac{2\sin(\theta/2)\cos(\theta/2)}{2\sin^2(\theta/2)} = \cot\left(\frac{\theta}{2}\right).$$

Their product is

$$\cot\phi_1 \cdot \cot\phi_2 = \left(-\tan\frac{\theta}{2}\right)\cdot\left(\cot\frac{\theta}{2}\right) = -1.$$

Since $\cot\phi_1 \cdot \cot\phi_2 = -1$ holds independently of θ, the two cardioids intersect **orthogonally** ($\alpha = 90°$) at all intersection points. ■

2.2.3 Lengths of Polar Tangent, Normal, Subtangent, and Subnormal

Draw a line through the pole O perpendicular to the radius vector OP. Let the tangent line PT and normal line PN at P meet this perpendicular line at T and N, respectively.

Definition 2.2.10 (Polar Lengths)**.** (i) **Polar Subtangent (**OT**):**

$$OT = r\tan\phi = r^2\frac{d\theta}{dr}.$$

(ii) **Polar Subnormal (**ON**):**

$$ON = r\cot\phi = \frac{dr}{d\theta}.$$

(iii) **Length of Polar Tangent (PT):**

$$PT = r\sec\phi = r\sqrt{1 + r^2\left(\frac{d\theta}{dr}\right)^2}.$$

(iv) **Length of Polar Normal (PN):**

$$PN = r\csc\phi = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}.$$

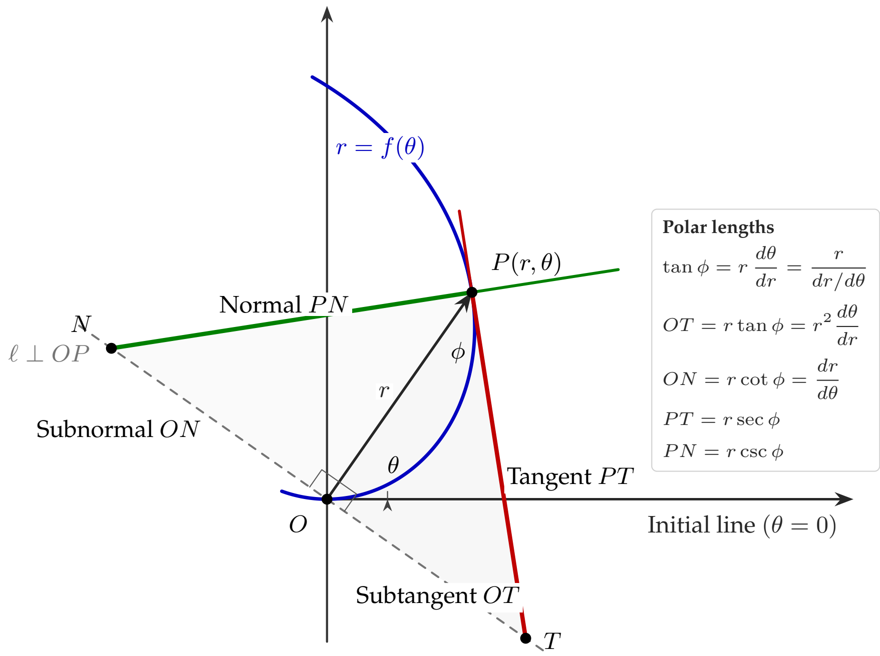


Figure 2.2: Geometric interpretation of the *polar tangent* PT, *polar normal* PN, *polar subtangent* OT, and *polar subnormal* ON for a polar curve $r = f(\theta)$. The line ℓ through the pole is perpendicular to the radius vector OP.

See Figure 14 in the Atlas for the polar tangent, polar normal, polar subtangent, and polar subnormal.

Example 2.2.11 (Angle Between Radius Vector and Tangent; See Figures 12 and 17 in the Atlas)**.** Find the angle ϕ between the radius vector and the tangent for the cardioid $r = a(1 + \cos\theta)$.

Solution. Given $r = a(1 + \cos\theta)$. Differentiating with respect to θ:

$$\frac{dr}{d\theta} = -a\sin\theta.$$

Evaluate $\cot\phi$:

$$\cot\phi = \frac{1}{r}\frac{dr}{d\theta} = \frac{-a\sin\theta}{a(1+\cos\theta)} = \frac{-2\sin(\theta/2)\cos(\theta/2)}{2\cos^2(\theta/2)} = -\tan\left(\frac{\theta}{2}\right) = \cot\left(\frac{\pi}{2} + \frac{\theta}{2}\right).$$

Therefore:

$$\phi = \frac{\pi}{2} + \frac{\theta}{2}.$$

■

Example 2.2.12 (Orthogonal Intersection of Polar Curves)**.** Show that the curves $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$ intersect orthogonally.

Solution. For Curve 1 ($r = a(1 + \cos\theta)$):

$$\cot\phi_1 = \frac{1}{r}\frac{dr}{d\theta} = \frac{-a\sin\theta}{a(1+\cos\theta)} = -\tan\left(\frac{\theta}{2}\right).$$

For Curve 2 ($r = b(1 - \cos\theta)$):

$$\cot\phi_2 = \frac{1}{r}\frac{dr}{d\theta} = \frac{b\sin\theta}{b(1-\cos\theta)} = \frac{2\sin(\theta/2)\cos(\theta/2)}{2\sin^2(\theta/2)} = \cot\left(\frac{\theta}{2}\right).$$

Compute the product $\cot\phi_1 \cdot \cot\phi_2$:

$$\cot\phi_1 \cdot \cot\phi_2 = \left(-\tan\frac{\theta}{2}\right) \cdot \left(\cot\frac{\theta}{2}\right) = -1.$$

Since $\cot\phi_1 \cdot \cot\phi_2 = -1$, the two curves intersect **orthogonally** ($\alpha = 90°$). ■

Example 2.2.13 (Pedal Equation of a Cardioid; See Figure 17 in the Atlas)**.** Find the pedal equation (p–r relation) of the cardioid $r = a(1 + \cos\theta)$.

Solution. From the previous example, $\phi = \frac{\pi}{2} + \frac{\theta}{2}$ for $r = a(1 + \cos\theta)$.

Using $p = r\sin\phi$:

$$p = r\sin\left(\frac{\pi}{2} + \frac{\theta}{2}\right) = r\cos\left(\frac{\theta}{2}\right).$$

Squaring both sides:

$$p^2 = r^2\cos^2\left(\frac{\theta}{2}\right) = r^2\left(\frac{1+\cos\theta}{2}\right).$$

Since $1 + \cos\theta = \frac{r}{a}$, substitute this into the equation:

$$p^2 = r^2\left(\frac{r}{2a}\right) \implies p^2 = \frac{r^3}{2a}.$$

This is the required pedal equation of the cardioid. ■

Example 2.2.14 (Pedal Equation of an Equiangular Spiral; See Figure 25 in the Atlas)**.** Find the pedal equation of the equiangular spiral $r = ae^{\theta\cot\alpha}$.

Solution. Differentiating $r = ae^{\theta\cot\alpha}$ with respect to θ:

$$\frac{dr}{d\theta} = a\cot\alpha e^{\theta\cot\alpha} = r\cot\alpha.$$

Compute $\cot\phi$:

$$\cot\phi = \frac{1}{r}\frac{dr}{d\theta} = \frac{r\cot\alpha}{r} = \cot\alpha \implies \phi = \alpha.$$

Since $\phi = \alpha$ is constant, substitute into $p = r\sin\phi$:

$$p = r\sin\alpha.$$

This is the pedal equation of the equiangular spiral. ■

Example 2.2.15 (Polar Subtangent of Hyperbolic Spiral; See Figures 14 and 26 in the Atlas)**.** Show that the polar subtangent for the hyperbolic spiral $r\theta = a$ is constant.

Solution. The given curve equation is $r\theta = a \implies r = \dfrac{a}{\theta}$.

Differentiating r with respect to θ:

$$\frac{dr}{d\theta} = -\frac{a}{\theta^2} \implies \frac{d\theta}{dr} = -\frac{\theta^2}{a}.$$

The length of the polar subtangent is:

$$\text{Polar Subtangent} = \left| r^2 \frac{d\theta}{dr} \right| = \left| \left(\frac{a}{\theta}\right)^2 \cdot \left(-\frac{\theta^2}{a}\right) \right| = |-a| = a.$$

Since a is a constant, the length of the polar subtangent is constant for all points on $r\theta = a$. ■

Example 2.2.16 (Polar Subnormal and Subtangent of a Logarithmic Spiral; See Figures 14 and 25 in the Atlas)**.** Find the lengths of the polar subtangent and polar subnormal for the spiral $r = ae^{\theta \cot \beta}$.

Solution. Given $r = ae^{\theta \cot \beta}$. Differentiating with respect to θ:

$$\frac{dr}{d\theta} = a \cot \beta e^{\theta \cot \beta} = r \cot \beta.$$

Now compute the required lengths:

(i) **Polar Subnormal:**

$$\text{Polar Subnormal} = \frac{dr}{d\theta} = r \cot \beta.$$

(ii) **Polar Subtangent:**

$$\text{Polar Subtangent} = r^2 \frac{d\theta}{dr} = r^2 \left(\frac{1}{r \cot \beta}\right) = r \tan \beta.$$

■

Example 2.2.17 (See Figures 14 and 27 in the Atlas)**.** Find the polar subnormal and polar subtangent for the parabola $\dfrac{2a}{r} = 1 - \cos \theta$.

Solution. Rewrite the equation as:

$$r = \frac{2a}{1 - \cos \theta}.$$

Differentiating with respect to θ:

$$\frac{dr}{d\theta} = -\frac{2a}{(1 - \cos \theta)^2} \cdot (\sin \theta) = -\left(\frac{2a}{1 - \cos \theta}\right) \frac{\sin \theta}{1 - \cos \theta} = -r \frac{\sin \theta}{1 - \cos \theta}.$$

Using half-angle identities:

$$\frac{\sin \theta}{1 - \cos \theta} = \frac{2 \sin(\theta/2) \cos(\theta/2)}{2 \sin^2(\theta/2)} = \cot\left(\frac{\theta}{2}\right).$$

Thus:

$$\frac{dr}{d\theta} = -r \cot\left(\frac{\theta}{2}\right).$$

Now evaluate the required lengths:

(i) **Polar Subnormal:**

$$\text{Polar Subnormal} = \left|\frac{dr}{d\theta}\right| = r\cot\left(\frac{\theta}{2}\right).$$

(ii) **Polar Subtangent:**

$$\text{Polar Subtangent} = \left|r^2\frac{d\theta}{dr}\right| = \left|r^2\left(\frac{1}{-r\cot(\theta/2)}\right)\right| = r\tan\left(\frac{\theta}{2}\right).$$

■

Exercise 2.2.18. Solve the following exercises:

1. Evaluate the limit:
$$\lim_{x\to 0}\frac{e^x+e^{-x}-2\cos x}{x\sin x}.$$

2. Evaluate the limit:
$$\lim_{x\to 0}\frac{x-\tan x}{x^3}.$$

3. Evaluate the limit:
$$\lim_{x\to\infty}\frac{\ln(1+e^{2x})}{x}.$$

4. Evaluate the limit:
$$\lim_{x\to 0^+} x^2\ln x.$$

5. Evaluate the limit:
$$\lim_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{\sin^2 x}\right).$$

6. Evaluate the limit:
$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{1/x^2}.$$

7. Evaluate the limit:
$$\lim_{x\to 0^+}(\tan x)^x.$$

8. Evaluate the limit:
$$\lim_{x\to\infty}(1+x)^{1/x}.$$

9. Show that for positive constants a and b:
$$\lim_{x\to 0}\left(\frac{a^x+b^x}{2}\right)^{1/x} = \sqrt{ab}.$$

10. Find the angle ϕ between the radius vector and the tangent for the cardioid $r = a(1+\sin\theta)$ at $\theta = \frac{\pi}{3}$.

11. Show that for the polar curve $r^n = a^n\cos n\theta$, the angle ϕ between the radius vector and the tangent is given by $\phi = \frac{\pi}{2} + n\theta$.

12. Derive the pedal equation (p–r relation) for the parabola $\frac{2a}{r} = 1+\cos\theta$.

13. Show that the pedal equation for the lemniscate $r^2 = a^2 \cos 2\theta$ is $p = \dfrac{r^3}{a^2}$.

14. Find the perpendicular distance p from the pole to the tangent for the curve $r^n = a^n \sin n\theta$.

15. Find the angle of intersection between the two circles $r = a \sin\theta$ and $r = a\cos\theta$.

16. Prove that the parabolas $r = \dfrac{a}{1+\cos\theta}$ and $r = \dfrac{b}{1-\cos\theta}$ intersect orthogonally.

17. Determine the angle of intersection between the polar curves $r^2 \sin 2\theta = 4$ and $r^2 \cos 2\theta = 4$.

18. For the cardioid $r = a(1-\cos\theta)$, calculate the lengths of the polar tangent, polar normal, polar subtangent, and polar subnormal at $\theta = \dfrac{\pi}{2}$.

19. Show that the length of the polar subnormal for the logarithmic spiral $r = ae^{\theta}$ is equal to its radius vector r at every point.

20. Show that the length of the polar subtangent for the curve $r^n\theta = a^n$ is inversely proportional to r^{n-1} (or proportional to r^{n+1} depending on n).

Chapter 3

Differential Geometry of Curves and Real Analysis

This unit explores the local geometry of plane curves alongside fundamental theorems of real analysis. It introduces differential measures of bending—curvature, radius of curvature, and osculating circles—along with pedal equations, arc lengths, asymptotes, and singular points. Furthermore, it establishes core global properties of continuous functions on closed intervals, including the Boundedness Theorem, Extreme Value Theorem, Intermediate Value Theorem, and Darboux's Theorem.

3.1 Curvature and Radius of Curvature

Definition 3.1.1 (Curvature)**. Curvature**, denoted by the Greek letter kappa (κ), measures the rate at which a curve changes direction with respect to arc length s.

- A straight line does not bend at all, so its curvature is $\kappa = 0$ everywhere.
- A circle bends uniformly. Smaller circles bend more sharply and thus have larger curvatures.

Formally, if $\mathbf{T} = \dfrac{d\mathbf{r}}{ds}$ is the unit tangent vector to the curve $s \mapsto \mathbf{r}(s)$, the curvature is defined as:

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| = \left| \frac{d\psi}{ds} \right|,$$

where ψ is the angle of inclination of the tangent line with the positive x-axis.

Definition 3.1.2 (Radius of Curvature)**.** The **radius of curvature**, denoted by the Greek letter rho (ρ), is the reciprocal of the curvature:

$$\rho = \frac{1}{\kappa} = \left| \frac{ds}{d\psi} \right| \quad (\text{for } \kappa \neq 0).$$

Geometrically, ρ is the radius of the **osculating circle** (or "kissing circle") that best fits the curve locally at a given point, sharing both the same tangent line and curvature.

3.1.1 Formulas for Computation in Various Coordinate Systems

1. **Cartesian Form $y = f(x)$:**

$$\kappa = \frac{|y''|}{[1 + (y')^2]^{3/2}} \implies \rho = \frac{[1 + (y')^2]^{3/2}}{|y''|},$$

where $y' = \dfrac{dy}{dx}$ and $y'' = \dfrac{d^2y}{dx^2}$. (If the tangent is vertical, use $\rho = \dfrac{[1+(x')^2]^{3/2}}{|x''|}$ where $x' = \dfrac{dx}{dy}$).

2. **Parametric Form** $x = x(t), y = y(t)$**:**

$$\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|},$$

where dots denote differentiation with respect to the parameter t.

3. **Polar Form** $r = f(\theta)$**:**

$$\rho = \frac{[r^2 + (r')^2]^{3/2}}{|r^2 + 2(r')^2 - rr''|},$$

where $r' = \dfrac{dr}{d\theta}$ and $r'' = \dfrac{d^2r}{d\theta^2}$.

4. **Pedal Form** $p = f(r)$**:** For a curve expressed in terms of its radius vector r and perpendicular distance p from the pole to the tangent line:

$$\rho = r\frac{dr}{dp}.$$

5. **Intrinsic Form** $s = f(\psi)$**:** When arc length s is given directly as a function of the tangent inclination angle ψ:

$$\rho = \left|\frac{ds}{d\psi}\right|.$$

Example 3.1.3 (Cartesian Form – Catenary Curve)**.** Find the radius of curvature at any point for the catenary $y = c\cosh\left(\frac{x}{c}\right)$, and show that $\rho = \dfrac{y^2}{c}$.

Solution. Given $y = c\cosh\left(\frac{x}{c}\right)$. First, compute the first derivative:

$$y' = \frac{dy}{dx} = \sinh\left(\frac{x}{c}\right).$$

Differentiating again to find the second derivative:

$$y'' = \frac{d^2y}{dx^2} = \frac{1}{c}\cosh\left(\frac{x}{c}\right).$$

Using the fundamental hyperbolic identity $1 + \sinh^2 u = \cosh^2 u$:

$$1 + (y')^2 = 1 + \sinh^2\left(\frac{x}{c}\right) = \cosh^2\left(\frac{x}{c}\right).$$

Substitute these expressions into the Cartesian radius of curvature formula:

$$\rho = \frac{[1+(y')^2]^{3/2}}{|y''|} = \frac{\left[\cosh^2(x/c)\right]^{3/2}}{\frac{1}{c}\cosh(x/c)} = c\frac{\cosh^3(x/c)}{\cosh(x/c)} = c\cosh^2\left(\frac{x}{c}\right).$$

Since $y = c\cosh(x/c) \implies \cosh(x/c) = \dfrac{y}{c}$, we substitute this back:

$$\rho = c\left(\frac{y}{c}\right)^2 = \frac{y^2}{c}.$$

■

Example 3.1.4 (Parametric Form – Cycloid; See Figures 28 and 31 in the Atlas)**.** Find the radius of curvature at any point t for the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

Solution. First, compute the first-order derivatives with respect to t:

$$\dot{x} = a(1 - \cos t), \quad \dot{y} = a \sin t.$$

Next, compute the second-order derivatives:

$$\ddot{x} = a \sin t, \quad \ddot{y} = a \cos t.$$

Evaluate the numerator factor $(\dot{x}^2 + \dot{y}^2)$:

$$\begin{aligned} \dot{x}^2 + \dot{y}^2 &= a^2(1 - \cos t)^2 + a^2 \sin^2 t \\ &= a^2(1 - 2\cos t + \cos^2 t + \sin^2 t) \\ &= a^2(2 - 2\cos t) = 2a^2(1 - \cos t) = 4a^2 \sin^2\left(\frac{t}{2}\right). \end{aligned}$$

Taking the $3/2$ power:

$$(\dot{x}^2 + \dot{y}^2)^{3/2} = \left[4a^2 \sin^2\left(\frac{t}{2}\right)\right]^{3/2} = 8a^3 \sin^3\left(\frac{t}{2}\right).$$

Now compute the cross-product term in the denominator:

$$\begin{aligned} \dot{x}\ddot{y} - \dot{y}\ddot{x} &= [a(1 - \cos t)][a \cos t] - [a \sin t][a \sin t] \\ &= a^2 \cos t - a^2 \cos^2 t - a^2 \sin^2 t \\ &= a^2(\cos t - 1) = -2a^2 \sin^2\left(\frac{t}{2}\right). \end{aligned}$$

Substituting these into the parametric radius of curvature formula:

$$\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|} = \frac{8a^3 \sin^3(t/2)}{|-2a^2 \sin^2(t/2)|} = 4a \sin\left(\frac{t}{2}\right).$$

■

Example 3.1.5 (Polar Form – General Cardioid; See Figures 18 and 28 in the Atlas)**.** Find the radius of curvature at any angle θ for the cardioid $r = a(1 - \cos\theta)$.

Solution. Differentiating $r = a(1 - \cos\theta)$ with respect to θ:

$$r' = a \sin\theta, \quad \text{and} \quad r'' = a \cos\theta.$$

Compute the term $r^2 + (r')^2$:

$$\begin{aligned} r^2 + (r')^2 &= a^2(1 - \cos\theta)^2 + a^2 \sin^2\theta \\ &= a^2(1 - 2\cos\theta + \cos^2\theta + \sin^2\theta) = 2a^2(1 - \cos\theta) = 2ar. \end{aligned}$$

Taking the $3/2$ power:

$$[r^2 + (r')^2]^{3/2} = (2ar)^{3/2} = 2\sqrt{2}a^{3/2}r^{3/2}.$$

Compute the denominator expression $r^2 + 2(r')^2 - rr''$:

$$\begin{aligned} r^2 + 2(r')^2 - rr'' &= a^2(1 - \cos\theta)^2 + 2a^2 \sin^2\theta - a(1 - \cos\theta)(a\cos\theta) \\ &= a^2\Big[(1 - 2\cos\theta + \cos^2\theta) + 2\sin^2\theta - (\cos\theta - \cos^2\theta)\Big] \\ &= a^2\Big[1 - 3\cos\theta + 2\cos^2\theta + 2\sin^2\theta\Big] \\ &= a^2[3 - 3\cos\theta] = 3a \cdot a(1 - \cos\theta) = 3ar. \end{aligned}$$

Substituting both terms into the polar radius of curvature formula:

$$\rho = \frac{[r^2 + (r')^2]^{3/2}}{|r^2 + 2(r')^2 - rr''|} = \frac{2\sqrt{2}a^{3/2}r^{3/2}}{3ar} = \frac{2}{3}\sqrt{2ar}.$$

Expressing ρ directly in terms of θ using $r = 2a\sin^2(\theta/2)$:

$$\rho = \frac{2}{3}\sqrt{2a \cdot 2a\sin^2(\theta/2)} = \frac{4a}{3}\sin\left(\frac{\theta}{2}\right).$$

■

Example 3.1.6 (Pedal Form Formula; See Figure 17 in the Atlas)**.** Find the radius of curvature for the cardioid whose pedal equation is $p^2 = \dfrac{r^3}{2a}$.

Solution. Given the pedal equation $p^2 = \dfrac{r^3}{2a} \implies p = \dfrac{r^{3/2}}{\sqrt{2a}}$.

Differentiating with respect to r:

$$\frac{dp}{dr} = \frac{3}{2\sqrt{2a}}r^{1/2}.$$

Taking the reciprocal:

$$\frac{dr}{dp} = \frac{2\sqrt{2a}}{3\sqrt{r}}.$$

Using the pedal form formula $\rho = r\dfrac{dr}{dp}$:

$$\rho = r\left(\frac{2\sqrt{2a}}{3\sqrt{r}}\right) = \frac{2}{3}\sqrt{2ar}.$$

This matches the result obtained from the polar form solution. ■

3.2 Pedal Equations

Definition 3.2.1 (Pedal Equation)**.** The **pedal equation** of a curve is the relation between the radius vector r and the perpendicular distance p from the pole to the tangent at the corresponding point. It is therefore an equation involving only p and r.

Theorem 3.2.2 (Pedal Equation Formula)**.** *Let the equation of the curve be given in polar form, $r = f(\theta)$. Then p and r satisfy*

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2 \tag{3.1}$$

Proof. Let ϕ be the angle between the radius vector and the tangent at the point $P(r,\theta)$. If p denotes the perpendicular distance from the pole to the tangent, then from the right-angled triangle formed by the pole, the point of contact, and the foot of the perpendicular, we have $p = r\sin\phi$.

Therefore, $\dfrac{1}{p^2} = \dfrac{1}{r^2\sin^2\phi} = \dfrac{\csc^2\phi}{r^2}$.

Using the identity $\csc^2\phi = 1 + \cot^2\phi$, we get

$$\frac{1}{p^2} = \frac{1}{r^2}\left(1 + \cot^2\phi\right).$$

For a polar curve $r = f(\theta)$, the angle between the radius vector and the tangent satisfies $\cot\phi = \dfrac{1}{r}\dfrac{dr}{d\theta}$.

Substituting this value of $\cot\phi$, we obtain

$$\frac{1}{p^2} = \frac{1}{r^2}\left[1 + \frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right].$$

Hence,

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2.$$

Thus, the pedal equation is obtained by eliminating θ between this relation and the given polar equation of the curve. ■

Note: To find the pedal equation, one must eliminate θ from this formula using the original curve's equation.

Example 3.2.3 (See Figure 17 in the Atlas)**.** Find the pedal equation of the cardioid $r = a(1 + \cos\theta)$.

Solution. For the given curve, $\dfrac{dr}{d\theta} = -a\sin\theta$. Using the pedal formula, $\dfrac{1}{p^2} = \dfrac{1}{r^2} + \dfrac{1}{r^4}\left(\dfrac{dr}{d\theta}\right)^2$, we get

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{a^2\sin^2\theta}{r^4}.$$

Since $r = a(1+\cos\theta)$, we have $\cos\theta = \dfrac{r}{a} - 1$, and hence $a^2\sin^2\theta = 2ar - r^2$. Therefore,

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{2ar - r^2}{r^4} = \frac{2a}{r^3}.$$

Thus, the required pedal equation is $p^2 = \dfrac{r^3}{2a}$. ■

Example 3.2.4 (See Figure 15 in the Atlas)**.** Find the pedal equation of the circle $r = 2a\cos\theta$.

Solution. Differentiating, $\dfrac{dr}{d\theta} = -2a\sin\theta$. Hence,

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{4a^2\sin^2\theta}{r^4}.$$

Since $r = 2a\cos\theta$, we have $4a^2\cos^2\theta = r^2$, so that $4a^2\sin^2\theta = 4a^2 - r^2$. Therefore,

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{4a^2 - r^2}{r^4} = \frac{4a^2}{r^4}.$$

Hence, the required pedal equation is $p = \dfrac{r^2}{2a}$, or equivalently, $p^2 = \dfrac{r^4}{4a^2}$. ■

Example 3.2.5 (See Figure 25 in the Atlas)**.** Find the pedal equation of the equiangular spiral $r = ae^{\theta\cot\alpha}$.

Solution. Differentiating, we obtain $\dfrac{dr}{d\theta} = r\cot\alpha$. Substitution in the pedal formula gives

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{r^2\cot^2\alpha}{r^4} = \frac{1+\cot^2\alpha}{r^2} = \frac{\csc^2\alpha}{r^2}.$$

Hence, $p = r\sin\alpha$.

Thus, for an equiangular spiral, the perpendicular from the pole to the tangent bears a constant ratio to the radius vector. ■

Example 3.2.6 (See Figure 21 in the Atlas)**.** Find the pedal equation of the lemniscate $r^2 = a^2\cos 2\theta$.

Solution. Differentiating the given equation, $2r\dfrac{dr}{d\theta} = -2a^2\sin 2\theta$, and therefore $\dfrac{dr}{d\theta} = -\dfrac{a^2\sin 2\theta}{r}$.

Since $\cos 2\theta = \dfrac{r^2}{a^2}$, we have $a^4\sin^2 2\theta = a^4 - r^4$. Consequently,

$$\left(\frac{dr}{d\theta}\right)^2 = \frac{a^4 - r^4}{r^2}.$$

Using the pedal formula,

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{a^4 - r^4}{r^6} = \frac{a^4}{r^6}.$$

Thus, the required pedal equation is $p^2 = \dfrac{r^6}{a^4}$. ■

Example 3.2.7 (See Figure 27 in the Atlas)**.** Find the pedal equation of the parabola $\dfrac{2a}{r} = 1 - \cos\theta$.

Solution. Writing the equation as $r = \dfrac{2a}{1-\cos\theta}$ and differentiating, we obtain

$$\frac{dr}{d\theta} = -\frac{2a\sin\theta}{(1-\cos\theta)^2} = -r\cot\frac{\theta}{2}.$$

Therefore,

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{r^2\cot^2(\theta/2)}{r^4} = \frac{\csc^2(\theta/2)}{r^2}.$$

Now $1-\cos\theta = 2\sin^2\dfrac{\theta}{2}$, while $\dfrac{2a}{r} = 1-\cos\theta$. Hence $\sin^2\dfrac{\theta}{2} = \dfrac{a}{r}$ and $\csc^2\dfrac{\theta}{2} = \dfrac{r}{a}$. It follows that

$$\frac{1}{p^2} = \frac{1}{r^2}\frac{r}{a} = \frac{1}{ar}.$$

Thus, the required pedal equation is $p^2 = ar$. ■

3.3 Lengths of Arcs

Definition 3.3.1 (Arc Length)**.** The **length of an arc** of a curve is the distance measured along the curve between two points. It is found by integrating a differential element of length, ds.

3.3.1 Formulas for Arc Length

The differential arc length ds and the total arc length S from a point A to a point B are given by:

1. **Cartesian Form:** For $y = f(x)$ from $x = a$ to $x = b$.

$$ds = \sqrt{1 + (y')^2}\, dx \qquad \Longrightarrow \qquad S = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

2. **Parametric Form:** For $x = x(t), y = y(t)$ from $t = t_1$ to $t = t_2$.

$$ds = \sqrt{(\dot{x})^2 + (\dot{y})^2}\, dt \qquad \Longrightarrow \qquad S = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

3. **Polar Form:** For $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$.

$$ds = \sqrt{r^2 + (r')^2}\, d\theta \qquad \Longrightarrow \qquad S = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

Example 3.3.2 (See Figures 17 and 30 in the Atlas)**.** Find the total length of the cardioid $r = a(1 + \cos\theta)$.

Solution. The curve is symmetric about the initial line, so we can find the length from $\theta = 0$ to $\theta = \pi$ and double it. First, find the derivative: $\frac{dr}{d\theta} = -a\sin\theta$. Now, set up the expression under the square root:

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= a^2(1 + \cos\theta)^2 + (-a\sin\theta)^2 \\ &= a^2(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta) \\ &= a^2(2 + 2\cos\theta) = 2a^2(1 + \cos\theta) \\ &= 2a^2(2\cos^2(\theta/2)) = 4a^2\cos^2(\theta/2) \end{aligned}$$

The integrand is $\sqrt{4a^2\cos^2(\theta/2)} = 2a|\cos(\theta/2)|$. For $\theta \in [0, \pi]$, $\cos(\theta/2) \geq 0$. The total length S is:

$$\begin{aligned} S &= 2\int_0^\pi 2a\cos(\theta/2)\, d\theta = 4a\left[\frac{\sin(\theta/2)}{1/2}\right]_0^\pi \\ &= 8a[\sin(\theta/2)]_0^\pi = 8a(\sin(\pi/2) - \sin(0)) = 8a(1 - 0) = 8a \end{aligned}$$

The total length of the cardioid is $8a$. ■

Example 3.3.3 (See Figure 31 in the Atlas)**.** Find the length of one arch of the cycloid given by $x(t) = a(t - \sin t)$ and $y(t) = a(1 - \cos t)$, where one arch corresponds to $t \in [0, 2\pi]$.

Solution. First, find the derivatives with respect to t:

$$\frac{dx}{dt} = a(1 - \cos t) \quad \text{and} \quad \frac{dy}{dt} = a\sin t$$

Next, set up the expression under the square root:

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= a^2(1-\cos t)^2 + (a\sin t)^2 \\ &= a^2(1 - 2\cos t + \cos^2 t + \sin^2 t) \\ &= a^2(2 - 2\cos t) = 2a^2(1-\cos t) \\ &= 2a^2(2\sin^2(t/2)) = 4a^2\sin^2(t/2)\end{aligned}$$

The integrand becomes $\sqrt{4a^2\sin^2(t/2)} = 2a|\sin(t/2)|$. For $t \in [0, 2\pi]$, $\sin(t/2) \geq 0$. The arc length is:

$$\begin{aligned}S &= \int_0^{2\pi} 2a\sin(t/2)\,dt = 2a\left[\frac{-\cos(t/2)}{1/2}\right]_0^{2\pi} \\ &= -4a[\cos(t/2)]_0^{2\pi} \\ &= -4a(\cos(\pi) - \cos(0)) = -4a(-1-1) = 8a\end{aligned}$$

The length of one arch of the cycloid is $8a$. ■

Example 3.3.4. Find the total length of the astroid given by $x(t) = a\cos^3 t$ and $y(t) = a\sin^3 t$.

Solution. The curve is symmetric. We will find the length of the arc in the first quadrant (from $t = 0$ to $t = \pi/2$) and multiply by 4. First, find the derivatives:

$$\frac{dx}{dt} = -3a\cos^2 t\sin t \quad \text{and} \quad \frac{dy}{dt} = 3a\sin^2 t\cos t$$

Next, set up the expression under the square root:

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= (-3a\cos^2 t\sin t)^2 + (3a\sin^2 t\cos t)^2 \\ &= 9a^2\cos^4 t\sin^2 t + 9a^2\sin^4 t\cos^2 t \\ &= 9a^2\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t) \\ &= 9a^2\cos^2 t\sin^2 t\end{aligned}$$

The integrand is $\sqrt{9a^2\cos^2 t\sin^2 t} = 3a|\cos t\sin t|$. For $t \in [0, \pi/2]$, this is positive. The total length S is:

$$\begin{aligned}S &= 4\int_0^{\pi/2} 3a\cos t\sin t\,dt = 12a\int_0^{\pi/2}\sin t\cos t\,dt \\ &= 12a\left[\frac{\sin^2 t}{2}\right]_0^{\pi/2} \\ &= 6a[\sin^2 t]_0^{\pi/2} = 6a(\sin^2(\pi/2) - \sin^2(0)) = 6a(1-0) = 6a\end{aligned}$$

The total length of the astroid is $6a$. ■

Example 3.3.5 (See Figures 25 and 30 in the Atlas)**.** Find the length of the spiral $r = e^{a\theta}$ from $\theta = 0$ to $\theta = 2\pi$.

Solution. First, find the derivative with respect to θ:

$$\frac{dr}{d\theta} = ae^{a\theta}$$

Next, set up the expression under the square root:

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= (e^{a\theta})^2 + (ae^{a\theta})^2 \\ &= e^{2a\theta} + a^2 e^{2a\theta} \\ &= e^{2a\theta}(1 + a^2) \end{aligned}$$

The integrand becomes $\sqrt{e^{2a\theta}(1+a^2)} = e^{a\theta}\sqrt{1+a^2}$. The arc length is:

$$\begin{aligned} S &= \int_0^{2\pi} e^{a\theta}\sqrt{1+a^2}\, d\theta = \sqrt{1+a^2}\int_0^{2\pi} e^{a\theta}\, d\theta \\ &= \sqrt{1+a^2}\left[\frac{e^{a\theta}}{a}\right]_0^{2\pi} \\ &= \frac{\sqrt{1+a^2}}{a}[e^{a\theta}]_0^{2\pi} = \frac{\sqrt{1+a^2}}{a}(e^{2a\pi} - e^0) \end{aligned}$$

The arc length is $\dfrac{\sqrt{1+a^2}}{a}(e^{2a\pi} - 1)$. ■

Example 3.3.6 (See Figures 15 and 30 in the Atlas)**.** Find the length of the circumference of the circle $r = 2a\cos\theta$.

Solution. The circle is traced once as θ goes from $-\pi/2$ to $\pi/2$. First, find the derivative:

$$\frac{dr}{d\theta} = -2a\sin\theta$$

Next, set up the expression under the square root:

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= (2a\cos\theta)^2 + (-2a\sin\theta)^2 \\ &= 4a^2\cos^2\theta + 4a^2\sin^2\theta \\ &= 4a^2(\cos^2\theta + \sin^2\theta) = 4a^2 \end{aligned}$$

The integrand is $\sqrt{4a^2} = 2a$. The arc length is:

$$\begin{aligned} S &= \int_{-\pi/2}^{\pi/2} 2a\, d\theta = 2a[\theta]_{-\pi/2}^{\pi/2} \\ &= 2a\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right) = 2a(\pi) = 2\pi a \end{aligned}$$

This confirms the circumference of a circle with radius a is $2\pi a$. ■

3.4 Asymptotes

Definition 3.4.1 (Asymptote)**.** An **asymptote** of a curve $y = f(x)$ is a straight line such that the perpendicular distance from a point P on the curve to the line approaches zero as P moves along the curve to infinity.

Asymptotes are classified into three geometric categories:

1. **Vertical Asymptotes:** The line $x = c$ is a vertical asymptote if:

$$\lim_{x\to c^-} |f(x)| = \infty \quad \text{or} \quad \lim_{x\to c^+} |f(x)| = \infty.$$

2. **Horizontal Asymptotes:** The line $y = L$ is a horizontal asymptote if:

$$\lim_{x \to +\infty} f(x) = L \quad \text{or} \quad \lim_{x \to -\infty} f(x) = L.$$

3. **Oblique (or Slant) Asymptotes:** The line $y = mx + c$ ($m \neq 0$) is an oblique asymptote if:

$$m = \lim_{x \to \pm\infty} \frac{f(x)}{x} \quad \text{and} \quad c = \lim_{x \to \pm\infty} [f(x) - mx].$$

3.4.1 Working Rule for Algebraic Curves

For a general algebraic curve $F(x, y) = 0$ of degree n, asymptotes are determined as follows:

1. Asymptotes Parallel to the Coordinate Axes

- **Parallel to y-axis (Vertical):** Equate the coefficient of the highest power of y in $F(x, y) = 0$ to zero. If this yields real linear factors $x = c_i$, these lines are vertical asymptotes.
- **Parallel to x-axis (Horizontal):** Equate the coefficient of the highest power of x in $F(x, y) = 0$ to zero. If this yields real linear factors $y = k_i$, these lines are horizontal asymptotes.

2. Oblique Asymptotes ($y = mx + c$)

Group terms of the algebraic curve by degree: $F_n(x, y) + F_{n-1}(x, y) + F_{n-2}(x, y) + \cdots = 0$. Substitute $x = 1$ and $y = m$ into these groups to obtain the functions $\phi_n(m), \phi_{n-1}(m), \phi_{n-2}(m), \ldots$

Solve the characteristic equation $\phi_n(m) = 0$ for m:

- **Case I: Non-repeated real root m:** The corresponding c value is given by:

$$c = -\frac{\phi_{n-1}(m)}{\phi_n'(m)} \quad \left(\text{provided } \phi_n'(m) \neq 0\right).$$

- **Case II: Repeated real roots ($m_1 = m_2 = m$):** Here $\phi_n'(m) = 0$. The two values of c are determined by solving the quadratic equation:

$$\frac{c^2}{2!}\phi_n''(m) + c\phi_{n-1}'(m) + \phi_{n-2}(m) = 0.$$

Example 3.4.2 (Folium of Descartes; See Figure 34 in the Atlas)**.** Find all asymptotes of the curve $x^3 + y^3 = 3axy$.

Solution. The given curve equation is $x^3 + y^3 - 3axy = 0$, which is of degree $n = 3$.

Asymptotes Parallel to Axes: The coefficient of the highest power of x (x^3) is 1 (a non-zero constant), so there are no asymptotes parallel to the x-axis. Similarly, the coefficient of y^3 is 1, so there are no asymptotes parallel to the y-axis.

Oblique Asymptotes ($y = mx + c$): Obtain $\phi_3(m)$ and $\phi_2(m)$ by setting $x = 1$ and $y = m$:

$$\begin{aligned} \phi_3(m) &= 1 + m^3 \quad (\text{from degree 3 terms } x^3 + y^3), \\ \phi_2(m) &= -3am \quad (\text{from degree 2 term } -3axy). \end{aligned}$$

Solve $\phi_3(m) = 0$:

$$m^3 + 1 = 0 \implies (m + 1)(m^2 - m + 1) = 0 \implies m = -1 \text{ (only real root)}.$$

Compute the derivative $\phi_3'(m) = 3m^2$. For $m = -1$:

$$c = -\frac{\phi_2(-1)}{\phi_3'(-1)} = -\frac{-3a(-1)}{3(-1)^2} = -\frac{3a}{3} = -a.$$

Substituting $m = -1$ and $c = -a$ into $y = mx + c$ yields the oblique asymptote $y = -x - a$, or:

$$x + y + a = 0.$$

■

Example 3.4.3 (Algebraic Curve with Axes Parallel and Oblique Asymptotes; See Figure 33 in the Atlas)**.** Find all the asymptotes of the curve $x^2y - xy^2 + xy + y^2 + x - y = 0$.

Solution. The degree of the given algebraic curve is $n = 3$.

Asymptotes Parallel to y-axis: The highest power of y present in the equation is y^2. Grouping its terms: $y^2(1 - x) + xy + x^2y + x - y = 0$. Equating the coefficient of y^2 to zero:

$$1 - x = 0 \implies x = 1.$$

Thus, $x = 1$ is a vertical asymptote.

Asymptotes Parallel to x-axis: The highest power of x present in the equation is x^2. Grouping its terms: $x^2(y) - xy^2 + xy + y^2 + x - y = 0$. Equating the coefficient of x^2 to zero:

$$y = 0.$$

Thus, $y = 0$ is a horizontal asymptote.

Oblique Asymptotes ($y = mx + c$): Substitute $x = 1$ and $y = m$ into the degree 3 and degree 2 terms:

$$\begin{aligned} \phi_3(m) &= m - m^2 \quad (\text{from } x^2y - xy^2), \\ \phi_2(m) &= m + m^2 \quad (\text{from } xy + y^2). \end{aligned}$$

Solve $\phi_3(m) = 0$:

$$m(1 - m) = 0 \implies m = 0 \quad \text{or} \quad m = 1.$$

For $m = 0$, we have $\phi_3'(m) = 1 - 2m \implies \phi_3'(0) = 1$. The intercept c is:

$$c = -\frac{\phi_2(0)}{\phi_3'(0)} = -\frac{0}{1} = 0,$$

which yields $y = 0$ (confirming the horizontal asymptote found above).

For $m = 1$, we have $\phi_3'(1) = 1 - 2(1) = -1$, and $\phi_2(1) = 1 + 1^2 = 2$. The intercept c is:

$$c = -\frac{\phi_2(1)}{\phi_3'(1)} = -\frac{2}{-1} = 2.$$

Substituting $m = 1$ and $c = 2$ gives the oblique asymptote $y = x + 2$.

Conclusion: The curve possesses three asymptotes: $x = 1$, $y = 0$, and $y = x + 2$. ■

Example 3.4.4 (Repeated Roots for Oblique Asymptotes)**.** Find all asymptotes of the curve $x^3 + x^2y - xy^2 - y^3 + 2x^2 - 2y^2 + x - 1 = 0$.

Solution. The degree of the given equation is $n = 3$.

Asymptotes Parallel to Axes: The coefficients of x^3 and y^3 are both non-zero constants (1 and -1), so there are no asymptotes parallel to the coordinate axes.

Oblique Asymptotes ($y = mx + c$): Obtain $\phi_3(m), \phi_2(m)$, and $\phi_1(m)$ by setting $x = 1$ and $y = m$:

$$\begin{aligned} &\phi_3(m) = 1 + m - m^2 - m^3 = (1 - m)(1 + m)^2, \\ &\phi_2(m) = 2 - 2m^2, \\ &\phi_1(m) = 1. \end{aligned}$$

Solve $\phi_3(m) = 0$:

$$(1-m)(1+m)^2 = 0 \implies m = 1 \quad \text{and} \quad m = -1 \text{ (repeated root)}.$$

For non-repeated root $m = 1$: Differentiating $\phi_3(m)$: $\phi_3'(m) = 1 - 2m - 3m^2 \implies \phi_3'(1) = -4$.

$$c = -\frac{\phi_2(1)}{\phi_3'(1)} = -\frac{2 - 2(1)^2}{-4} = 0.$$

This yields the first oblique asymptote: $y = x$.

For repeated root $m = -1$ (double root): Since $\phi_3'(-1) = 0$, we use the quadratic equation for c:

$$\frac{c^2}{2!}\phi_3''(-1) + c\phi_2'(-1) + \phi_1(-1) = 0.$$

Compute the required derivatives:

$$\begin{aligned}
\phi_3''(m) &= -2 - 6m \implies \phi_3''(-1) = 4, \\
\phi_2'(m) &= -4m \implies \phi_2'(-1) = 4, \\
\phi_1(-1) &= 1.
\end{aligned}$$

Substituting these derivatives into the quadratic equation:

$$\frac{c^2}{2}(4) + c(4) + 1 = 0 \implies 2c^2 + 4c + 1 = 0.$$

Solving for c using the quadratic formula:

$$c = \frac{-4 \pm \sqrt{16 - 8}}{4} = \frac{-4 \pm 2\sqrt{2}}{4} = -1 \pm \frac{\sqrt{2}}{2}.$$

This yields two parallel oblique asymptotes:

$$y = -x - 1 + \frac{\sqrt{2}}{2} \quad \text{and} \quad y = -x - 1 - \frac{\sqrt{2}}{2}.$$

Conclusion: The three asymptotes of the curve are $y = x$, $y = -x - 1 + \frac{\sqrt{2}}{2}$, and $y = -x - 1 - \frac{\sqrt{2}}{2}$. ■

3.5 Singular Points

Definition 3.5.1 (Singular Point)**.** A **singular point** on an algebraic curve $f(x, y) = 0$ is a point where the curve behaves in an extraordinary way. Mathematically, it is a point (x_0, y_0) on the curve where both partial derivatives are simultaneously zero:

$$f(x_0, y_0) = 0, \quad \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = 0, \quad \text{and} \quad \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = 0$$

At a singular point, the tangent to the curve is not uniquely defined.

3.5.1 Classification of Singular Points (Double Points)

A common type of singular point is a **double point**, where two branches of the curve pass. We classify double points by examining the quantity:

$$\Delta = \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 - \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right)$$

evaluated at the singular point.

1. If $\Delta > 0$, the point is a **node**. The curve has two distinct real tangents at this point.
2. If $\Delta = 0$, the point is a **cusp**. The curve has two coincident real tangents.
3. If $\Delta < 0$, the point is a **conjugate point** or isolated point. The tangents are imaginary.

Example 3.5.2. Find and classify the singular points on the curve $y^2 = x^2(x+1)$.

Solution. Let $f(x,y) = y^2 - x^3 - x^2 = 0$. Find partial derivatives:

$$\frac{\partial f}{\partial x} = -3x^2 - 2x \qquad \frac{\partial f}{\partial y} = 2y$$

Set them to zero: $2y = 0 \implies y = 0$. And $-3x^2 - 2x = -x(3x+2) = 0 \implies x = 0$ or $x = -2/3$. We check which of these potential points, $(0,0)$ and $(-2/3, 0)$, are on the curve.

- For $(0,0)$: $0^2 - 0^3 - 0^2 = 0$. This point is on the curve. It is a singular point.
- For $(-2/3, 0)$: $0^2 - (-2/3)^3 - (-2/3)^2 = 8/27 - 4/9 = -4/27 \neq 0$. Not on the curve.

The only singular point is $(0,0)$. To classify it, find second derivatives at $(0,0)$:

$$\frac{\partial^2 f}{\partial x^2} = -6x - 2 \implies -2 \qquad \frac{\partial^2 f}{\partial y^2} = 2 \qquad \frac{\partial^2 f}{\partial x \partial y} = 0$$

Compute $\Delta = (0)^2 - (-2)(2) = 4$. Since $\Delta > 0$, the point $(0,0)$ is a **node**. ■

Example 3.5.3. Find and classify the singular points on the curve $y^2 = x^3$.

Solution. Let the curve be $f(x,y) = y^2 - x^3 = 0$.

$$\frac{\partial f}{\partial x} = -3x^2 \implies -3x^2 = 0 \implies x = 0$$
$$\frac{\partial f}{\partial y} = 2y \implies 2y = 0 \implies y = 0$$

The only potential singular point is $(0,0)$. This point satisfies the original equation ($0^2 - 0^3 = 0$), so it is a singular point.

The second-order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = -6x \qquad \frac{\partial^2 f}{\partial y^2} = 2 \qquad \frac{\partial^2 f}{\partial x \partial y} = 0$$

Evaluate these at the singular point $(0,0)$:

$$\left.\frac{\partial^2 f}{\partial x^2}\right|_{(0,0)} = -6(0) = 0$$

$$\left.\frac{\partial^2 f}{\partial y^2}\right|_{(0,0)} = 2$$

$$\left.\frac{\partial^2 f}{\partial x \partial y}\right|_{(0,0)} = 0$$

Therefore,

$$\Delta = (0)^2 - (0)(2) = 0$$

Since $\Delta = 0$, the singular point $(0, 0)$ is a **cusp**. ■

3.6 Maxima and Minima of Functions

Definition 3.6.1 (Local Extrema)**.** A function $f(x)$ has a **local maximum** at $x = c$ if $f(c) \geq f(x)$ for all x in some open interval containing c. A function $f(x)$ has a **local minimum** at $x = c$ if $f(c) \leq f(x)$ for all x in some open interval containing c.

3.6.1 Finding Local Extrema

1. **Find Critical Points:** These are points where the first derivative is zero or undefined, i.e., $f'(x) = 0$ or $f'(x)$ does not exist.
2. **First Derivative Test:**
   - If $f'(x)$ changes from positive to negative at a critical point c, then f has a local maximum at c.
   - If $f'(x)$ changes from negative to positive at a critical point c, then f has a local minimum at c.
   - If $f'(x)$ does not change sign, there is no local extremum at c.
3. **Second Derivative Test:** Let c be a critical point such that $f'(c) = 0$.
   - If $f''(c) < 0$, then f has a local maximum at c.
   - If $f''(c) > 0$, then f has a local minimum at c.
   - If $f''(c) = 0$, the test is inconclusive. Use the First Derivative Test.

Example 3.6.2 (See Figure 38 in the Atlas)**.** Find the local maxima and minima for the function $f(x) = x^3 - 6x^2 + 5$.

Solution.

$$f'(x) = 3x^2 - 12x$$

Set $f'(x) = 0 \implies 3x(x - 4) = 0$. The critical points are $x = 0$ and $x = 4$.

We use the Second Derivative Test.

$$f''(x) = 6x - 12$$

- At $x = 0$: $f''(0) = 6(0) - 12 = -12 < 0$. So there is a **local maximum** at $x = 0$. The maximum value is $f(0) = 0^3 - 6(0)^2 + 5 = 5$.
- At $x = 4$: $f''(4) = 6(4) - 12 = 12 > 0$. So there is a **local minimum** at $x = 4$. The minimum value is $f(4) = 4^3 - 6(4)^2 + 5 = 64 - 96 + 5 = -27$.

■

Example 3.6.3. Find the local extrema of $f(x) = x + \frac{1}{x}$.

Solution.

$$f'(x) = 1 - \frac{1}{x^2}$$

Set $f'(x) = 0 \implies 1 - \frac{1}{x^2} = 0 \implies x^2 = 1 \implies x = \pm 1$. The derivative is undefined at $x = 0$, but $x = 0$ is not in the domain of $f(x)$. The critical points are $x = 1$ and $x = -1$.

We use the Second Derivative Test.

$$f''(x) = \frac{2}{x^3}$$

- At $x = 1$: $f''(1) = \frac{2}{1^3} = 2 > 0$. So there is a **local minimum** at $x = 1$. The minimum value is $f(1) = 1 + \frac{1}{1} = 2$.
- At $x = -1$: $f''(-1) = \frac{2}{(-1)^3} = -2 < 0$. So there is a **local maximum** at $x = -1$. The maximum value is $f(-1) = -1 + \frac{1}{-1} = -2$.

■

3.7 Bounded Functions

Definition 3.7.1 (Bounded Function)**.** A function f is said to be **bounded** on an interval I if there exists a real number $M > 0$ such that $|f(x)| \leq M$ for all $x \in I$. This means the function's range is contained within the finite interval $[-M, M]$.

- f is **bounded above** if there is a number K such that $f(x) \leq K$ for all $x \in I$.
- f is **bounded below** if there is a number k such that $f(x) \geq k$ for all $x \in I$.

A function is bounded if and only if it is bounded both above and below.

Example 3.7.2 (Bounded Function; See Figure 39 in the Atlas)**.** The function $f(x) = \sin(x)$ is bounded on $\mathbb{R}$ because for all x, we have $-1 \leq \sin(x) \leq 1$. We can choose $M = 1$, since $|\sin(x)| \leq 1$.

Example 3.7.3 (Unbounded Function)**.** The function $f(x) = \frac{1}{x}$ is unbounded on the interval $(0, 1]$. As x approaches 0 from the right, $f(x)$ approaches ∞, so there is no upper bound M that can contain all function values.

3.8 Boundedness Theorem

Continuous functions defined on a closed and bounded interval $[a, b]$ exhibit special and powerful properties that are not guaranteed on open intervals.

Theorem 3.8.1 (The Boundedness Theorem)**.** *If a function f is continuous on a closed interval $[a, b]$, then f is bounded on $[a, b]$.*

Proof. Suppose, to the contrary, that f is unbounded on $[a, b]$. Bisect $[a, b]$. At least one of the two closed halves must be an interval on which f is unbounded; denote such a half by I_1. Bisect I_1 and choose a closed half $I_2 \subset I_1$ on which f is unbounded. Continuing in this way, we obtain nested closed intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots$$

on each of which f is unbounded, while the length of I_n is $(b-a)/2^n$.

By the nested interval property, there is a unique point $c \in [a, b]$ belonging to every I_n. Since f is continuous at c, there exists $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < 1$. Consequently, $|f(x)| \le |f(c)| + 1$ whenever $|x - c| < \delta$.

For sufficiently large n, the interval I_n is contained in $(c - \delta, c + \delta)$. Thus f is bounded on I_n, contradicting the choice of I_n. Hence f is bounded on $[a, b]$. ■

2nd Proof. We prove by contradiction. Assume f is not bounded on $[a, b]$. This means for any natural number n, there exists a point $x_n \in [a, b]$ such that $|f(x_n)| > n$.

Thus $\{x_n\}$ is a sequence in the bounded interval $[a, b]$. By the Bolzano–Weierstrass theorem, it has a convergent subsequence $\{x_{n_k}\}$ with $x_{n_k} \to c$ for some $c \in [a, b]$.

Continuity at c gives $f(x_{n_k}) \to f(c)$, so the sequence $\{f(x_{n_k})\}$ is bounded.

However, by our initial construction, $|f(x_{n_k})| > n_k$. As $k \to \infty$, $n_k \to \infty$, which means the sequence $\{f(x_{n_k})\}$ is unbounded.

This contradiction proves that f is bounded on $[a, b]$. ■

Alternative Proof (Advanced). For each $x \in [a, b]$, by continuity, there exists an open interval U_x containing x such that f is bounded on $U_x \cap [a, b]$. The collection $\{U_x\}_{x \in [a,b]}$ forms an open cover of $[a, b]$. Since $[a, b]$ is compact, there exists a finite subcover $\{U_{x_1}, \ldots, U_{x_k}\}$. Since f is bounded on each $U_{x_i} \cap [a, b]$, and $[a, b]$ is covered by the finite union of these sets, f must be bounded on $[a, b]$. ■

Example 3.8.2 (See Figure 39 in the Atlas). $f(x) = x^2$ is continuous on $[-2, 3]$. The theorem guarantees it is bounded. Indeed, its values range from $f(0) = 0$ to $f(3) = 9$, so it is bounded by $M = 9$.

3.9 Extreme Value Theorem

Theorem 3.9.1 (The Extreme Value Theorem (EVT))**.** *If a function f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value and an absolute minimum value on $[a, b]$. This means there exist numbers $c, d \in [a, b]$ such that $f(c) \le f(x) \le f(d)$ for all $x \in [a, b]$.*

Proof. It is enough to prove the existence of an absolute maximum; applying the same argument to $-f$ gives the minimum.

By the Boundedness Theorem, the set of values $S = \{f(x) : x \in [a, b]\}$ is bounded above. Therefore, by the completeness axiom of real numbers, S has a least upper bound (supremum). Let $M = \sup S$.

We must show that there exists a point $d \in [a, b]$ such that $f(d) = M$. By the definition of a supremum, for any natural number n, there exists an element $y_n \in S$ such that $M - \dfrac{1}{n} < y_n \le M$. For each y_n, there is a corresponding $x_n \in [a, b]$ such that $f(x_n) = y_n$. Thus we obtain a sequence $\{x_n\} \subset [a, b]$.

By the Bolzano-Weierstrass theorem, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ that converges to some point $d \in [a, b]$.

Since f is continuous at d, we have $\lim_{k\to\infty} f(x_{n_k}) = f(d)$. Also, from our construction, we have $M - \frac{1}{n_k} < f(x_{n_k}) \leq M$. As $k \to \infty$, $\frac{1}{n_k} \to 0$. By the Squeeze Theorem, $\lim_{k\to\infty} f(x_{n_k}) = M$.

The two limits are equal, so $f(d) = M$. Hence f attains its absolute maximum at d. ■

2nd Proof. By the previous theorem, f is bounded on $[a, b]$. Let $M = \sup_{x\in[a,b]} f(x)$. Assume, for contradiction, that $f(x) < M$ for all $x \in [a, b]$. Then $M - f(x) > 0$ for all $x \in [a, b]$. Consider the function $g(x) = \frac{1}{M - f(x)}$. Since f is continuous and $M - f(x) \neq 0$, $g(x)$ is continuous on $[a, b]$. By the previous theorem, $g(x)$ must be bounded on $[a, b]$. So there exists $K > 0$ such that $g(x) \leq K$. This implies $\frac{1}{M - f(x)} \leq K$, so $M - f(x) \geq \frac{1}{K}$. Therefore, $f(x) \leq M - \frac{1}{K}$ for all $x \in [a, b]$. But $M - \frac{1}{K} < M$, which contradicts M being the supremum of f. Hence, there must exist $c \in [a, b]$ such that $f(c) = M$. The proof for the minimum value is analogous by considering $-f(x)$. ■

Example 3.9.2 (See Figure 40 in the Atlas)**.** $f(x) = x^3 - 3x$ is continuous on $[0, 2]$. The theorem guarantees it attains its absolute max and min. By checking critical points and endpoints, we find the absolute minimum is $f(1) = -2$ and the absolute maximum is $f(2) = 2$.

3.10 The Intermediate Value Theorem (IVT)

The Intermediate Value Theorem expresses the fundamental fact that a continuous function assumes every value lying between two of its values.

Theorem 3.10.1 (Intermediate Value Theorem)**.** *If a function f is continuous on the closed interval $[a, b]$, and N is any number between $f(a)$ and $f(b)$ (where $f(a) \neq f(b)$), then there exists at least one number c in the open interval (a, b) such that $f(c) = N$.*

Proof. Assume without loss of generality that $f(a) < N < f(b)$. Consider the set $S = \{x \in [a, b] : f(x) \leq N\}$.

The set S is non-empty since $a \in S$. The set S is also bounded above by b. By the completeness axiom, S has a least upper bound (supremum). Let $c = \sup S$. Since $a \in S$ and b is an upper bound, we have $a \leq c \leq b$.

We will show that $f(c) = N$. By the definition of a supremum, for any natural number n, there exists $x_n \in S$ such that $c - \frac{1}{n} < x_n \leq c$. This implies that $\lim_{n\to\infty} x_n = c$. Since f is continuous, $\lim_{n\to\infty} f(x_n) = f(c)$. As each $x_n \in S$, we have $f(x_n) \leq N$. Taking the limit, we get $f(c) \leq N$.

Now, since $f(b) > N$, $c \neq b$, so $c < b$. Consider the sequence $z_n = c + \frac{1}{n}$ for sufficiently large n so that $z_n < b$. None of these z_n can be in S because c is the supremum. Thus, $f(z_n) > N$ for all such n. As $n \to \infty$, $z_n \to c$. By continuity, $\lim_{n\to\infty} f(z_n) = f(c)$. From $f(z_n) > N$, we conclude $f(c) \geq N$.

Combining the two inequalities, $f(c) \leq N$ and $f(c) \geq N$, we must have $f(c) = N$. Since $f(a) < N$ and $f(b) > N$, we know c cannot be a or b, so $c \in (a, b)$. ■

Example 3.10.2. Show that the equation $x^3 + 2x - 1 = 0$ has a root in the interval $[0, 1]$.

Solution. Let $f(x) = x^3 + 2x - 1$. The function f is a polynomial, so it is continuous everywhere, including on $[0, 1]$. We evaluate the function at the endpoints:

- $f(0) = 0^3 + 2(0) - 1 = -1$.

- $f(1) = 1^3 + 2(1) - 1 = 2$.

Let $N = 0$. Since $f(0) < 0 < f(1)$, the IVT guarantees that there must be a number $c \in (0, 1)$ such that $f(c) = 0$. This c is a root of the equation. ■

3.11 Darboux's Theorem

Darboux's Theorem shows that derivatives possess the intermediate value property even when they are not continuous.

Theorem 3.11.1 (Darboux's Theorem). ***Statement:*** *If a function f is differentiable on the closed interval $[a, b]$, and k is any number between $f'(a)$ and $f'(b)$, then there exists at least one number c in the open interval (a, b) such that $f'(c) = k$.*

Proof. Assume without loss of generality that $f'(a) < k < f'(b)$. Define a new function $g(x) = f(x) - kx$. This function is differentiable on $[a, b]$ because $f(x)$ and kx are.

The derivative of $g(x)$ is $g'(x) = f'(x) - k$. At the endpoints, we have:

- $g'(a) = f'(a) - k < 0$.
- $g'(b) = f'(b) - k > 0$.

Since $g(x)$ is differentiable on the closed interval $[a, b]$, it must also be continuous on $[a, b]$. By the Extreme Value Theorem, $g(x)$ must attain an absolute minimum value at some point $c \in [a, b]$.

Because $g'(a) < 0$, the function is decreasing at $x = a$, so the minimum cannot occur at a. Because $g'(b) > 0$, the function is increasing at $x = b$, so the minimum cannot occur at b.

Therefore, the minimum must occur at an interior point $c \in (a, b)$. At an interior minimum, the derivative must be zero. So, $g'(c) = 0$. By definition of $g'(x)$, this means:

$$f'(c) - k = 0 \implies f'(c) = k$$

Thus, we have found a point $c \in (a, b)$ where the derivative equals k. ■

Example 3.11.2 (See Figure 42 in the Atlas). Consider the function $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Its derivative is $f'(x) = 2x \sin(1/x) - \cos(1/x)$ for $x \neq 0$, and $f'(0) = 0$. The derivative $f'(x)$ is not continuous at $x = 0$. Let's apply Darboux's Theorem on $[0, 2/\pi]$.

Solution. The function f is differentiable on $[0, 2/\pi]$. We evaluate the derivative at the endpoints:

- $f'(0) = 0$.
- $f'(2/\pi) = 2(2/\pi) \sin(\pi/2) - \cos(\pi/2) = 4/\pi \approx 1.27$.

Let's choose an intermediate value $k = 1/2$. Since $0 < 1/2 < 4/\pi$, Darboux's Theorem guarantees that there exists a $c \in (0, 2/\pi)$ such that $f'(c) = 1/2$, even though $f'(x)$ oscillates wildly and is not continuous at the endpoint $x = 0$. ■

Exercise 3.11.3. Solve the following exercises:

1. Find the radius of curvature ρ for the folium of Descartes $x^3 + y^3 = 3axy$ at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.

2. Show that for the astroid $x = a\cos^3 t,\ y = a\sin^3 t$, the radius of curvature at any parameter t is given by:

$$\rho = \frac{3a}{2} \sin 2t.$$

3. Find the radius of curvature ρ for the polar curve $r^n = a^n \cos n\theta$, and show that:

$$\rho = \frac{a^n}{(n+1)r^{n-1}}.$$

4. Show that for the intrinsic equation of a catenary $s = c\tan\psi$, the radius of curvature is $\rho = c\sec^2\psi$.

5. Find all the asymptotes of the algebraic curve:

$$x^3 + 2x^2y - xy^2 - 2y^3 + 4x^2 - y^2 + 1 = 0.$$

6. Find all the asymptotes (parallel to axes and oblique) of the curve:

$$y^3 - x^2y + 2y^2 + 4x + 1 = 0.$$

7. Find the oblique asymptotes of the curve $x^3 - 3x^2y + 3xy^2 - y^3 + 2x^2 - 2xy + 1 = 0$, taking into account the presence of repeated real roots for m.

8. Determine the polar asymptotes of the curve $r\cos\theta = a\cos 2\theta$.

9. Determine the nature of the singular point at the origin for the curve $y^2 = x^2(x+1)$ (distinguish whether it is a node, cusp, or conjugate point).

10. Calculate the total perimeter (arc length) of the cardioid $r = a(1+\cos\theta)$.

11. Find the length of the arc of the parabola $y^2 = 4ax$ cut off by its latus rectum ($x = a$).

12. Find the local maximum and local minimum values of the two-variable function:

$$f(x,y) = x^3 + y^3 - 3axy \quad (a > 0).$$

13. Using the Intermediate Value Theorem, prove that the polynomial equation $x^5 - 3x - 1 = 0$ has at least one real root in the open interval $(1,2)$.

14. State Darboux's Theorem (Intermediate Value Property for Derivatives), and apply it to show that if $f(x) = x^2\sin(1/x)$ for $x \neq 0$ and $f(0) = 0$, $f'(x)$ assumes every value between -1 and 1 on any neighborhood of $x = 0$.

15. State the Extreme Value Theorem, and verify that the continuous function $f(x) = \dfrac{x}{x^2+1}$ attains its absolute maximum and absolute minimum on the closed interval $[-2,2]$.

Chapter 4

Mean Value Theorems and their Applications

Unit IV explores fundamental mean value theorems, infinite series representations, and the geometry of curve envelopes. It establishes Rolle's Theorem, Lagrange's Mean Value Theorem, and Cauchy's Mean Value Theorem alongside their geometrical interpretations and rigorous applications. Furthermore, it covers Taylor's and Maclaurin's series expansions with Lagrange and Cauchy forms of remainders for elementary functions, concluding with the theory of envelopes for one-parameter and two-parameter families of curves.

4.1 Rolle's Theorem

Theorem 4.1.1 (Rolle's Theorem)**.** *Let a function f be:*

1. *continuous on the closed interval $[a, b]$,*
2. *differentiable on the open interval (a, b), and*
3. *such that $f(a) = f(b)$.*

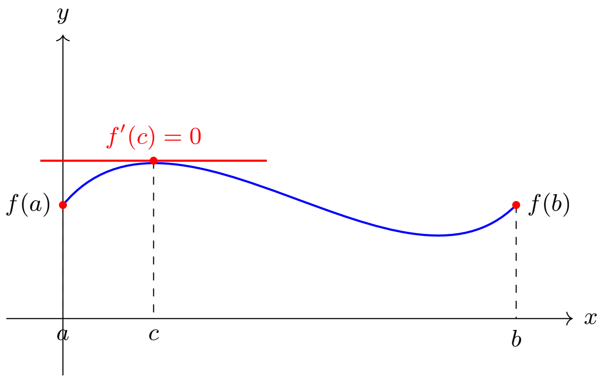
Then there exists at least one number c in the open interval (a, b) such that $f'(c) = 0$.

Proof. By the Extreme Value Theorem, f attains both an absolute maximum and an absolute minimum on $[a, b]$.

If these two values are equal, then f is constant on $[a, b]$, and hence $f'(c) = 0$ for every $c \in (a, b)$.

Otherwise the maximum and minimum are distinct. Because $f(a) = f(b)$, at least one of these extrema must occur at an interior point $c \in (a, b)$. Fermat's theorem for interior extrema then gives $f'(c) = 0$. ■

Interpretation 4.1.2. Geometrically, Rolle's Theorem states that if a smooth curve starts and ends at the same height, there must be at least one point between them where the tangent line is horizontal. See Figure 43 in the Atlas.



See Figure 43 in the Atlas for the geometric interpretation of Rolle's Theorem.

Example 4.1.3 (See Figure 43 in the Atlas)**.** Verify Rolle's Theorem for the function $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

Solution. We verify the hypotheses of Rolle's Theorem.

1. **Continuity:** $f(x)$ is a polynomial, so it is continuous on $[1, 3]$.
2. **Differentiability:** $f'(x) = 2x - 4$, which exists for all $x \in (1, 3)$.
3. **Equal Endpoints:** $f(1) = 1^2 - 4(1) + 3 = 0$. And $f(3) = 3^2 - 4(3) + 3 = 9 - 12 + 3 = 0$. So, $f(1) = f(3)$.

Hence the hypotheses are satisfied, and the theorem guarantees a point $c \in (1, 3)$ where $f'(c) = 0$. To find c, we set the derivative to zero:

$$f'(c) = 2c - 4 = 0 \implies 2c = 4 \implies c = 2$$

Since $c = 2$ is in the open interval $(1, 3)$, Rolle's Theorem is verified. ■

Example 4.1.4 (See Figure 43 in the Atlas)**.** Verify Rolle's Theorem for the polynomial function $f(x) = x^3 - 6x^2 + 11x - 6$ on $[1, 3]$.

Solution. Since $f(x)$ is a polynomial function, it is continuous on $[1, 3]$ and differentiable on $(1, 3)$.

Evaluating $f(x)$ at the endpoints $a = 1$ and $b = 3$:

$$f(1) = 1^3 - 6(1)^2 + 11(1) - 6 = 0,$$

$$f(3) = 3^3 - 6(3)^2 + 11(3) - 6 = 27 - 54 + 33 - 6 = 0.$$

Since $f(1) = f(3) = 0$, all three conditions of Rolle's Theorem are satisfied.

To find $c \in (1, 3)$ such that $f'(c) = 0$, differentiate $f(x)$:

$$f'(x) = 3x^2 - 12x + 11.$$

Set $f'(c) = 0$:

$$3c^2 - 12c + 11 = 0.$$

Solving using the quadratic formula:

$$c = \frac{12 \pm \sqrt{144 - 132}}{6} = \frac{12 \pm \sqrt{12}}{6} = 2 \pm \frac{\sqrt{3}}{3} \approx 2 \pm 0.577.$$

This yields two values: $c_1 = 2 - \frac{1}{\sqrt{3}} \approx 1.423$ and $c_2 = 2 + \frac{1}{\sqrt{3}} \approx 2.577$. Both c_1 and c_2 lie within the open interval $(1, 3)$. Hence, Rolle's Theorem is verified. ■

Example 4.1.5 (See Figure 43 in the Atlas)**.** Verify Rolle's Theorem for $f(x) = e^x(\sin x - \cos x)$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

Solution. The function $f(x)$ is the product of exponential and trigonometric functions, which are continuous and differentiable everywhere. Thus, $f(x)$ is continuous on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ and differentiable on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$.

Evaluating $f(x)$ at the endpoints:

$$f\left(\frac{\pi}{4}\right) = e^{\pi/4}\left(\sin\frac{\pi}{4} - \cos\frac{\pi}{4}\right) = e^{\pi/4}\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) = 0,$$

$$f\left(\frac{5\pi}{4}\right) = e^{5\pi/4}\left(\sin\frac{5\pi}{4} - \cos\frac{5\pi}{4}\right) = e^{5\pi/4}\left(-\frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right)\right) = 0.$$

Since $f(\pi/4) = f(5\pi/4) = 0$, all hypotheses of Rolle's Theorem hold.

Differentiating $f(x)$ using the product rule:

$$f'(x) = e^x(\sin x - \cos x) + e^x(\cos x + \sin x) = 2e^x \sin x.$$

Setting $f'(c) = 0$:

$$2e^c \sin c = 0 \implies \sin c = 0 \quad (\text{since } e^c \neq 0).$$

Solving for c in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ gives $c = \pi$. Since $\pi \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$, Rolle's Theorem is verified. ∎

4.2 Lagrange's Mean Value Theorem (MVT)

This is a generalization of Rolle's Theorem.

Theorem 4.2.1 (Lagrange's Mean Value Theorem)**.** *Let a function f be:*

1. *continuous on the closed interval $[a, b]$, and*
2. *differentiable on the open interval (a, b).*

Then there exists at least one number c in the open interval (a, b) such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. Let

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

The function g is continuous on $[a, b]$ and differentiable on (a, b). Moreover,

$$g(a) = f(a), \qquad g(b) = f(b) - [f(b) - f(a)] = f(a),$$

so $g(a) = g(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. Since

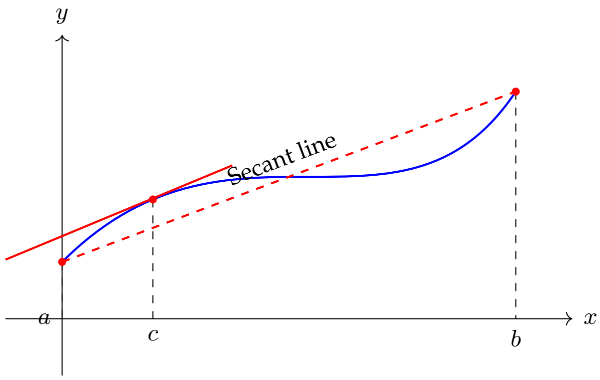
$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

we obtain

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

∎

Interpretation 4.2.2. Geometrically, the MVT states that for any smooth curve, there is at least one point where the instantaneous rate of change (slope of the tangent line) is equal to the average rate of change over the interval (slope of the secant line connecting the endpoints).



See Figure 44 in the Atlas for the geometric interpretation of Lagrange's Mean Value Theorem.

Example 4.2.3 (See Figure 44 in the Atlas)**.** Verify the Mean Value Theorem for the function $f(x) = \sqrt{x}$ on the interval $[1, 4]$.

Solution. We first verify the hypotheses of the Mean Value Theorem.

1. **Continuity:** $f(x) = \sqrt{x}$ is continuous on $[1, 4]$.
2. **Differentiability:** $f'(x) = \dfrac{1}{2\sqrt{x}}$, which exists for all $x \in (1, 4)$.

Hence the hypotheses are satisfied, and the theorem guarantees a point $c \in (1, 4)$ such that $f'(c) = \dfrac{f(4) - f(1)}{4 - 1}$. First, compute the slope of the secant line:

$$\frac{f(4) - f(1)}{4 - 1} = \frac{\sqrt{4} - \sqrt{1}}{3} = \frac{2 - 1}{3} = \frac{1}{3}$$

Equating the derivative to this secant slope gives

$$f'(c) = \frac{1}{2\sqrt{c}} = \frac{1}{3} \implies 2\sqrt{c} = 3 \implies \sqrt{c} = \frac{3}{2} \implies c = \frac{9}{4}$$

Since $c = 9/4 = 2.25$ is in the open interval $(1, 4)$, the Mean Value Theorem is verified. ■

Example 4.2.4 (See Figure 44 in the Atlas)**.** Verify Lagrange's Mean Value Theorem for $f(x) = x^3 - x$ on $[0, 2]$.

Solution. As a polynomial function, $f(x)$ is continuous on $[0, 2]$ and differentiable on $(0, 2)$.
Evaluating $f(x)$ at $a = 0$ and $b = 2$:

$$f(0) = 0^3 - 0 = 0, \quad \text{and} \quad f(2) = 2^3 - 2 = 6.$$

The average rate of change on $[0, 2]$ is:

$$\frac{f(2) - f(0)}{2 - 0} = \frac{6 - 0}{2} = 3.$$

Differentiating $f(x)$ gives $f'(x) = 3x^2 - 1$. Setting $f'(c) = 3$:

$$3c^2 - 1 = 3 \implies 3c^2 = 4 \implies c = \pm\frac{2}{\sqrt{3}}.$$

Selecting the value lying within the open interval $(0, 2)$:

$$c = \frac{2}{\sqrt{3}} \approx 1.155 \in (0, 2).$$

Hence, Lagrange's Mean Value Theorem is verified. ■

Example 4.2.5 (See Figure 44 in the Atlas)**.** Verify Lagrange's Mean Value Theorem for $f(x) = \ln x$ on $[1, e]$.

Solution. The logarithmic function $f(x) = \ln x$ is continuous on $[1, e]$ and differentiable on $(1, e)$.

Evaluating $f(x)$ at $a = 1$ and $b = e$:

$$f(1) = \ln 1 = 0, \quad \text{and} \quad f(e) = \ln e = 1.$$

The average rate of change on $[1, e]$ is:

$$\frac{f(e) - f(1)}{e - 1} = \frac{1 - 0}{e - 1} = \frac{1}{e - 1}.$$

Differentiating $f(x)$ gives $f'(x) = \dfrac{1}{x}$. Setting $f'(c) = \dfrac{1}{e-1}$:

$$\frac{1}{c} = \frac{1}{e - 1} \implies c = e - 1.$$

Since $e \approx 2.718$, we have $c = e - 1 \approx 1.718$, which lies strictly in the interval $(1, e)$. This verifies Lagrange's Mean Value Theorem. ■

4.3 Cauchy's Mean Value Theorem (CMVT)

This is a further generalization of the MVT and is crucial for proving L'Hôpital's Rule.

Theorem 4.3.1 (Cauchy's Mean Value Theorem)**.** *Let two functions f and g be:*

1. *continuous on the closed interval $[a, b]$,*
2. *differentiable on the open interval (a, b), and*
3. *such that $g'(x) \neq 0$ for all $x \in (a, b)$.*

Then there exists at least one number c in the open interval (a, b) such that:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Proof. Note that since $g'(x) \neq 0$ on (a, b), by Rolle's Theorem, we must have $g(b) \neq g(a)$, so the denominator on the right is not zero. Define an auxiliary function $h(x) = f(x) - \left[\dfrac{f(b) - f(a)}{g(b) - g(a)}\right] g(x)$. This function $h(x)$ is continuous on $[a, b]$ and differentiable on (a, b). Let's check the values of $h(x)$ at the endpoints. Let $K = \dfrac{f(b) - f(a)}{g(b) - g(a)}$.

$$\begin{aligned} h(a) &= f(a) - K \cdot g(a) \\ h(b) &= f(b) - K \cdot g(b) \\ h(b) - h(a) &= (f(b) - f(a)) - K(g(b) - g(a)) \\ &= (f(b) - f(a)) - \frac{f(b) - f(a)}{g(b) - g(a)}(g(b) - g(a)) = 0 \end{aligned}$$

So, $h(a) = h(b)$. The function $h(x)$ satisfies all conditions of Rolle's Theorem. Therefore, there exists a $c \in (a, b)$ such that $h'(c) = 0$. The derivative of $h(x)$ is $h'(x) = f'(x) - K \cdot g'(x)$. At $x = c$, we have $h'(c) = f'(c) - K \cdot g'(c) = 0$. This implies $f'(c) = K \cdot g'(c)$, and since $g'(c) \neq 0$, we can write:

$$\frac{f'(c)}{g'(c)} = K = \frac{f(b) - f(a)}{g(b) - g(a)}$$

■

Interpretation 4.3.2. The CMVT relates the ratio of the derivatives of two functions at an interior point to the ratio of the changes in the functions over the whole interval. It can be seen as a parametric version of the MVT. If we consider a curve defined parametrically by $x = g(t)$ and $y = f(t)$, then $\frac{f'(c)}{g'(c)}$ is the slope of the tangent at $t = c$, while $\frac{f(b) - f(a)}{g(b) - g(a)}$ is the slope of the secant line connecting the endpoints of the parametric curve. The theorem guarantees these slopes are equal for some $t = c$.

Example 4.3.3 (See Figure 45 in the Atlas)**.** Verify Cauchy's Mean Value Theorem for the functions $f(x) = x^2$ and $g(x) = x^3$ on the interval $[1, 2]$.

Solution. First, we check the three conditions of the CMVT.

1. **Continuity:** $f(x)$ and $g(x)$ are polynomials, so they are continuous on $[1, 2]$.
2. **Differentiability:** $f'(x) = 2x$ and $g'(x) = 3x^2$ exist for all $x \in (1, 2)$.
3. **Non-zero derivative:** $g'(x) = 3x^2$ is not zero for any $x \in (1, 2)$.

Hence the hypotheses are satisfied, and the theorem guarantees a point $c \in (1, 2)$ such that $\frac{f'(c)}{g'(c)} = \frac{f(2) - f(1)}{g(2) - g(1)}$. First, compute the ratio of the differences:

$$\frac{f(2) - f(1)}{g(2) - g(1)} = \frac{2^2 - 1^2}{2^3 - 1^3} = \frac{4 - 1}{8 - 1} = \frac{3}{7}$$

Next compute the ratio of the derivatives:

$$\frac{f'(c)}{g'(c)} = \frac{2c}{3c^2} = \frac{2}{3c}$$

Now, we set these two expressions equal to find c:

$$\frac{2}{3c} = \frac{3}{7} \implies 9c = 14 \implies c = \frac{14}{9}$$

Since $c = 14/9 \approx 1.556$ is in the open interval $(1, 2)$, Cauchy's Mean Value Theorem is verified.

■

Example 4.3.4 (See Figure 45 in the Atlas)**.** Verify Cauchy's Mean Value Theorem for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$.

Solution. Both $f(x) = x^2$ and $g(x) = x^3$ are polynomials, so they are continuous on $[1, 2]$ and differentiable on $(1, 2)$. Furthermore, $g'(x) = 3x^2 \neq 0$ for all $x \in (1, 2)$.

Evaluating the functions at the endpoints:

$$\begin{aligned} f(1) &= 1, & f(2) &= 4 \implies f(2) - f(1) = 3, \\ g(1) &= 1, & g(2) &= 8 \implies g(2) - g(1) = 7. \end{aligned}$$

The quotient of the increments is:

$$\frac{f(2) - f(1)}{g(2) - g(1)} = \frac{3}{7}.$$

Now compute the derivatives $f'(x) = 2x$ and $g'(x) = 3x^2$. Setting the ratio at $x = c$:

$$\frac{f'(c)}{g'(c)} = \frac{2c}{3c^2} = \frac{2}{3c}.$$

Equating the two expressions:

$$\frac{2}{3c} = \frac{3}{7} \implies 9c = 14 \implies c = \frac{14}{9} \approx 1.556.$$

Since $c = \dfrac{14}{9} \in (1, 2)$, Cauchy's Mean Value Theorem is verified. ■

Example 4.3.5 (See Figure 45 in the Atlas)**.** Verify Cauchy's Mean Value Theorem for $f(x) = \sin x$ and $g(x) = \cos x$ on $\left[0, \frac{\pi}{2}\right]$.

Solution. The trigonometric functions $f(x) = \sin x$ and $g(x) = \cos x$ are continuous on $\left[0, \frac{\pi}{2}\right]$ and differentiable on $\left(0, \frac{\pi}{2}\right)$. Also, $g'(x) = -\sin x \neq 0$ for all $x \in \left(0, \frac{\pi}{2}\right)$.

Evaluating the functions at $a = 0$ and $b = \frac{\pi}{2}$:

$$\begin{aligned} f(0) &= 0, & f\left(\frac{\pi}{2}\right) &= 1 \implies f\left(\frac{\pi}{2}\right) - f(0) = 1, \\ g(0) &= 1, & g\left(\frac{\pi}{2}\right) &= 0 \implies g\left(\frac{\pi}{2}\right) - g(0) = -1. \end{aligned}$$

The quotient of the increments is:

$$\frac{f(\pi/2) - f(0)}{g(\pi/2) - g(0)} = \frac{1}{-1} = -1.$$

Compute the derivatives $f'(x) = \cos x$ and $g'(x) = -\sin x$. Setting the ratio at $x = c$:

$$\frac{f'(c)}{g'(c)} = \frac{\cos c}{-\sin c} = -\cot c.$$

Equating the two expressions:

$$-\cot c = -1 \implies \cot c = 1 \implies c = \frac{\pi}{4}.$$

Since $c = \dfrac{\pi}{4} \in \left(0, \frac{\pi}{2}\right)$, Cauchy's Mean Value Theorem is verified. ■

4.4 Introduction to Taylor's Theorem

Taylor's theorem is a powerful result that allows us to approximate a sufficiently differentiable function by a polynomial, called the **Taylor polynomial**. The theorem also provides a precise formula for the error, or **remainder**, in this approximation.

Let f be a function such that its first n derivatives exist in a neighborhood of a point a. The Taylor polynomial of degree n for f at a is:

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n$$

Taylor's Theorem states that $f(x) = P_n(x) + R_n(x)$, where $R_n(x)$ is the remainder term. The specific form of $R_n(x)$ depends on which version of the theorem is used.

4.4.1 Taylor's Theorem with Lagrange's Form of Remainder

This is the most common form of the remainder and is a direct generalization of the Mean Value Theorem.

Theorem 4.4.1 (Taylor's Theorem with Lagrange's Remainder). *Let f be a function such that its first n derivatives are continuous on a closed interval $[a, x]$ and its $(n+1)$-th derivative, $f^{(n+1)}$, exists on the open interval (a, x). Then there exists at least one number $c \in (a, x)$ such that:*

$$f(x) = P_n(x) + R_n(x)$$

where the remainder $R_n(x)$ is given by:

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

Proof. Define an auxiliary function $F(t) = f(x) - f(t) - f'(t)(x-t) - \frac{f''(t)}{2!}(x-t)^2 - \cdots - \frac{f^{(n)}(t)}{n!}(x-t)^n$. The derivative of $F(t)$ with respect to t is:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n$$

This is found by applying the product rule to each term, which results in a telescoping cancellation.

Now define another function $G(t) = F(t) - \left(\frac{x-t}{x-a}\right)^{n+1} F(a)$. This function is continuous on $[a, x]$ and differentiable on (a, x). We check the values at the endpoints:

- $G(x) = F(x) - \left(\frac{x-x}{x-a}\right)^{n+1} F(a) = 0 - 0 = 0$. (Note that $F(x) = 0$ by definition).
- $G(a) = F(a) - \left(\frac{x-a}{x-a}\right)^{n+1} F(a) = F(a) - F(a) = 0$.

Since $G(a) = G(x) = 0$, by Rolle's Theorem, there exists a $c \in (a, x)$ such that $G'(c) = 0$. Differentiating $G(t)$:

$$G'(t) = F'(t) + (n+1)\frac{(x-t)^n}{(x-a)^{n+1}}F(a)$$

At $t = c$, we have $G'(c) = 0$, so $F'(c) = -(n+1)\frac{(x-c)^n}{(x-a)^{n+1}}F(a)$. Substituting the expression for $F'(c)$:

$$-\frac{f^{(n+1)}(c)}{n!}(x-c)^n = -(n+1)\frac{(x-c)^n}{(x-a)^{n+1}}F(a)$$

Solving for $F(a)$ (and noting $x \neq c$):

$$F(a) = \frac{f^{(n+1)}(c)}{n!}\frac{(x-a)^{n+1}}{n+1} = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

By definition, $F(a)$ is the remainder $R_n(x) = f(x) - P_n(x)$. This completes the proof. ■

Example 4.4.2. Use the second-degree Taylor polynomial for $f(x) = \ln(1+x)$ centered at $a = 0$ to approximate $\ln(1.1)$. Then use Lagrange's remainder to find an upper bound for the error in this approximation.

Solution. The derivatives of $f(x) = \ln(1+x)$ at $a = 0$ are $f(0) = 0$, $f'(0) = 1$, and $f''(0) = -1$. The second-degree Taylor polynomial is therefore $P_2(x) = x - \frac{x^2}{2}$. To approximate $\ln(1.1)$, we set $x = 0.1$:

$$\ln(1.1) \approx P_2(0.1) = 0.1 - \frac{(0.1)^2}{2} = 0.095$$

The error in this approximation is given by the Lagrange remainder term $R_2(x) = \frac{f'''(c)}{3!}x^3$ for some c between 0 and x. The third derivative is $f'''(x) = \frac{2}{(1+x)^3}$. For $x = 0.1$, the error is:

$$R_2(0.1) = \frac{1}{3!} \cdot \frac{2}{(1+c)^3} \cdot (0.1)^3 = \frac{0.001}{3(1+c)^3} \quad \text{for some } c \in (0, 0.1)$$

To find an upper bound for the magnitude of this error, we must maximize the expression. This occurs when the denominator is minimized, which is when $c = 0$.

$$|R_2(0.1)| \leq \frac{0.001}{3(1+0)^3} = \frac{0.001}{3} \approx 0.000333$$

The error in our approximation is less than 0.000334. ∎

4.4.2 Taylor's Theorem with Cauchy's Form of Remainder

This form of the remainder is often used in theoretical contexts, such as proving the convergence of Taylor series.

Theorem 4.4.3 (Taylor's Theorem with Cauchy's Remainder). *Let f be a function satisfying the same conditions as for Lagrange's form. Then there exists at least one number $c \in (a, x)$ such that the remainder $R_n(x)$ is given by:*

$$R_n(x) = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a)$$

Proof. We use the same auxiliary function $F(t)$ as in the previous proof, where $F'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n$. Now, consider the simple function $\phi(t) = x - t$. Applying Cauchy's Mean Value Theorem to the functions $F(t)$ and $\phi(t)$ on the interval $[a, x]$, there exists a $c \in (a, x)$ such that:

$$\frac{F(x) - F(a)}{\phi(x) - \phi(a)} = \frac{F'(c)}{\phi'(c)}$$

We evaluate each piece:

- $F(x) = 0$
- $\phi(x) = x - x = 0$
- $\phi(a) = x - a$
- $\phi'(t) = -1$, so $\phi'(c) = -1$

Substituting these into the CMVT equation:

$$\frac{0 - F(a)}{0 - (x-a)} = \frac{F'(c)}{-1} \implies \frac{-F(a)}{-(x-a)} = -F'(c)$$

$$\frac{F(a)}{x-a} = \frac{f^{(n+1)}(c)}{n!}(x-c)^n$$

Solving for $F(a)$, which is the remainder $R_n(x)$:

$$R_n(x) = F(a) = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a)$$

This completes the proof. ■

Example 4.4.4. Consider the same approximation as before: $f(x) = \ln(1+x)$ approximated by $P_2(x) = x - x^2/2$ at $x = 0.1$. Use Cauchy's remainder to find an upper bound for the error.

Solution. The remainder $R_n(x)$ in Cauchy's form is $R_n(x) = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a)$. For our problem, $n = 2$, $a = 0$, $x = 0.1$, and $f'''(x) = \frac{2}{(1+x)^3}$. The error is:

$$R_2(0.1) = \frac{f'''(c)}{2!}(0.1-c)^2(0.1-0) = \frac{2/(1+c)^3}{2}(0.1-c)^2(0.1)$$

$$R_2(0.1) = \frac{(0.1-c)^2}{(1+c)^3}(0.1) \quad \text{for some } c \in (0, 0.1)$$

To bound this error, we can find an upper bound for the fractional term $g(c) = \frac{(0.1-c)^2}{(1+c)^3}$ on the interval $c \in (0, 0.1)$. The numerator $(0.1-c)^2$ is maximized when c is at its minimum ($c = 0$), and the denominator $(1+c)^3$ is also minimized at $c = 0$. Therefore, the entire fraction is bounded by its value at $c = 0$:

$$\frac{(0.1-c)^2}{(1+c)^3} \leq \frac{(0.1-0)^2}{(1+0)^3} = 0.01$$

The error is thus bounded by:

$$|R_2(0.1)| \leq (0.01) \cdot (0.1) = 0.001$$

This bound is correct, though less tight than the one found using Lagrange's form. ■

Taylor's Series and Maclaurin's Series

Definition 4.4.5 (Taylor's Series)**.** If a function $f(x)$ has derivatives of all orders at a point $x = a$, then the **Taylor series** for $f(x)$ centered at a is the power series given by:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

Definition 4.4.6 (Maclaurin's Series)**.** A **Maclaurin series** is a special case of a Taylor series centered at the point $a = 0$.

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

This is the most common type of Taylor series used for elementary functions.

4.4.3 Standard Maclaurin Series Expansions

Below are the standard Maclaurin series for several fundamental functions, along with their intervals of convergence.

Exponential Function: e^x

For $f(x) = e^x$, we have $f^{(n)}(x) = e^x$ for all n. At $x = 0$, $f^{(n)}(0) = e^0 = 1$.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Sine Function: $\sin x$

For $f(x) = \sin x$, the derivatives at $x = 0$ follow a pattern: $0, 1, 0, -1, \dots$ This results in a series with only odd powers of x.

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Cosine Function: $\cos x$

For $f(x) = \cos x$, the derivatives at $x = 0$ follow a pattern: $1, 0, -1, 0, \dots$ This results in a series with only even powers of x.

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Logarithmic Function: $\log(1 + x)$

The Maclaurin series for $\log x$ itself does not exist because $\log(0)$ is undefined. We find the series for $f(x) = \log(1 + x)$ instead. The derivatives at $x = 0$ are $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, $f'''(0) = 2$, etc.

$$\log(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Binomial Series: $(1 + x)^m$

For $f(x) = (1 + x)^m$, where m is any real number, the derivatives are $f^{(n)}(x) = m(m - 1) \dots (m - n + 1)(1 + x)^{m-n}$. At $x = 0$, this gives $f^{(n)}(0) = m(m - 1) \dots (m - n + 1)$. This leads to the generalized binomial coefficient $\binom{m}{n} = \dfrac{m(m-1)\dots(m-n+1)}{n!}$.

$$(1 + x)^m = \sum_{n=0}^{\infty} \binom{m}{n} x^n = 1 + mx + \frac{m(m-1)}{2!} x^2 + \frac{m(m-1)(m-2)}{3!} x^3 + \dots$$

Interval of Convergence: $(-1, 1)$. The convergence at the endpoints $x = \pm 1$ depends on the value of m. If m is a non-negative integer, the series is finite (a polynomial) and converges everywhere.

4.5 Envelope of a One-Parameter Family of Curves

Definition 4.5.1 (Envelope)**.** An **envelope** of a one-parameter family of curves is a curve that is tangent to each member of the family at some point. The family of curves is represented by an equation of the form $F(x, y, \alpha) = 0$, where α is the parameter.

4.5.1 Method for Finding the Envelope

To find the envelope, we treat the parameter α as a variable and solve the system of equations formed by the family and its partial derivative with respect to the parameter.

1. The family of curves: $F(x, y, \alpha) = 0$.
2. The partial derivative: $\frac{\partial F}{\partial \alpha}(x, y, \alpha) = 0$.

The equation of the envelope is found by eliminating the parameter α from these two equations.

Example 4.5.2 (See Figure 49 in the Atlas)**.** Find the envelope of the family of straight lines $y = mx + \frac{a}{m}$, where m is the parameter.

Solution. Write the family in implicit form:

$$F(x, y, m) \equiv y - mx - \frac{a}{m} = 0$$

Differentiate with respect to the parameter m:

$$\frac{\partial F}{\partial m} = -x - a\left(-\frac{1}{m^2}\right) = -x + \frac{a}{m^2} = 0$$

The derivative condition gives

$$-x + \frac{a}{m^2} = 0 \implies m^2 = \frac{a}{x} \implies m = \pm\sqrt{\frac{a}{x}}$$

We now eliminate m. From the family equation, From $y = mx + a/m$, we can write $y = m(x + a/m^2)$. From the derivative, we know $x = a/m^2$, so:

$$y = m(x + x) = 2mx$$

Now we solve for m to get $m = y/(2x)$. Substitute this into the relation $m^2 = a/x$:

$$\left(\frac{y}{2x}\right)^2 = \frac{a}{x} \implies \frac{y^2}{4x^2} = \frac{a}{x}$$

Multiplying by $4x^2$ (assuming $x \neq 0$) gives the envelope equation:

$$y^2 = 4ax$$

The envelope is a parabola. ■

4.6 Envelope of a Two-Parameter Family of Curves

Definition 4.6.1. A two-parameter family of curves is given by an equation of the form $F(x, y, \alpha, \beta) = 0$. For an envelope to exist, the two parameters must be related by a constraint equation, $g(\alpha, \beta) = 0$.

4.6.1 Method for Finding the Envelope

The envelope is found by eliminating the parameters α and β from a system of three equations. This method is related to Lagrange multipliers.

1. The family of curves: $F(x, y, \alpha, \beta) = 0$.

2. The constraint equation: $g(\alpha, \beta) = 0$.
3. The condition of proportionality of partial derivatives:

$$\frac{\partial F/\partial \alpha}{\partial g/\partial \alpha} = \frac{\partial F/\partial \beta}{\partial g/\partial \beta}$$

Example 4.6.2. Find the envelope of the family of lines $\frac{x}{\alpha} + \frac{y}{\beta} = 1$, where the parameters α and β are related by the constraint $\alpha + \beta = c$, for a constant c.

Solution. The family of curves and the constraint are:

$$F(x, y, \alpha, \beta) \equiv \frac{x}{\alpha} + \frac{y}{\beta} - 1 = 0 \quad \text{and} \quad g(\alpha, \beta) \equiv \alpha + \beta - c = 0$$

We find the four necessary partial derivatives:

$$\frac{\partial F}{\partial \alpha} = -\frac{x}{\alpha^2} \qquad \frac{\partial F}{\partial \beta} = -\frac{y}{\beta^2}$$
$$\frac{\partial g}{\partial \alpha} = 1 \qquad \frac{\partial g}{\partial \beta} = 1$$

Now, set up the proportionality equation:

$$\frac{-x/\alpha^2}{1} = \frac{-y/\beta^2}{1} \implies \frac{x}{\alpha^2} = \frac{y}{\beta^2}$$

This implies $\frac{\sqrt{x}}{\alpha} = \frac{\sqrt{y}}{\beta}$. Let this common ratio be k.

$$\frac{\sqrt{x}}{\alpha} = k \implies \alpha = \frac{\sqrt{x}}{k} \quad \text{and} \quad \frac{\sqrt{y}}{\beta} = k \implies \beta = \frac{\sqrt{y}}{k}$$

Substitute these expressions for α and β into the constraint $\alpha + \beta = c$:

$$\frac{\sqrt{x}}{k} + \frac{\sqrt{y}}{k} = c \implies \frac{\sqrt{x} + \sqrt{y}}{k} = c \implies k = \frac{\sqrt{x} + \sqrt{y}}{c}$$

Finally, substitute α and β into the original family equation $\frac{x}{\alpha} + \frac{y}{\beta} = 1$:

$$x\left(\frac{k}{\sqrt{x}}\right) + y\left(\frac{k}{\sqrt{y}}\right) = 1 \implies k\sqrt{x} + k\sqrt{y} = 1 \implies k(\sqrt{x} + \sqrt{y}) = 1$$

Now substitute the expression for k we found:

$$\left(\frac{\sqrt{x} + \sqrt{y}}{c}\right)(\sqrt{x} + \sqrt{y}) = 1 \implies (\sqrt{x} + \sqrt{y})^2 = c$$

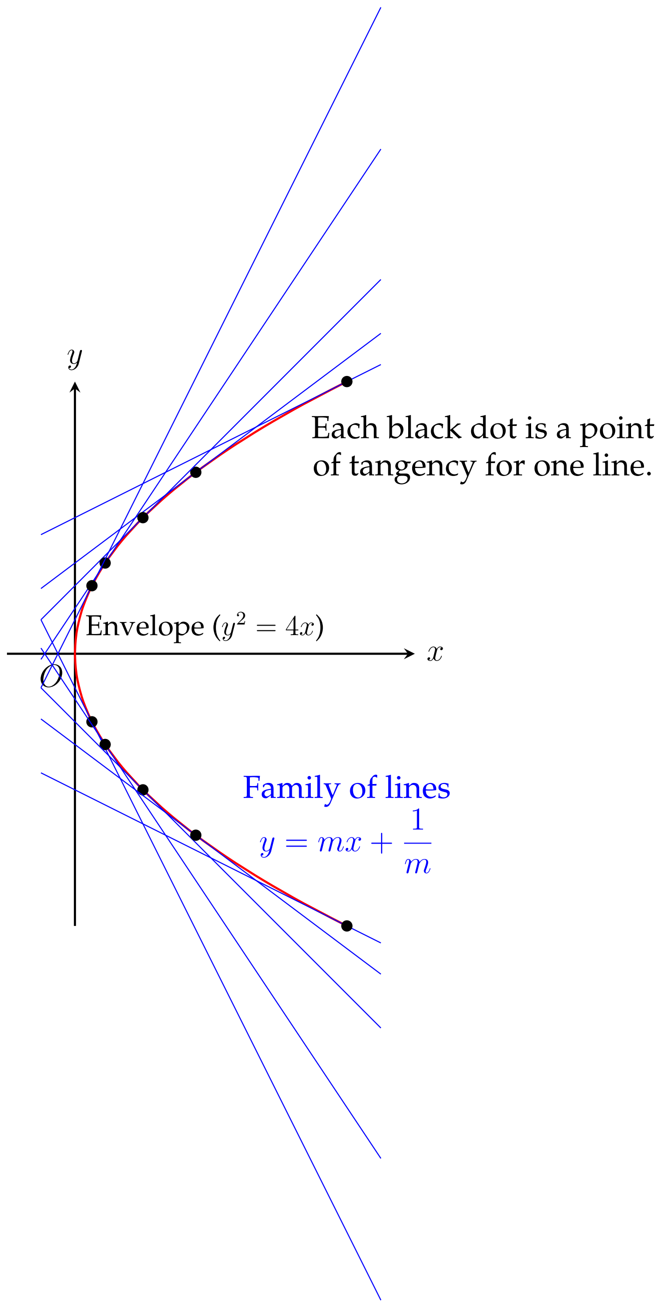
The equation of the envelope is $\sqrt{x} + \sqrt{y} = \sqrt{c}$. This is also a parabola. ■

Visualizing an Envelope

The concept of an envelope can be understood visually. Consider a family of curves, such as the set of straight lines given by the equation $y = mx + \frac{1}{m}$. Each value of the parameter

m defines a different line. The **envelope** is a new curve that is tangent to every one of these lines at some point.

In Figure 49 in the Atlas, several lines from the family are shown together with their envelope. As you can see, they collectively outline a distinct shape. This shape, shown as a thick red curve, is the envelope. The envelope itself is not part of the original family, but it is "carved out" by the family of lines. For this specific family of lines, the envelope is the parabola $y^2 = 4x$.



See Figure 49 in the Atlas for the family of lines and its parabolic envelope.

Exercise 4.6.3. Solve the following exercises:

1. Verify Rolle's Theorem for the function $f(x) = (x-a)^m(x-b)^n$ on $[a, b]$, where $m, n \in \mathbb{N}$.

2. Using Rolle's Theorem, prove that between any two real roots of $e^x \cos x = 1$, there lies at least one real root of $e^x \sin x = 1$.

3. Verify Rolle's Theorem for $f(x) = e^{-x} \sin x$ on $[0, \pi]$.

4. Prove using Rolle's Theorem that the polynomial equation $x^3 - 3x + k = 0$ cannot have two distinct real roots in the closed interval $[0, 1]$ for any real constant k.

5. Verify Lagrange's Mean Value Theorem for $f(x) = \sqrt{x^2 - 4}$ on $[2, 4]$.

6. Using Lagrange's Mean Value Theorem, establish the inequality:

$$\frac{x}{1+x} < \ln(1+x) < x \quad \forall x > 0.$$

7. Verify Cauchy's Mean Value Theorem for $f(x) = e^x$ and $g(x) = e^{-x}$ on $[a, b]$.

8. Verify Cauchy's Mean Value Theorem for $f(x) = \dfrac{1}{x^2}$ and $g(x) = \dfrac{1}{x}$ on $[a, b]$, where $0 < a < b$.

9. Expand $f(x) = e^x$ in powers of $(x - 1)$ up to the term containing $(x - 1)^4$ using Taylor's series.

10. Derive Maclaurin's series expansion for $f(x) = \ln(1 + x)$ and determine its interval of convergence.

11. Expand $f(x) = \sin x$ in powers of $\left(x - \frac{\pi}{2}\right)$ using Taylor's series.

12. State Taylor's theorem with Lagrange's form of remainder, and write the n-term Maclaurin expansion of $f(x) = (1 + x)^m$.

13. Using the Maclaurin series for $f(x) = \cos x$, estimate $\cos(0.1)$ correct to four decimal places.

14. Obtain Maclaurin's series expansion for $f(x) = \tan^{-1} x$ up to the term involving x^5.

15. Find the envelope of the one-parameter family of straight lines $y = mx + \dfrac{a}{m}$, where m is the parameter ($m \neq 0$).

16. Find the envelope of the one-parameter family of circles $(x - a)^2 + y^2 = 4a$, where a is the parameter.

17. Find the envelope of the family of straight lines $x \cos \alpha + y \sin \alpha = p$, where α is the parameter and p is a constant.

18. Find the envelope of the two-parameter family of ellipses $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ under the linear parameter constraint $a + b = c$, where c is a constant.

19. Find the envelope of the two-parameter family of straight lines $\dfrac{x}{a} + \dfrac{y}{b} = 1$ subject to the constant area constraint $ab = k^2$.

A Concluding Note to Students

"Galileo once wrote that the book of Nature is written in the language of mathematics. If that is true, then Calculus is the dynamic grammar of that language—it is the art of capturing motion, growth, curvature, and change in a single, harmonious framework."

DEAR STUDENTS,

Congratulations on completing Calculus – I!
Take a moment to reflect on how far we have come. We began by stepping into the rigorous world of ϵ–δ definitions, mastering the delicate boundary between continuity and differentiability. From there, we learned to untangle indeterminate forms, navigate the graceful geometry of polar curves, study intrinsic curvature and use Taylor and Maclaurin series expansions.
What we have acquired in this course is far more than a collection of computational formulas. We have developed a **mathematical mindset**—a way of seeing the world through continuous change, breaking complex problems down into infinitesimals, and reconstructing them into profound truths.
Calculus is the dynamic language powering modern physics, artificial intelligence, quantum computing, aerospace engineering, and finance. As we step forward into advanced mathematics and applied sciences, the foundational tools we forged here will serve as our permanent compass.

Keep your curiosity alive! Never fear a difficult problem—every mathematical challenge you encounter is simply a hidden beauty waiting to be unveiled. Einstein once said, "I am thankful for all of those who said NO to me. It's because of them I'm doing it myself."

DR. AIJAZ
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JK Higher Education Department
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Atlas of Figures

Standard Curves and Geometrical Results

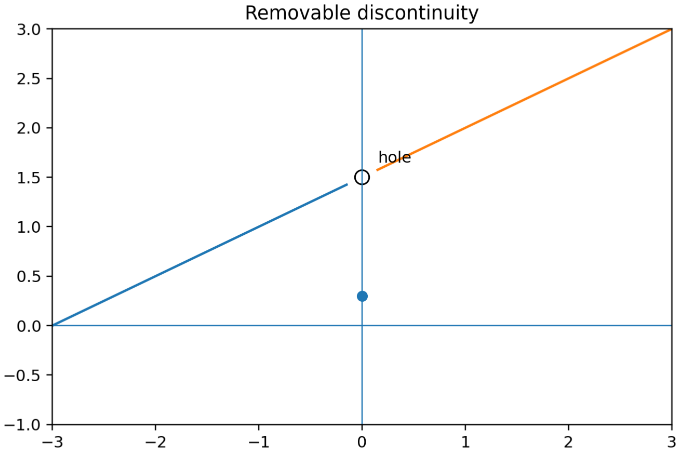

Fig. 1. Removable discontinuity

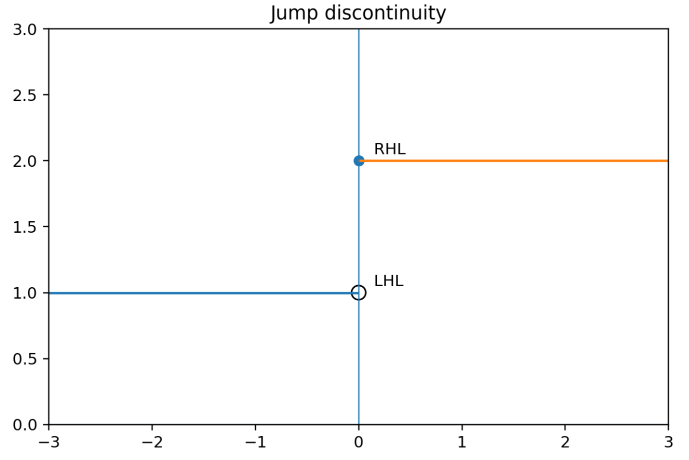

Fig. 2. Jump discontinuity

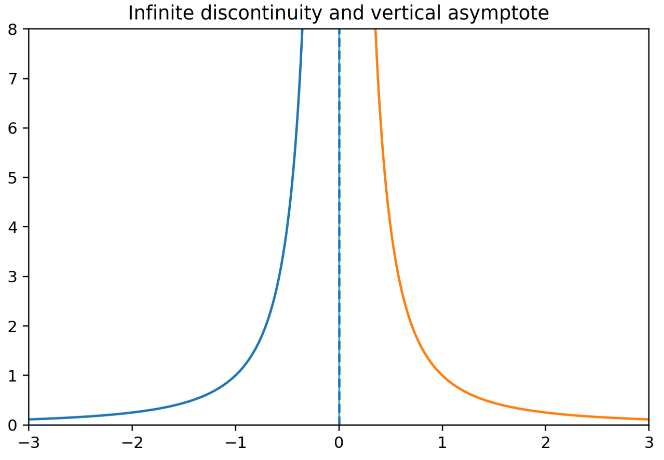

Fig. 3. Infinite discontinuity

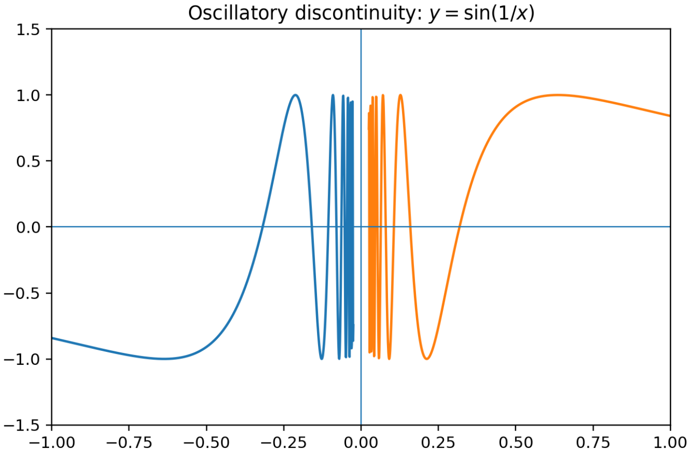

Fig. 4. Oscillatory discontinuity

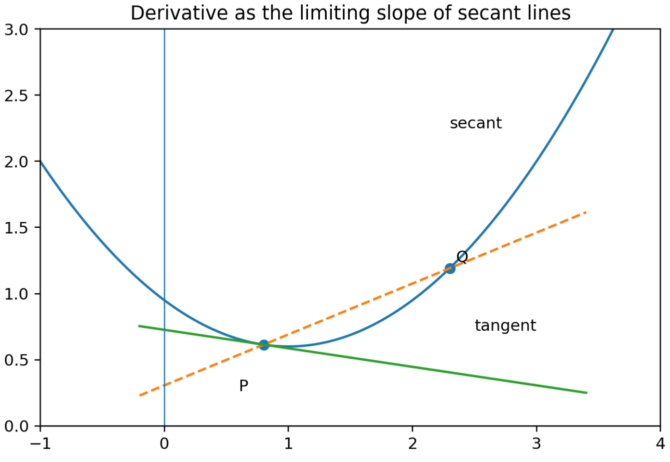

Fig. 5. Derivative: secant to tangent

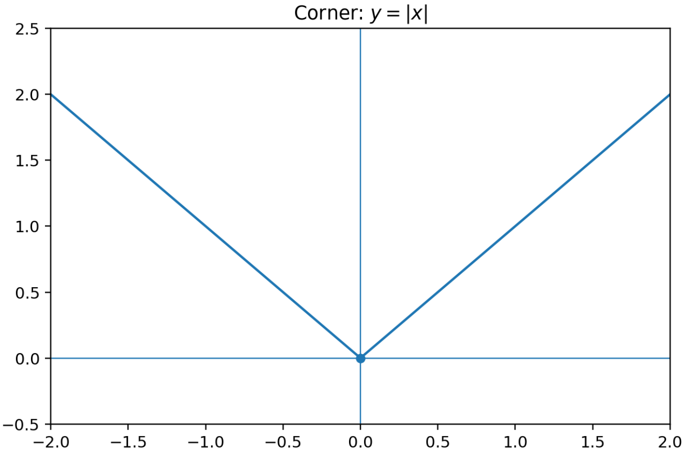

Fig. 6. Corner point

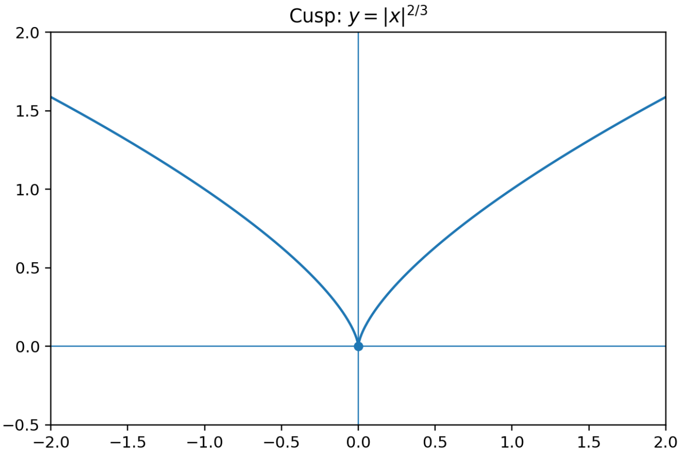

Fig. 7. Cusp

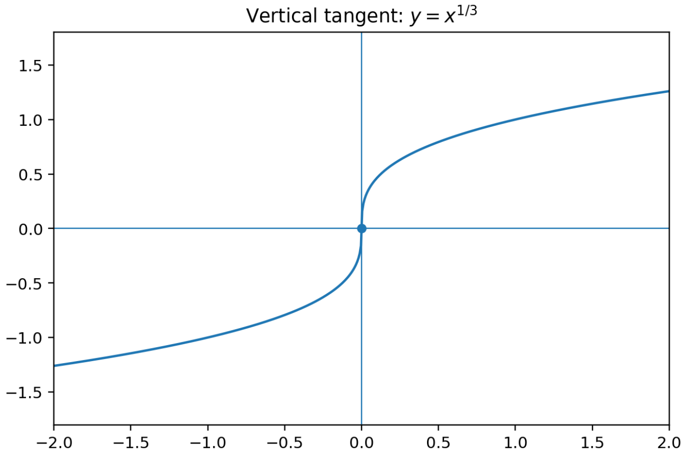

Fig. 8. Vertical tangent

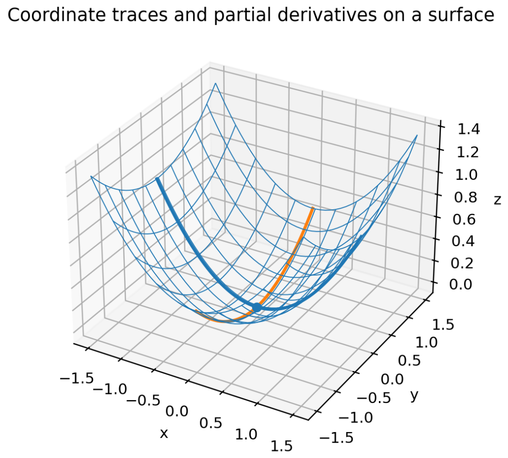

Fig. 9. Partial derivatives on a surface

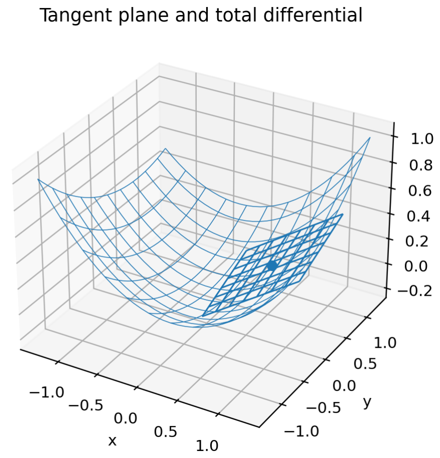

Fig. 10. Tangent plane and total differential

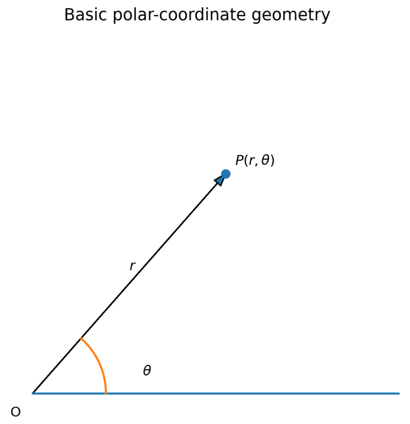

Fig. 11. Basic polar-coordinate geometry

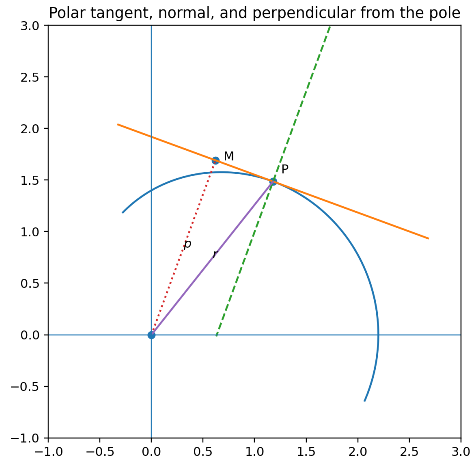

Fig. 12. Polar tangent, normal and perpendicular

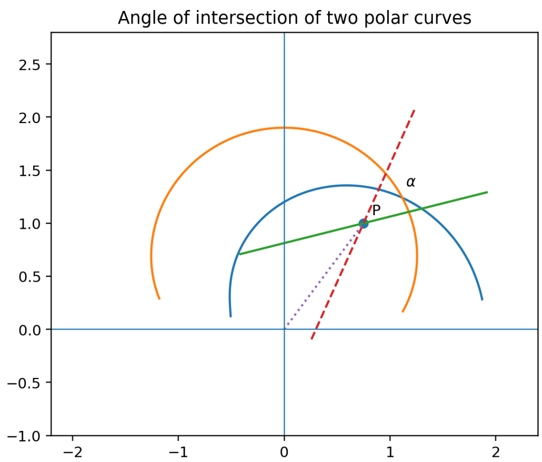

Fig. 13. Angle of intersection of polar curves

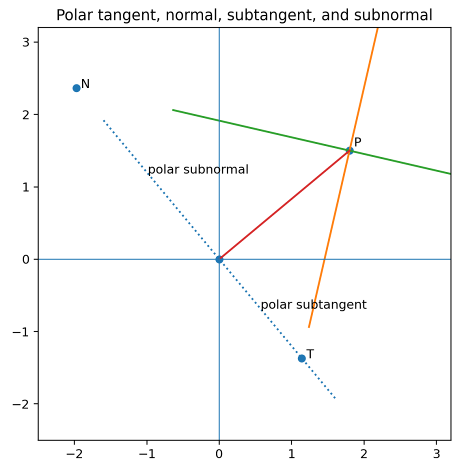

Fig. 14. Polar tangent, normal, subtangent and subnormal

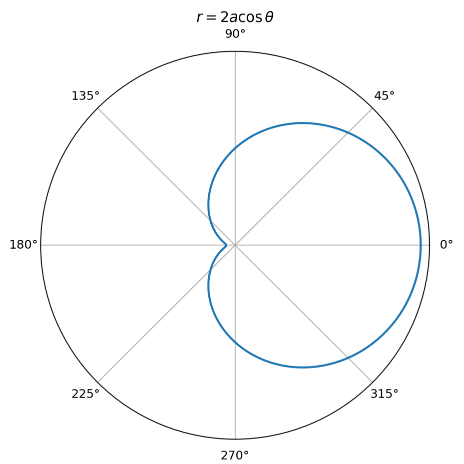

Fig. 15. Circle: r = 2a cos(theta)

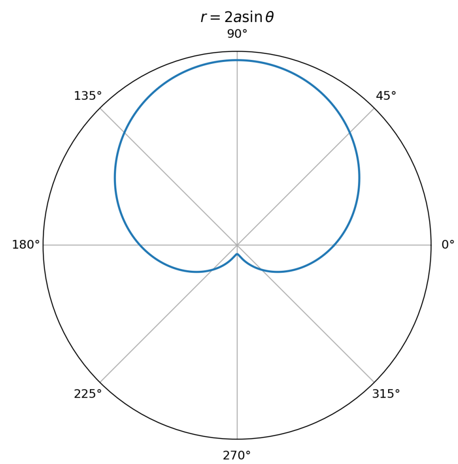

Fig. 16. Circle: r = 2a sin(theta)

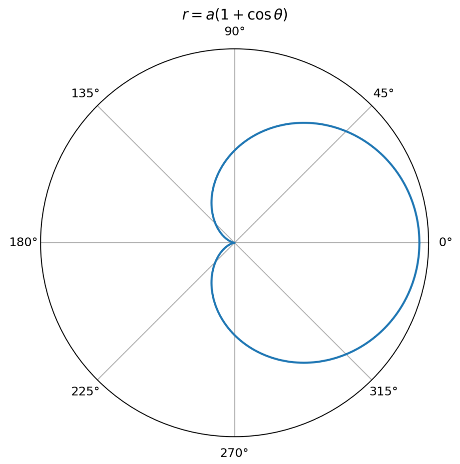

Fig. 17. Cardioid: r = a(1 + cos(theta))

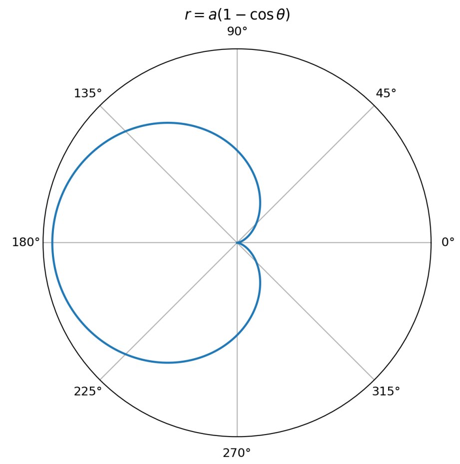

Fig. 18. Cardioid: r = a(1 - cos(theta))

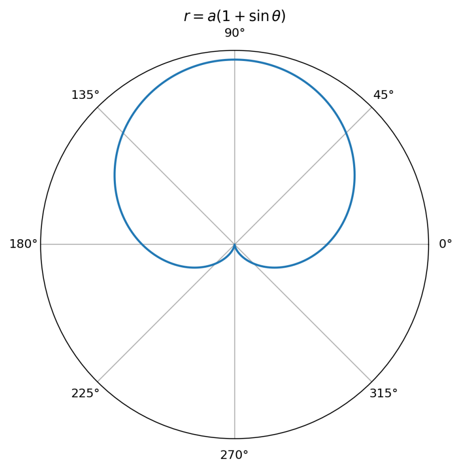

Fig. 19. Cardioid: r = a(1 + sin(theta))

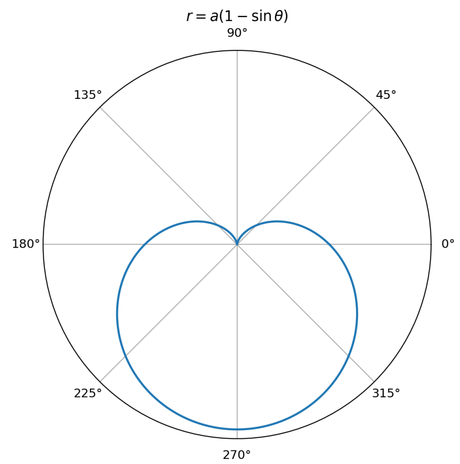

Fig. 20. Cardioid: r = a(1 - sin(theta))

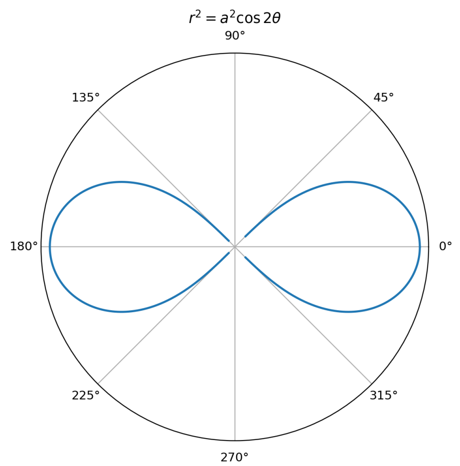

Fig. 21. Lemniscate: r^2 = a^2 cos(2 theta)

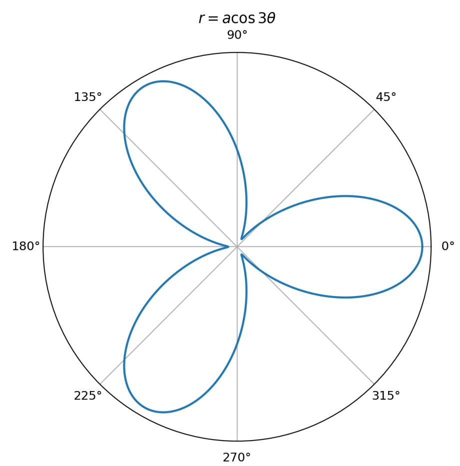

Fig. 22. Three-petalled rose

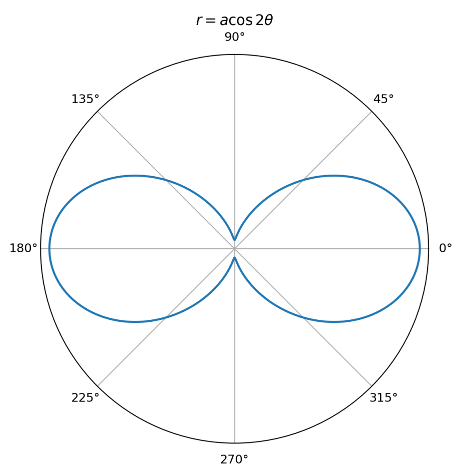

Fig. 23. Four-petalled rose

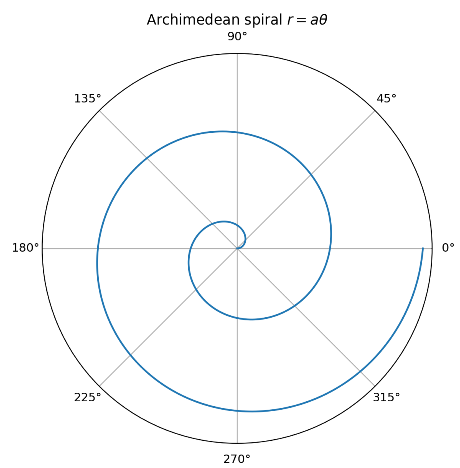

Fig. 24. Archimedean spiral

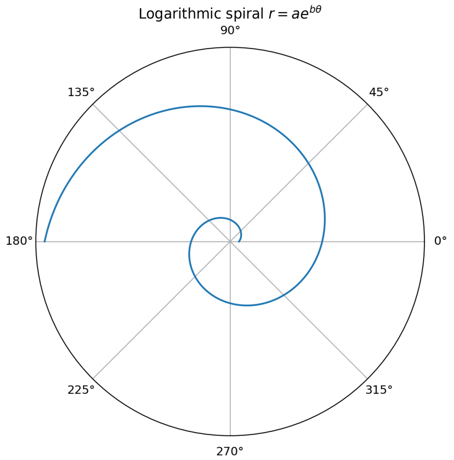

Fig. 25. Logarithmic spiral

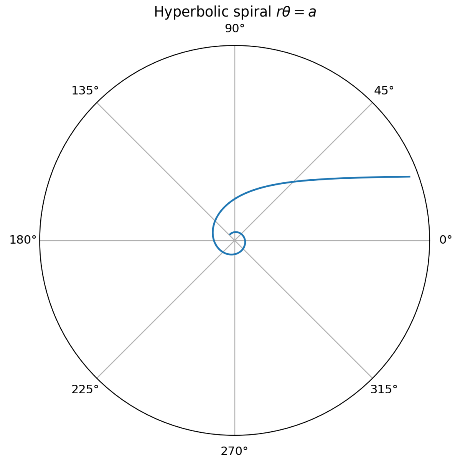

Fig. 26. Hyperbolic spiral

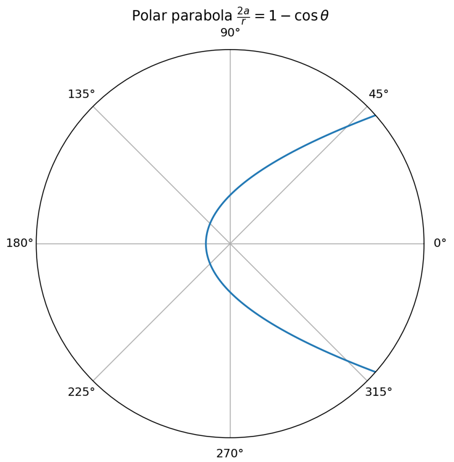

Fig. 27. Polar parabola

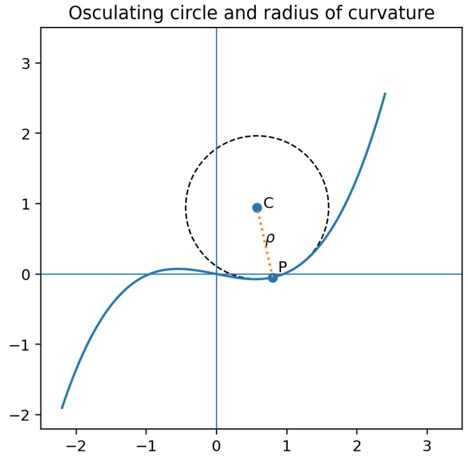

Fig. 28. Osculating circle and radius of curvature

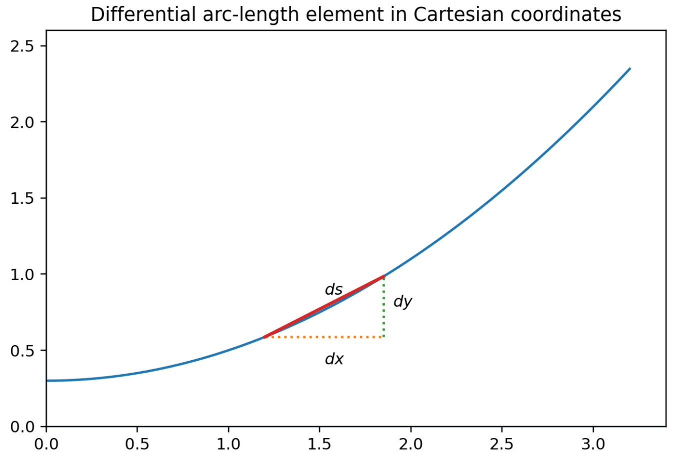

Fig. 29. Cartesian arc-length element

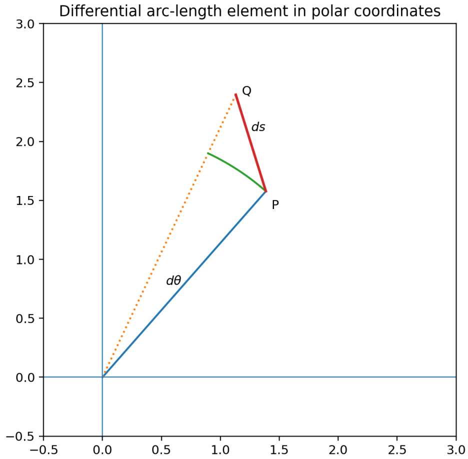

Fig. 30. Polar arc-length element

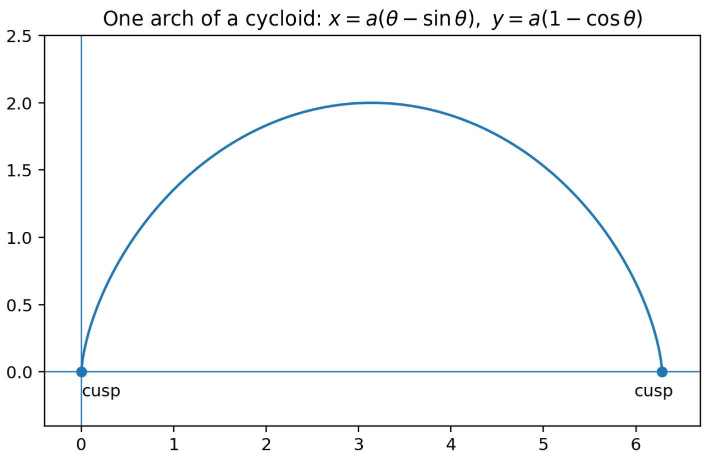

Fig. 31. Cycloid

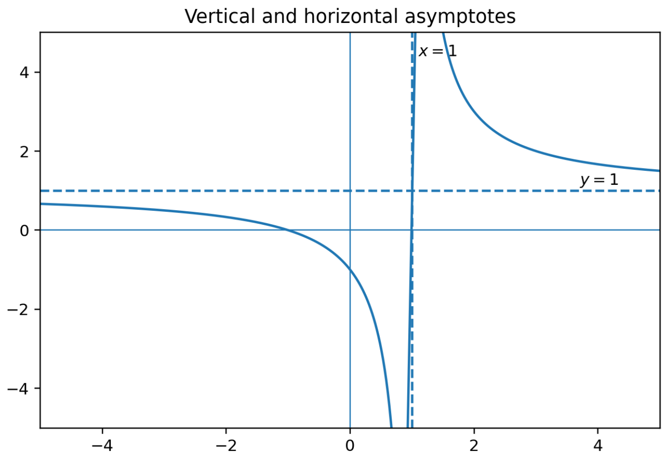

Fig. 32. Vertical and horizontal asymptotes

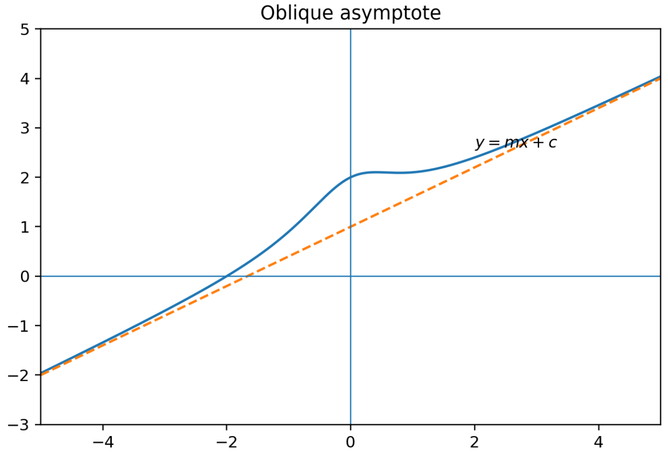

Fig. 33. Oblique asymptote

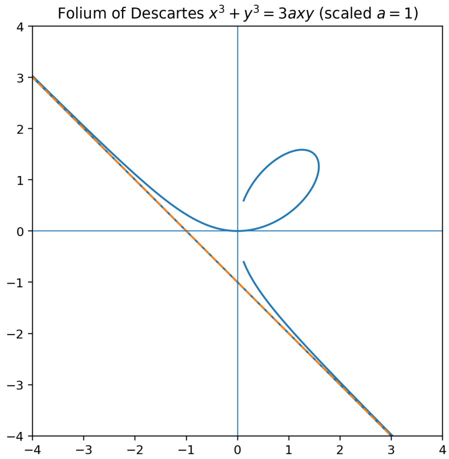

Fig. 34. Folium of Descartes

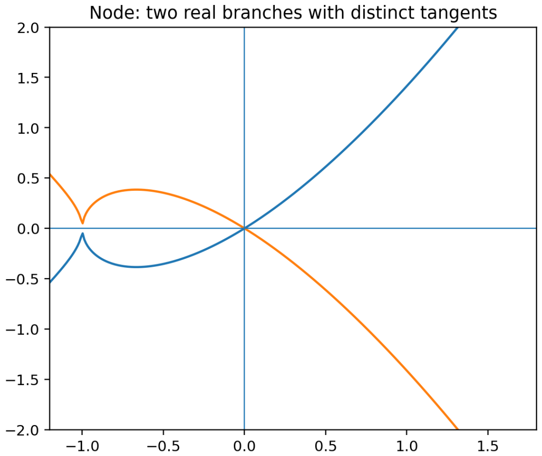

Fig. 35. Singular point: node

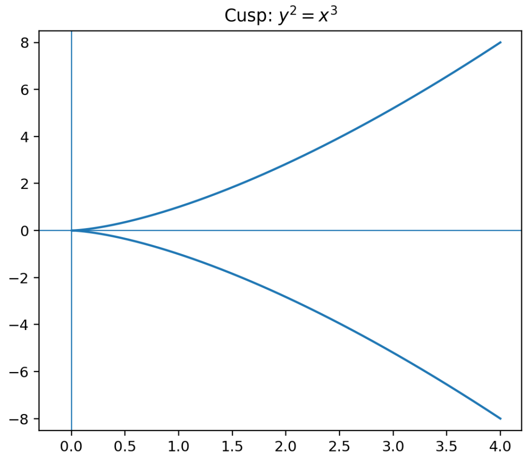

Fig. 36. Singular point: cusp

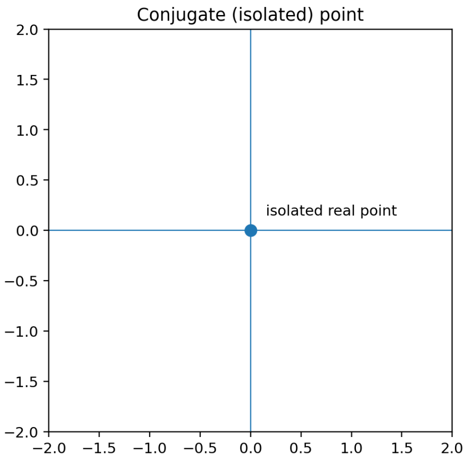

Fig. 37. Singular point: isolated/conjugate point

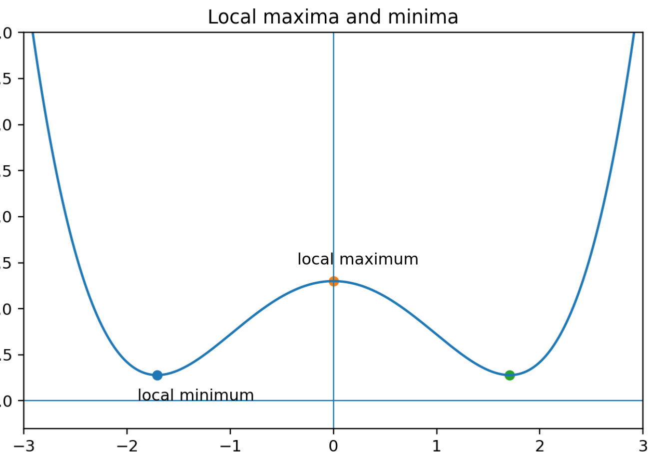

Fig. 38. Local maxima and minima

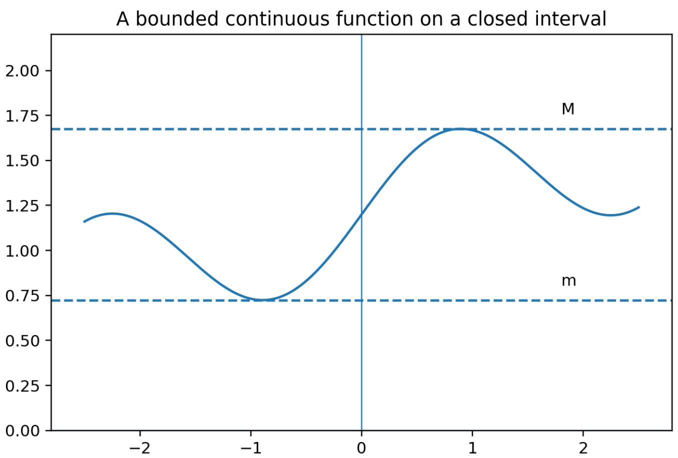

Fig. 39. Boundedness theorem

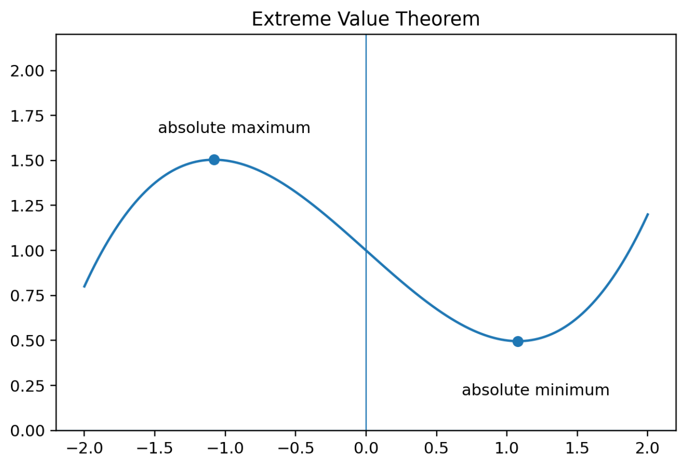

Fig. 40. Extreme Value Theorem

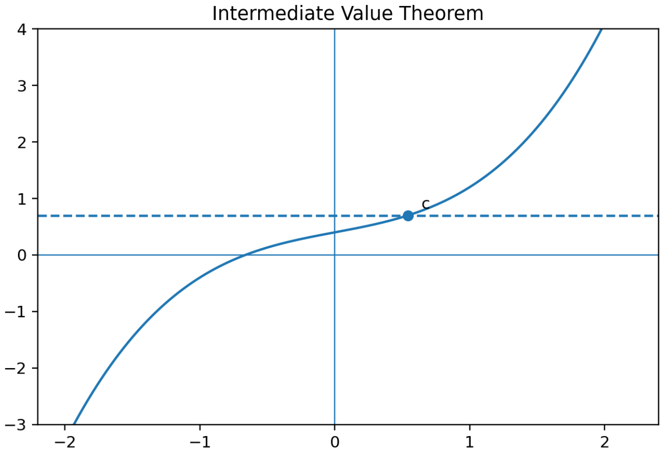

Fig. 41. Intermediate Value Theorem

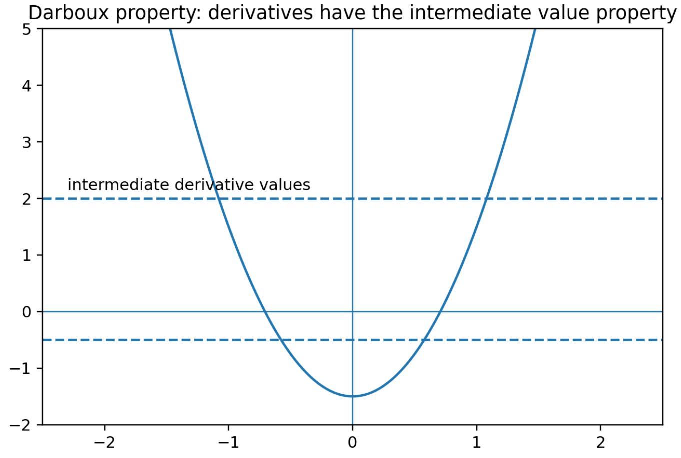

Fig. 42. Darboux theorem

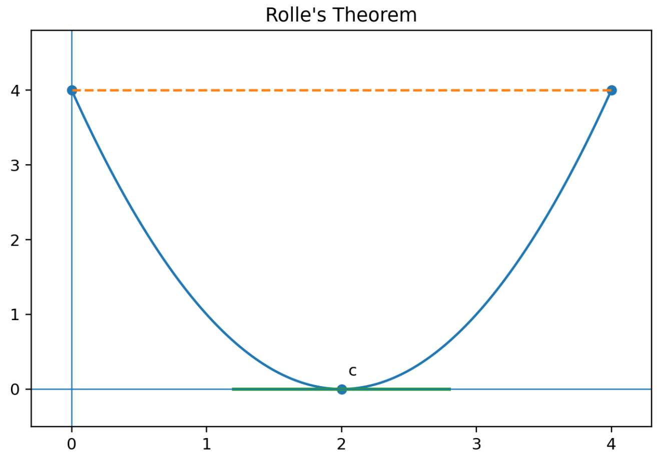

Fig. 43. Rolle's Theorem

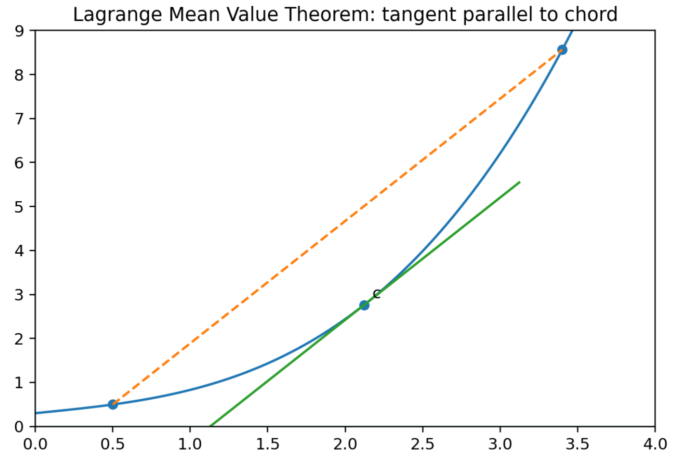

Fig. 44. Lagrange Mean Value Theorem

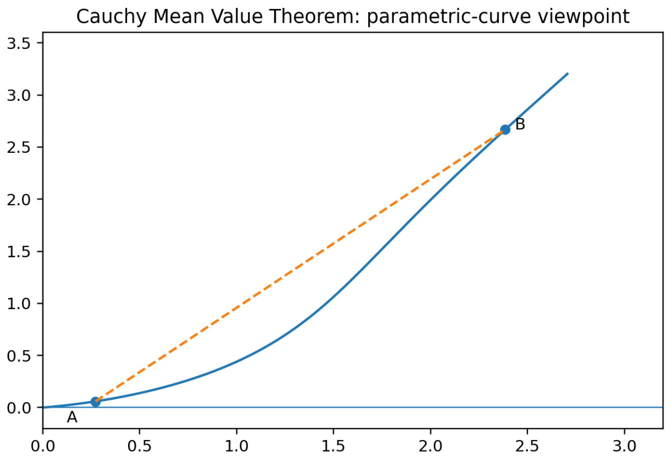

Fig. 45. Cauchy Mean Value Theorem

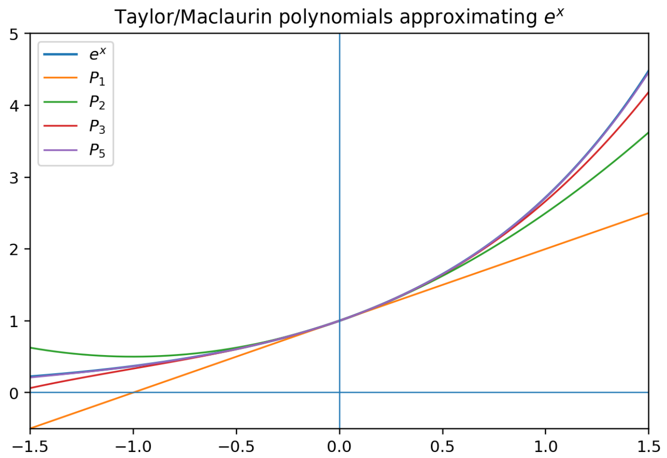

Fig. 46. Taylor approximation to exp(x)

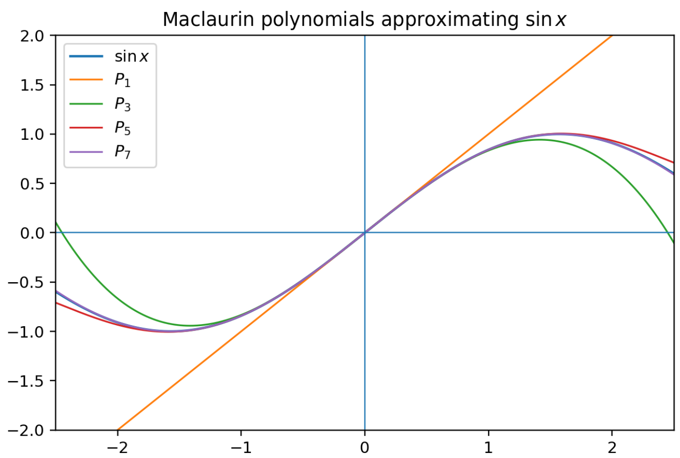

Fig. 47. Maclaurin approximation to sin(x)

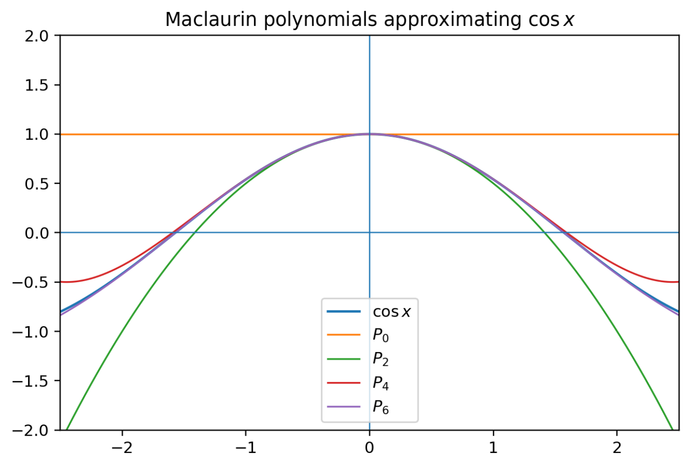

Fig. 48. Maclaurin approximation to cos(x)

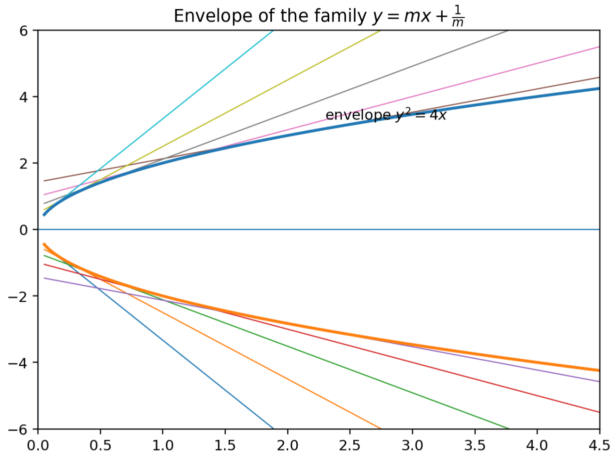

Fig. 49. Envelope of y = mx + 1/m

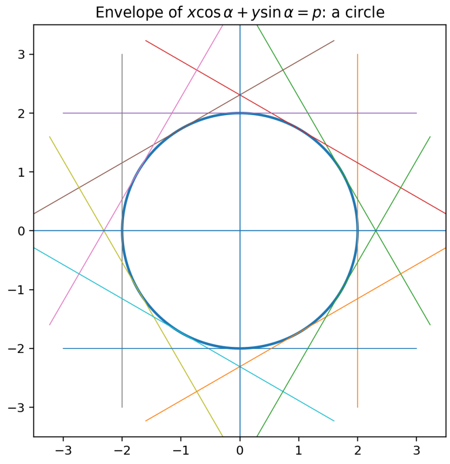

Fig. 50. Envelope of tangent lines to a circle